



WHOAMI

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  /mohamedeid404/



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IT & Cloud Fundamentals

💡 All From [Eisa Abo Sherif](#) & [Stephane Maarek](#)



• IT Fundamental:



- اي اجهزة احنا بنستخدمها اسمها Computing Device

- هتلاقى فى الاجهزة دي حاجات مشتركة كتير وهيا:

1. CPU (Central Processing Unit):

- وحدة المعالجة المركزية .. دا المخ اللى بي process كل حاجة
- بي process , decode , execute , fetch كل ال instructions اللى جباله من ال programs
- الاداء بتاعه بيتقاس بعدد ال cores وال clock speed علاقة طردية
- جوا ال CPU حاجة اسمها Registers ودي عباره عن small fast storage بيستخدمها بديلة عن ال RAM لانها اسرع منها .. بيستخدمها ف حالة انو عاوز ي quick access على حاجة.
- فى ال CPU الجديدة هتلاقى small amount of high-speed memory تستخدم فال cashing علشان موضوع ال processing مياخدش وقت اطول



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- فى ال CPU الجديدة بيكون فيها Multi-core Processing يعنى تقدر ت perform multiple operations فى نفس الوقت
- سرعته بتتقاس بال GHz

2. GPU (Graphics Processing Unit):

- عبارة عن processor مخصص انو ي process ال graphic renders واي processing task خاصة بالجرافيك زي ال images , animation , videos, ال 3D environment

3. RAM (Random Access Memory):

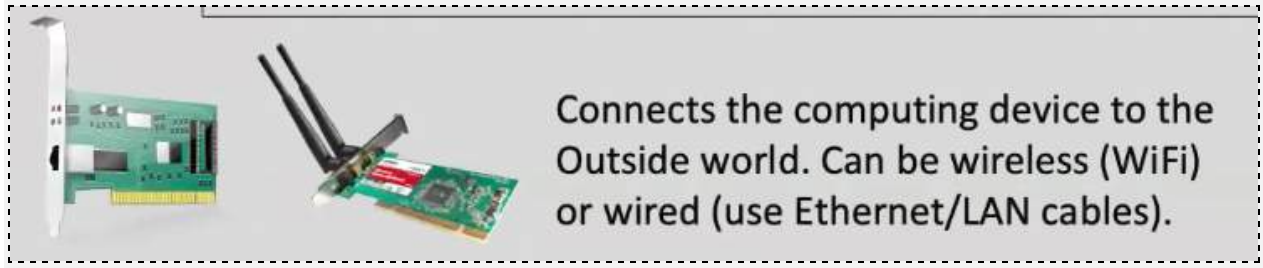
- اسرع من ال SSD , HDD
- عبارة عن temporary storage لل data اللي بيستخدمها ال CPU وي process it .. بتخلي ال CPU ي Access الداتا وي process it بطريقة سريعة من خلالها زي انك ت run app او ت open file
- كل ما تكون ال RAM اكبر كل ما تكون قادر انك تعمل Multi tasking يعنى تفتح برامج وتعمل حاجات كتير فى نفس الوقت بدون ما الجهاز يهجن
- بتستخدم برودو فى ال buffer , caching
- لو ال RAM مش كافية ممكن انك تاخد من ال HDD او SSD جزء portion ك virtual memory
- سرعتها بتتقاس بال Giga byte

4. HDD & SSD

- بيخزنو الداتا العادية وال SSD اسرع من ال SSD
- سرعتها بتتقاس بـ Giga byte او tera



5. Network Cards / Interface



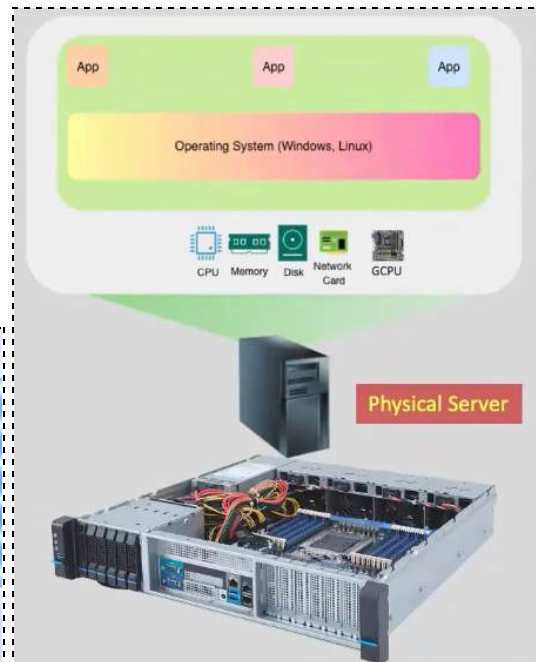
- عن طريقها بنوصل لل internet سواء wireless زي اللاب كذا او لو كمبيوتر ممكن بكابل او ممكن ب وصلة تخليه يلقط واي فاي.
- سرعتها بتتقاس بال **Mbps or Gbps** .. ميغا او جيجا بايت / ثانية

اتكلمنا عن كل اللي فات دة لان دول نفس مكونات ال Servers اللي بتبقى موجودة في ال

Data Centers

➡ Cloud

● Physical Server (Powerful Computers):



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- سيكون عبارة عن نفس المكونات بتاعة الكمبيوتر العادي ولكن اكبر بمراحل ومحطوتين في Racks زي اللي فى الصورة دي وال Racks دي فى اماكن اسمها Data Centers.
- بين ال server hardware وال Applications سيكون موجود ال Operating System اللي بيوصل ال requests لل hardware علشان يتعملها process ويرجع تاني ال Applications لل responds.

● Data Centers:



- دي الاماكن المجهزة اللي بيكون فيها ال servers بتاعتنا .. حماية وكاميرات وانظمة مراقبة وتكيفات وكل حاجة تحافظ على ان ال servers دي تفضل شغالة ومحدث يخرقها.
- هتلاقى جوا يمكن شوارع ويمكن levels فوق بعض طوابق .. ف بيبقى عندك كميات رهيبه .. بتتقسم فى كابينات كذا اللي هيا بالطول دي اسمها racks بيكون جواها ال servers وعلي كل حاجة stickers بحيث نعرف ال server دة بتاع مين وتابع لاي

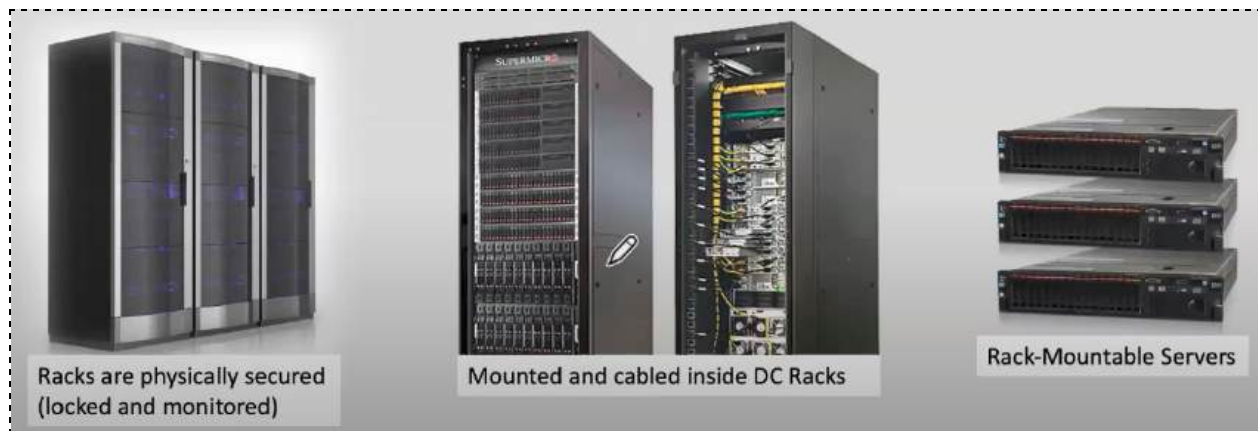


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تيم ولا اي business unit وممكن يكون مكتوب عليه كمان ال IP address ..
الارضيه حتي دي بتبقي تحتها كابلات.

- كل ال servers دي بتبقي معزولة عن بعضها .. يقدر يتكلمو من خلال ال network لو انت سامحلهم او محتاج دة.
- ممكن تاخذ server او rack او room كاملة .. كل حاجة متاحة على حسب احتياجاتك

● المكونات بالتفصيل :



- **Rack-Mountable Servers** : دي ال Physical servers وال
Rack-Mountable دة نوع معين .. ال servers دي بناخدها نخطها في ال Racks
ال Racks دي من الخلف بيكون فيها ال network interface وال storage وال power وكدا ..
ال servers دي بقي اللى بيكون عليها ممكن تطبيقات عادية زي اوبر او مثلا data bases ..
ال Racks دي كل واحدة فيهم بتكون secured ومراقبة والمبني كمان مترقب وكله تحت الكنترول 😊

● أنواع ال Data Centers :

- **On-Premises** : بيكون مملوك للهيئة او الشركة او الحكومة وهو اللى بيتحمل كل التكاليف
من الصفر .. OS , Virtualization , hardware , storage , network , building , Data , App
- **On-Cloud** : عالكلاود زي AWS , Azure ومن مميزاته ..



● Cloud Types

1. Public & Private Cloud

- **Public** : زي AWS , Azure متاح للعامة
- **Private** : ال Data centers باللى فيها بتاعتنا ومحدث بيوصلها غيرنا ودة مهم فى ال critical use case زي البنوك او كذا

2. Hybrid Cloud

- لو هتستخدم ال Private وال Public مع بعض ف دة اسمو **Hybrid Cloud** سواء ال connection بينهم by the internet او private connectivity .. هيئة حكومية مثلا وعندها بيانات عملاء ف دي هتخليها private وبقيت البيانات دي عادي تكون عال public ف بتخليها عال public ويبقى ده **Hybrid**

التحدي بيكون فى اننا نـ Authenticate ال users مابين ال Public , private

Cloud .. يعني لو فيه user بيـ Access حاجة عال Public ومحتاج يـ Access حاجة عال

Private ف ازاي نخليه يقدر يعمل دة.

3. Multi Cloud

- اني اقسم ال work load بتاعي علي كذا cloud provider
- لو عندنا تطبيقات خاصة ب ال E-Commerce ف افضل استخدم AWS
- لو عندنا تطبيقات خاصة ب Microsoft ف افضل Azure
- لو بنك ف افضل IBM
- لو AI/Machine Learning ف Google افضل
- لو عندي جزئية خاصة بال Virtualization ف هتلاقي جزأ خاص بال VMware على كل ال

Cloud Providers



يمكن استخدام Private & Public & Multi Cloud مع بعض .. شوية data على 

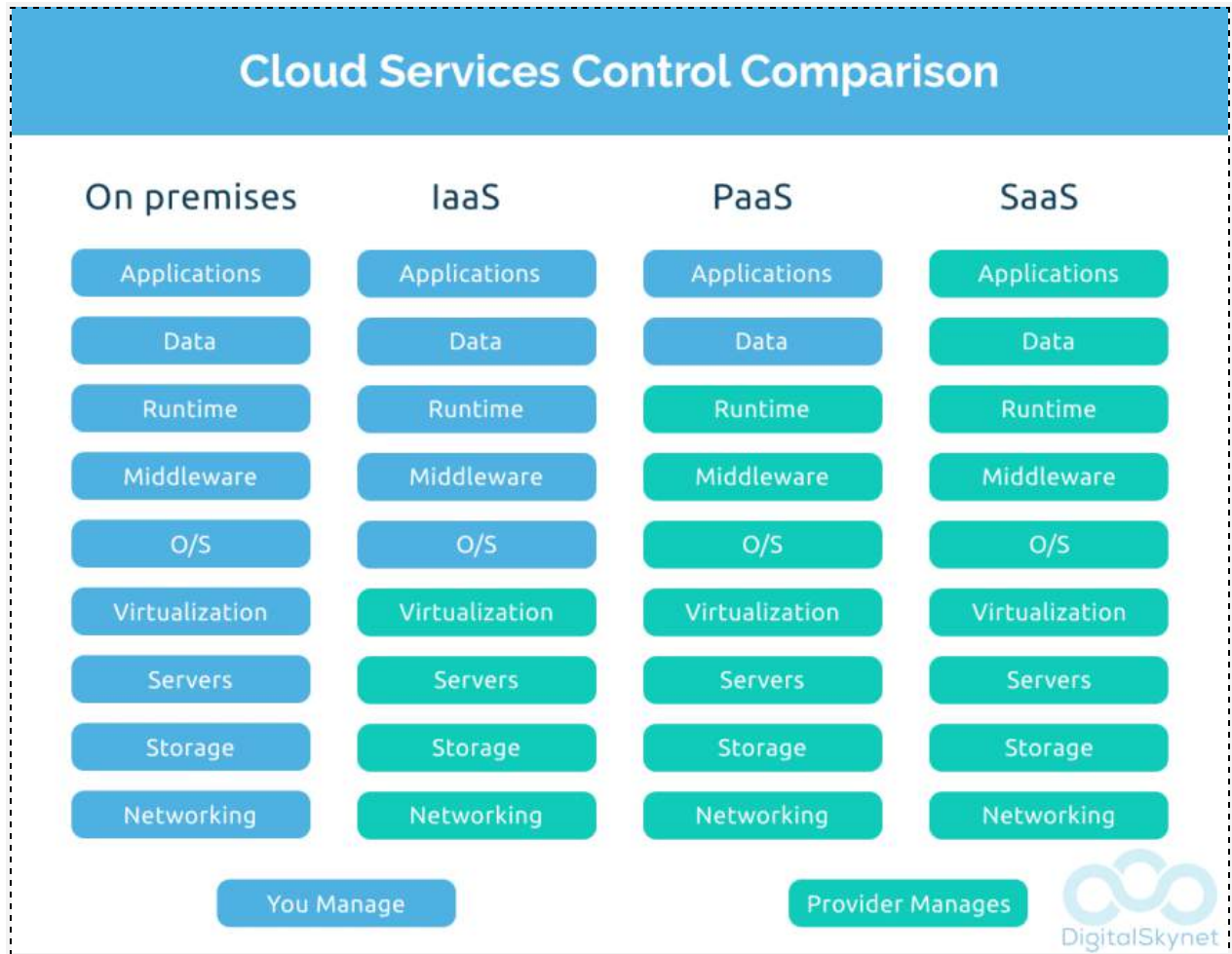
AWS وشوية على Azure وشوية على ال Private data center .. كذا دة اسمو Hybrid & Multi Cloud

● Shift from CAPEX to OPEX model :

- بوجود ال Cloud اتحولنا من CAPEX إلى OPEX .. يعني بدل ما انا همدف واكلف واعمل Data Center واصرف عليه من الصفر .. لأ انا بس همدف تكاليف التشغيل فقط ودة اللي وفرتو ال cloud providers زي AWS , Azure واي حاجة تانية بتكون من مسؤولية ال Cloud Provider
 - **capital expenditure (النفقات الرأسمالية : CAPEX)** : ودة راس المال اللي بحتاجة علشان ابدأ بيه المشروع او ابني بيه ال data center
 - **Operational expenditure (النفقات التشغيلية : OPEX)** : ودي هيا المصاريف التشغيلية بعد ما بنيت ال data centers
 - **Consumption Based Model** : دة يعني هنشغل بمبدأ Pay As You Go ادفع على قد استخدامك.
-



- **Cloud Services:**



- **On-Premises :**

- انت اللي شايل كل حاجة من البداية للنهاية.

- **Infrastructure as a service (IaaS)**

- بتاخد فقط ال Virtual Machines زي ال EC2 instance فى AWS .. والباقي من عند ال Providers

- **Run Time** : هتشغل python ولا java ولا بتستخدم انهي interpreter

- **Middleware** : ال software اللي بين ال Applications وال OS زي Message brokers like RabbitMQ, Apache Kafka

- AWS هي الرائدة فى الجزء دة



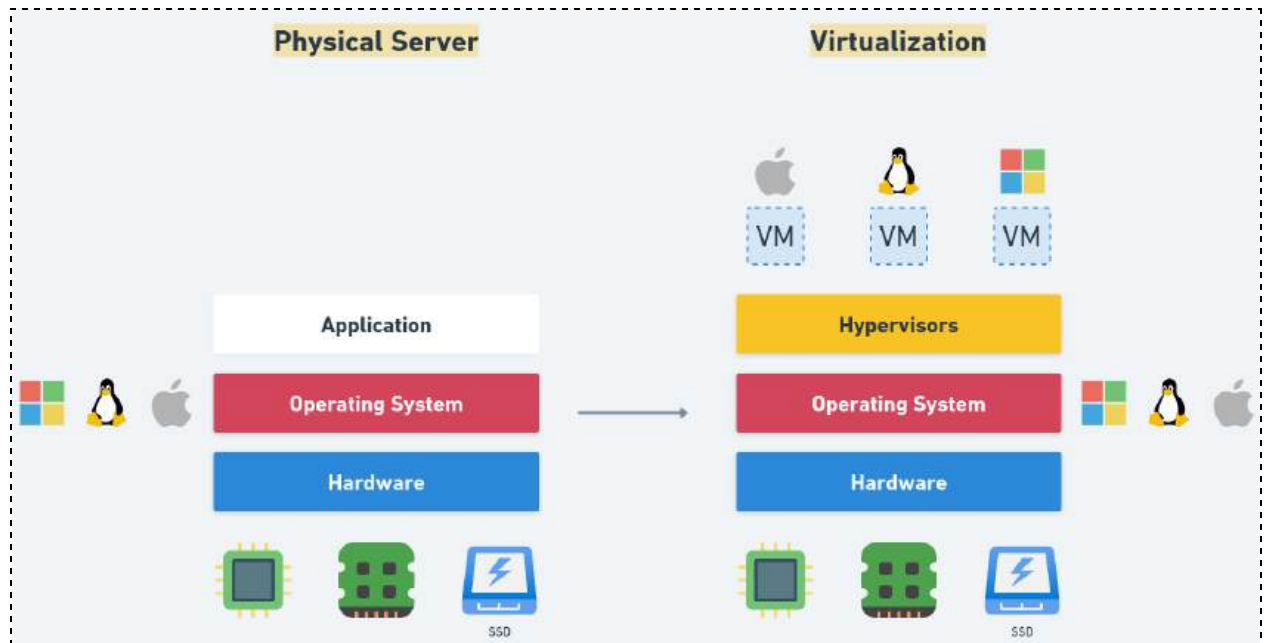
- **Platform as a service (PaaS)**

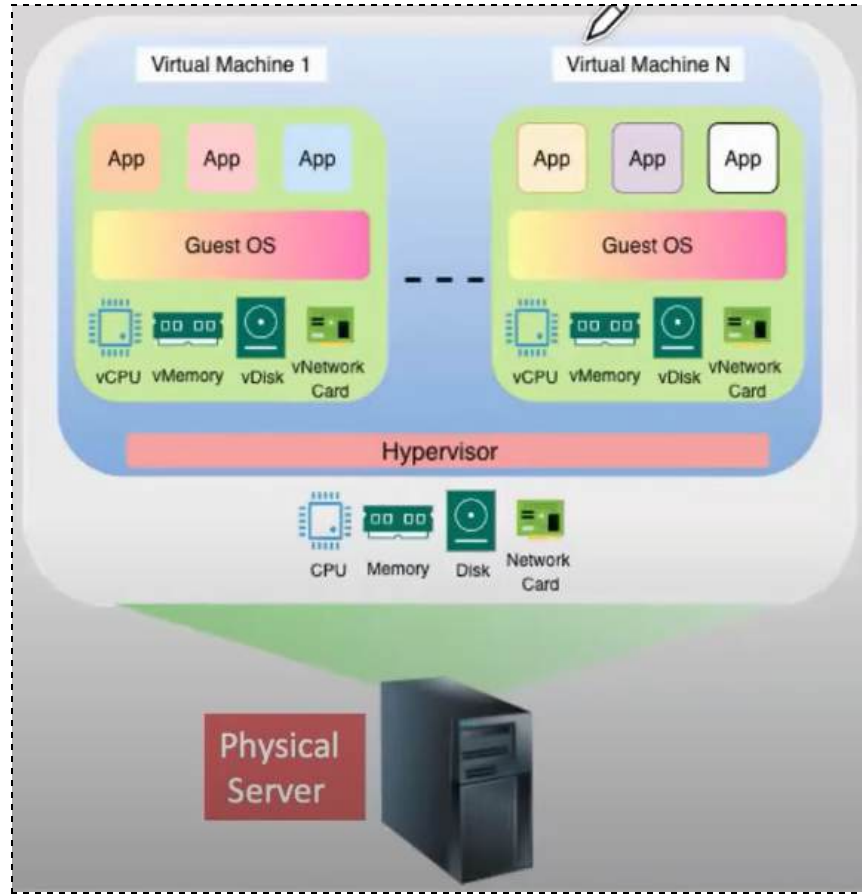
- ف ال model دة ال customer بيأجر platform جاهزة بي run عليها ال app بتاعه.

- **Software as a service (SaaS)**

- ف ال model دة ال customer بيأجر software جاهز , managed by hosted , cloud provider

- **Virtualizations:**



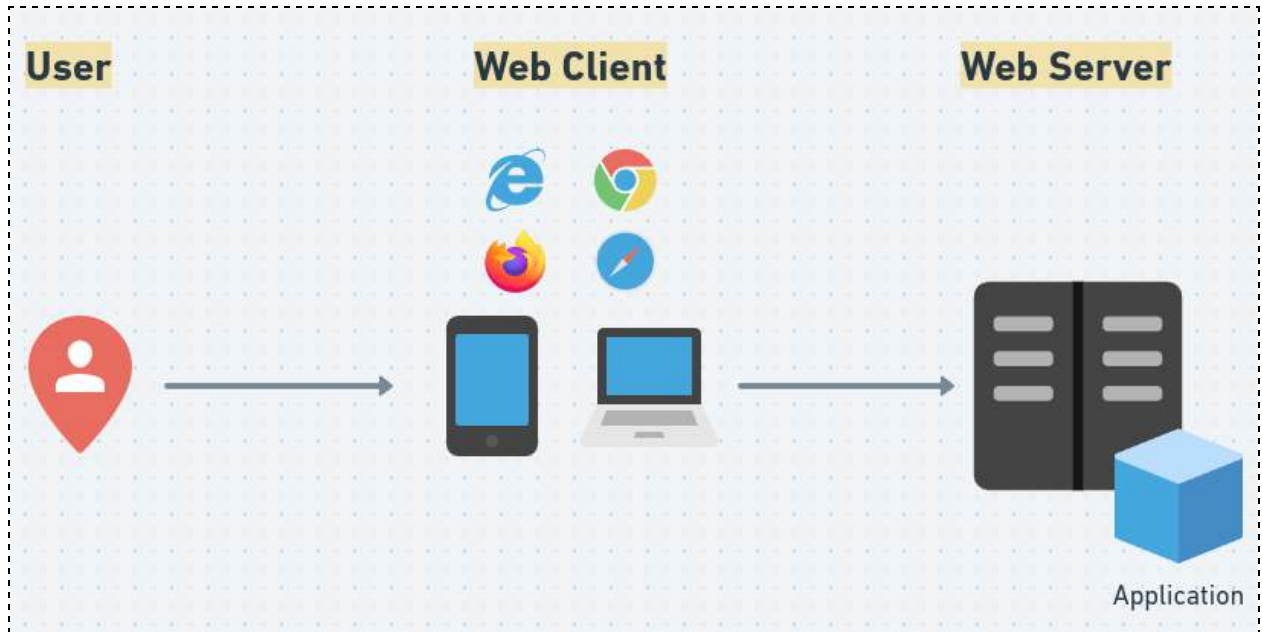


- اشتريت server .. كمبيوتر قوي جدا ومستشغلتش امكانياته كلها يبقي كدا غلط
- ال **Hypervisor** عبارة عن Software .. من خلال ال Hypervisor اللي هو المحاكى او ال Emulator بتتشارك ال Virtual Machines .. VM فى ال Hardware اللي تحت ومن امثله VMware , Virtual box , HyperV
- وانت بتكرى ال VM اللي فوق دي بتحدد لها ال Storage وال Ram وكل حاجة من ال hardware اللي هتحتاجها لانك بتكون عارف ال VM دي هينزل عليها تطبيقات اية بالظبط وعارف هيا عاوزة متطلبات اية
- كل ال VM دي بتبقى معزولة عن بعضها .. يقدر يتكلمو من خلال ال network لو انت سامحلهم ب دة.



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● Web Application:



- **Web Clients** : هي الاجهزة اللى ال user بيخس من خلالها عال web
- **Web Server** : هو ال server اللى بيستضيف ال Application بتاعنا .. يعنى لو ال server ده معانا ال ip بتاعه ودخلنا عليه هيدخلنا علطول عال Application
- اشهر تطبيقات web server ف ويندوز iis وف لينكس nginx , apache
- علشان تبعت ملفاتك لل server البعيدة بتستخدم اي software من ال FTP زي filezilla

● After Access URL:

- ال url بيتحول ل ip عن طريق ال DNS
- ال ip ده بيروح للسيرفر اللى هو ف الحالة دي ويب سيرفر ومن ال ip بيعرف انت عاوز انهى صفحة ويب او موقع يعنى ويا بيكون فيه السورس كود بتاعها او بي generate ده ويعرضها لك.

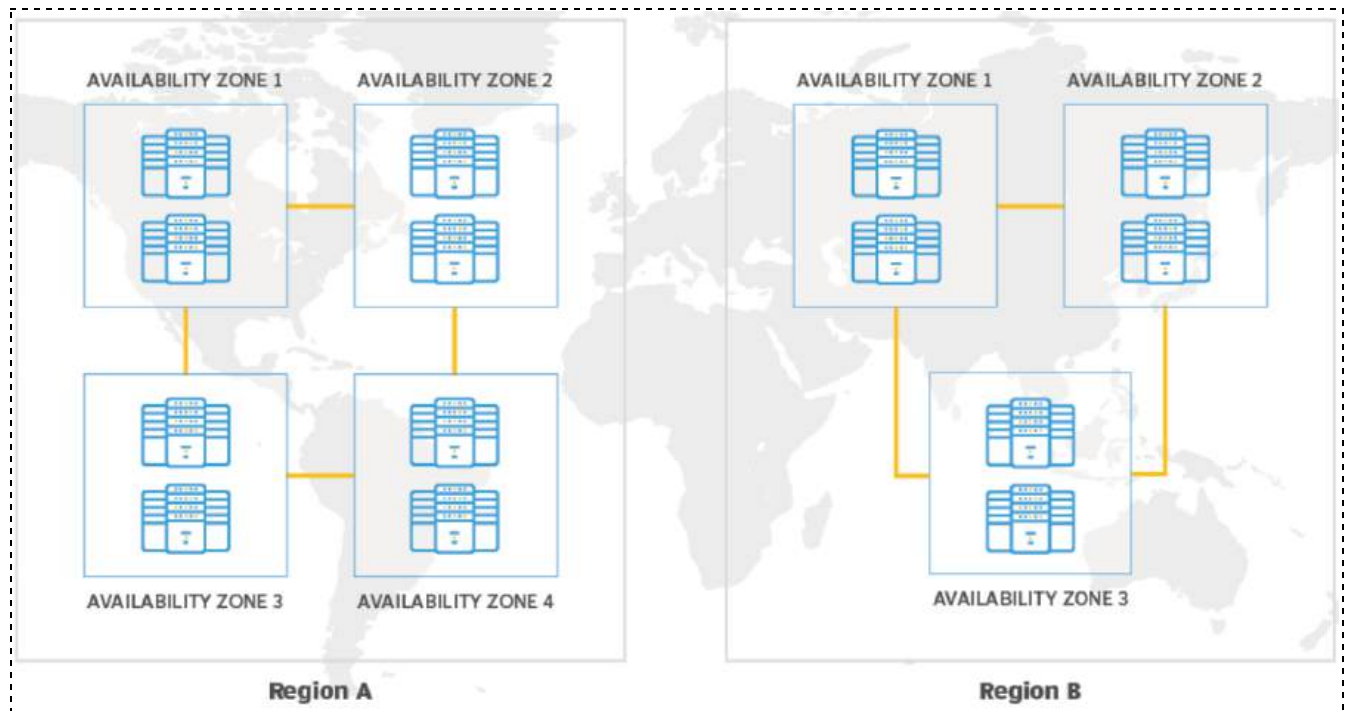


- **DNS & Hosts File:**

- ملف الـهوسـتس دة بيكون فيه الـip بتاع المواقع بمعنى انا لو عندي موقع `example.com` ف بيكون ف ملف الـهوسـتس دة لازم لية الـip معين بحيث مثلا لو بنجت عليه هيرد عليا بالـip دة او لو كتبـتـو كـ url ف بيتحول عن طريق الـdns لنفس الـip الموجود ف ملف الـهوسـتس.

CH-03 - Getting started with AWS

➔ Region:



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- منطقة جغرافية يكون جواها Availability Zones اللى بيكون جواها cluster of data centers اللى ب deploy عليها ال applications بتاعتي
- بتبقي منتشرة فى العالم وكل region منفصلة عن الثانية وكل شوية بيظهر regions جديدة بيحصل توسع يعني.
- كل region بيبقي ليها كود Sydney مثلا الكود بتاعها ap-southeast-2

● **بختار ال Region دي بناء على الآتي:**

● **Latency**

- لو عندي مثلا website و ال end user بتاعي اللى هيستخدم ال website من Europe مينفعش اروح استخدم region ف Asia مثلا هيكون فيه Latency عليه وهو بي Access ال website
- لو ال end user دة هي Access latency sensitive applications or services زي ال video gaming or real time videos مثلا مينفعش يكون فيه تاخير عليه

● **Services**

- مش كل ال regions فيها نفس ال services ممكن تلاقي service موجودة ف region وف region تانية مش موجودة

● **Cost**

- مش كل ال services ليها نفس التكلفة ف كل ال regions .. تكلفة ال services بتختلف من region للتانية

● **Compliance or Data Sovereignty (سيادة)**

- ممكن دولة يكون ليها قوانين معينة تشترط بقاء ال data داخل حدود الدولة دي مينفعش تخرج براها.



➡ Availability Zones (AZ):

- عبارة عن data centers او اكثر ومنفصلين عن بعضهم physically .. ومنفصلين وبعاد عن بعض الى حد ما يكفي ان لو حصل كارثة زي مثلا زلزال او قطع تيار كهربى او كذا ال Availability Zones بتاعتي متناثرش كلها .. الى تقع التانية تشيل مكانها وهكذا
- كل AZ ليها ال power,cooling and networking منفصلة.
- كل AZ تحتوي على عدد من ال Data Centers مش معروف
- بيكونو متوصلين ببعض ب high bandwidth, ultra-low latency networking
- من 3 اقل حاجة ل 6 اقصى حاجة فى كل region

● ف دة بيضمنك حاجتين من مميزات ال cloud :

● Disaster recovery:

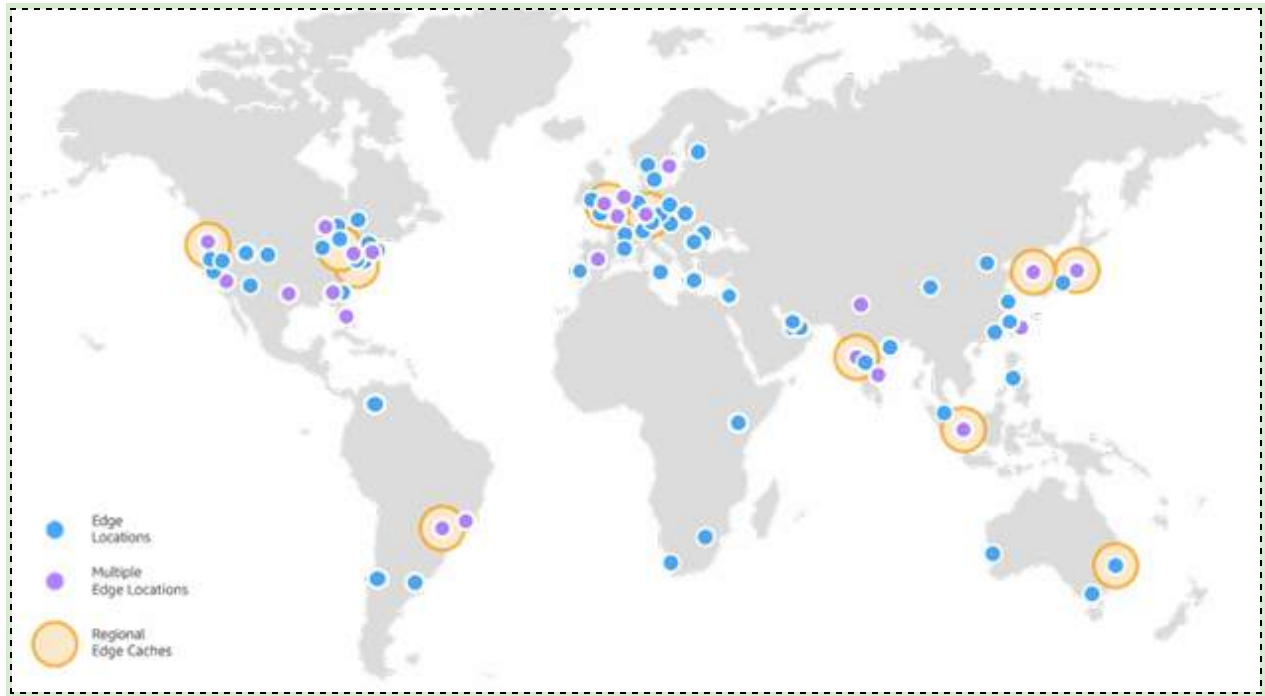
- ان لو حصل failure ف ال services بتاعتي بقدر ا recover بشكل سريع و automatic عن طريق ان بيكون عندي ال services دي فى data center تاني ف AZ تانية

● Fault tolerance:

- ودي عبارة عن ان لو عندي services ف مكان و redundant زيها ف مكان تاني لو حصل مشكلة ف المكان الاولاني ال services اللي ف المكان التاني بتاخذ مكانها من غير ما ال end user يحس



➡ AWS Points of Presence (Edge Locations)



- Amazon has 400+ Points of Presence (400+ Edge Locations & 10+ Regional Caches) in 90+ cities across 40+ countries.
 - Content is delivered to end users with lower latency.
-

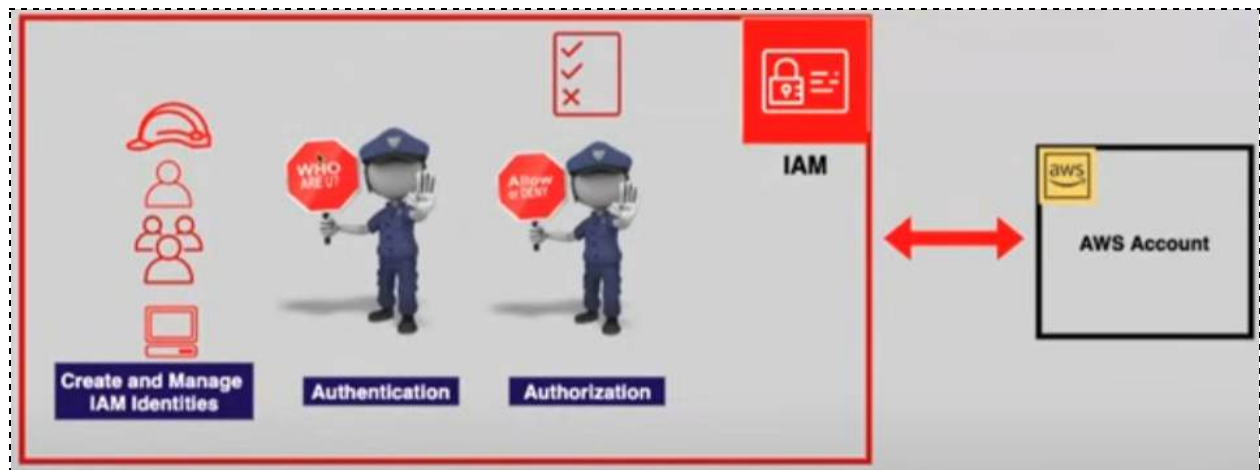


CH-04 - IAM & AWS CLI

➔ IAM Introduction (Identity and Access

Management): Users, Groups, Policies

- عبارة عن Free Service موجودة في AWS ونوعها Global Service
- ال Global Services يتبقي services متاحة في كل ال regions .. وفيه Services تانيه بتكون متاحة في regions معينه بس عندك مثلا EC2 instance لازم تحدد معاها ال Region علشان لو عملت Launch ل EC2 instance في Region معينه مش هتلاقىها ف ال Region التانية علي عكس ال Global Services
- بتمكنك من انك تكريت users & groups وتديهم permissions وكدا
- انت By Default بتخش بال Root User ولكن دة مش مفضل انك تستخدمه في العادي بتستخدم Admin Or Normal User
- علشان تعرف ال Service دي متاحة عندك في ال region ولا لا من [هنا](#)



➡ IAM Users, Groups Hands ON.

Search >> IAM >> User Name (Stephan), Password >>
Permissions

Permissions options

☒ **Add user to group**
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ **Copy permissions**
Copy all group memberships, attached managed policies, and inline policies from an existing user.

☐ **Attach policies directly**
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

 **Get started with groups**
Create a group and select policies to attach to the group. We recommend using groups to manage user permissions by job function, AWS service access, or custom permissions. [Learn more](#)

Create group

► **Set permissions boundary - optional**

"تقدر تدخل ال user جروب انت عامله وال permissions فيه جاهزة او ت create

group أو ت Copy ال Permissions او ت Attach Policy

"بما ان هنا مفيش Groups قديمة ف احنا هنكريت جروب جديد"



Create user group

×

Create a user group and select policies to attach to the group. We recommend using groups to manage user permissions by job function, AWS service access, or custom permissions. [Learn more](#)

User group name
Enter a meaningful name to identify this group.

admin

Maximum 128 characters. Use alphanumeric and '+=, @-_' characters.

Permissions policies (1/884)

↻

Create policy

Filter by Type

Search

All ty... ▼

< 1 2 3 4 5 6 7 ... 45 >

⚙

	Policy name	Type	Use...	Description
☒	AdministratorAccess	AWS managed ...	None	Provides full access to AWS service

"في الحالة بتاعتنا عملنا user وعاوزينو Admin ف عملنا create group وال policy بتاعته خليناها Administrator Access بحيث يكون واخد كل ال Access عال AWS Services"

Permissions options

☒ Add user to group
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ Copy permissions
Copy all group memberships, attached managed policies, and inline policies from an existing user.

☐ Attach policies directly
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

User groups (1/1)

↻

Create group

Search

< 1 >

⚙

	Group name	Users	Attached policies	Created
☒	admin	0	AdministratorAccess	2023-10-24 (Now)



"كدا بقا الجروب جاهز .. ف ظهرلنا فى ال groups ف عملنا check عالجروب اللى لسة عاملينه علشان نضيف ال user فيه."

"تقدر وانت بتعمل ال user تضيف Tags ودي بتسهل عليه ت Access ال users مثلا تقسمهم : department, project, environment, or cost center."

Search >> Users , User Groups

"هتلاقى عندك بقا ال user وال permissions اللى واخدها اللى هيا بتاعة الجروب وهتلاقى ال groups اللى عندك وكل ال Configurations اللى انت عملتها وانت بت create ال user , group"

AWS Account

Account ID

 219637129727

Account Alias

aws-stephane-v5 [Edit](#) | [Delete](#)

Sign-in URL for IAM users in this account

 <https://aws-stephane-v5.signin.aws.amazon.com/console>

"ال user اللى انت عملته بيكون ليه Account id و Account Alias من خلالهم بتخش بيهم عالاكونت بجانب ال password"



I'm Here

Sign in

☐ **Root user**
Account owner that performs tasks requiring unrestricted access. [Learn more](#)

☒ **IAM user**
User within an account that performs daily tasks. [Learn more](#)

Account ID (12 digits) or account alias

Next

"بالنسبة لل url دة من خلاله بيدخلك مباشر على ال user دة بتكون محتاج ال password بجانب ال alias او ال id"

Amazon Web Services Sign-in

https://signin.aws.amazon.com/signin?redirect_url=https%3A%2F%2Fconsole.aws.amazon.com

aws

Sign in as IAM user

Account ID (12 digits) or account alias

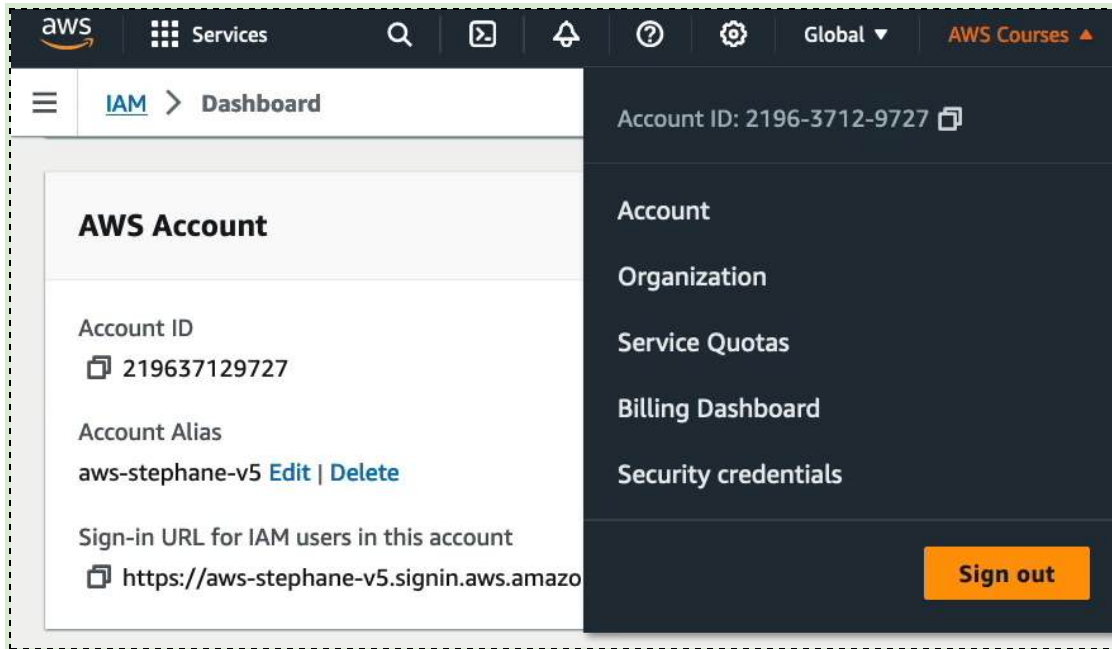
IAM user name

Password

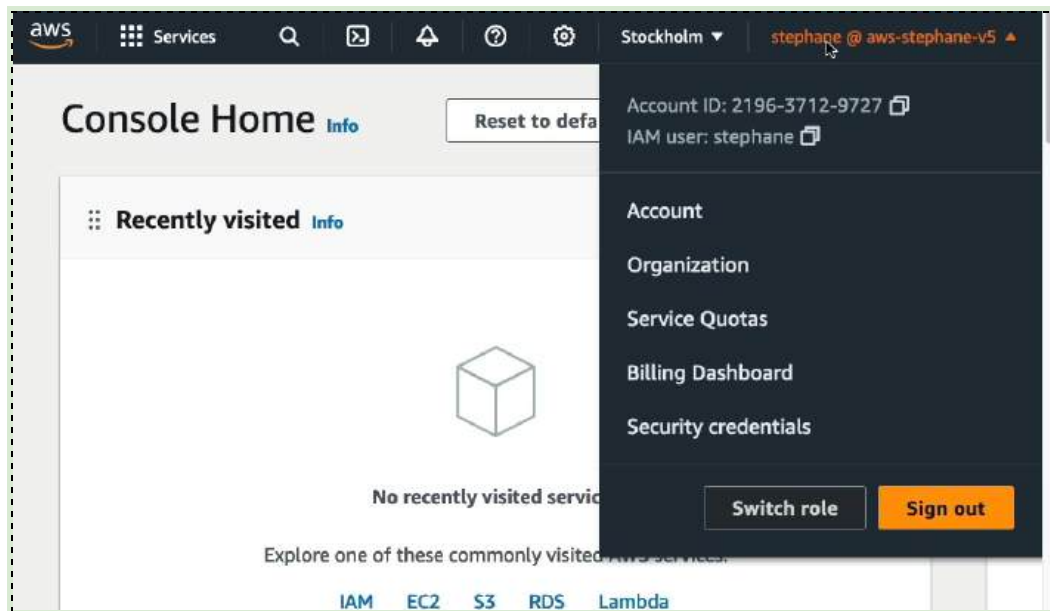
☐ Remember this account

"دة بيكون شكل ال root اللي انت دخلت بيه فى البداية خالص .. Account ID فقط"



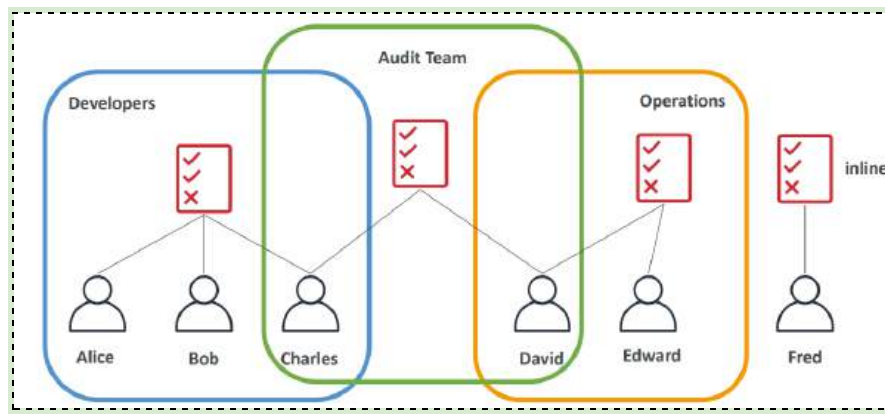


"دە بىكون شكل ال IAM User"



" ان حصلت مشكلة فى ال IAM Users Accounts هتندلها من خلال ال Root الاصلي ..
 إنما لو حصلت مشكلة فى ال Root Account مشكلة كبيرة و هتكلم ال Support."
 " من خلال ال Root user تقدر ت delete stephane user وميكونش ليه access علي
 اي حاجة"

➔ IAM Policies:



- عبارته عن JSON Files بتحدد ال Permissions لل User & Group
- ال Policy اللى واخدها الجروب بتطبق علي كل user فيه.

● Policy Structure:

- بيكون عبارة عن JSON file .. من خلاله بت define ال permissions وبت specify ال actions هل allowed or denied ل resources معينه .. ال Structure دة بيتكون من :



```

{
  "Version": "2012-10-17",
  "Id": "S3-Account-Permissions",
  "Statement": [
    {
      "Sid": "1",
      "Effect": "Allow",
      "Principal": {
        "AWS": ["arn:aws:iam::123456789012:root"]
      },
      "Action": [
        "s3:GetObject",
        "s3:PutObject"
      ],
      "Resource": ["arn:aws:s3:::mybucket/*"]
    }
  ]
}

```

- **Version :**

- دة version ال policy language ودة حاليا 17-10-2012

- **ID (Optional):**

- عبارة عن identifier لل policy

- **Statement:**

- بتحتوي علي Statement او اكثر ال Statement عبارة عن rule وبتحتوي علي كذا عنصر:

- **SID (Optional):**

- عبارة عن identifier لل Statement

- **Effect:**

- هل ال Statement دي Allow or Deny

- **Action:**

- ال account or user or which role اللى ال policy دي هتطبق عليه.



- **Resources:**

- ال Resources اللى ال Actions هتطبق عليها.

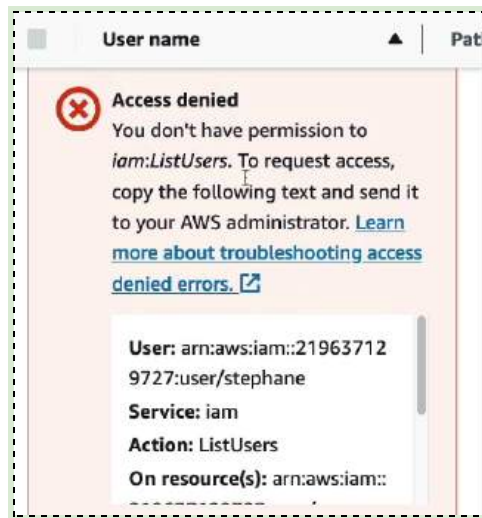
- **Condition(Optional):**

- شروط تفعيل ال policies دي.

➔ IAM Policies Hands On:

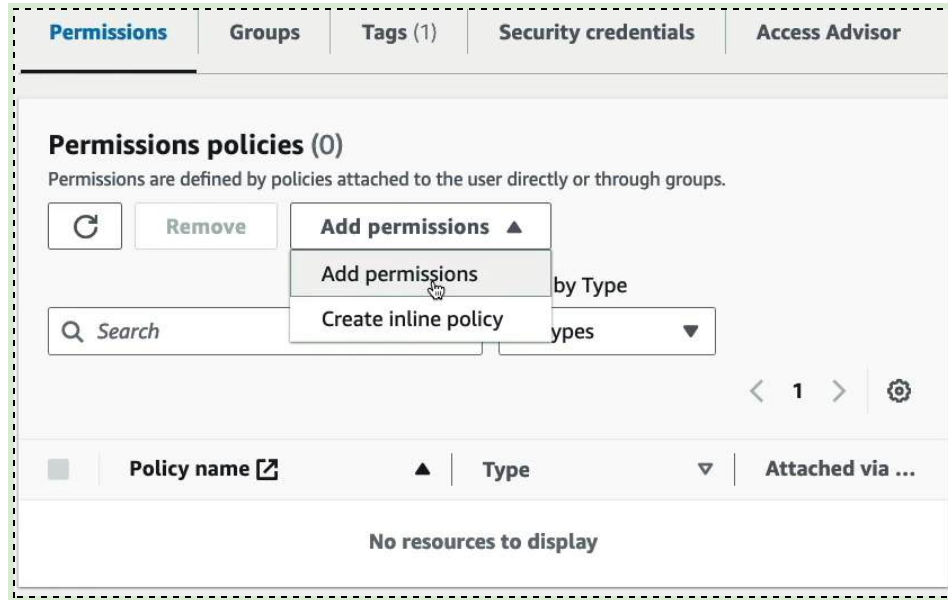
"لو شيلت stephane من ال group كدا هيفقد ال admin permissions اللى كان واخدها

لأنها تبع الجروب.. ف مش هيقدر مثلا انوي list ال users"

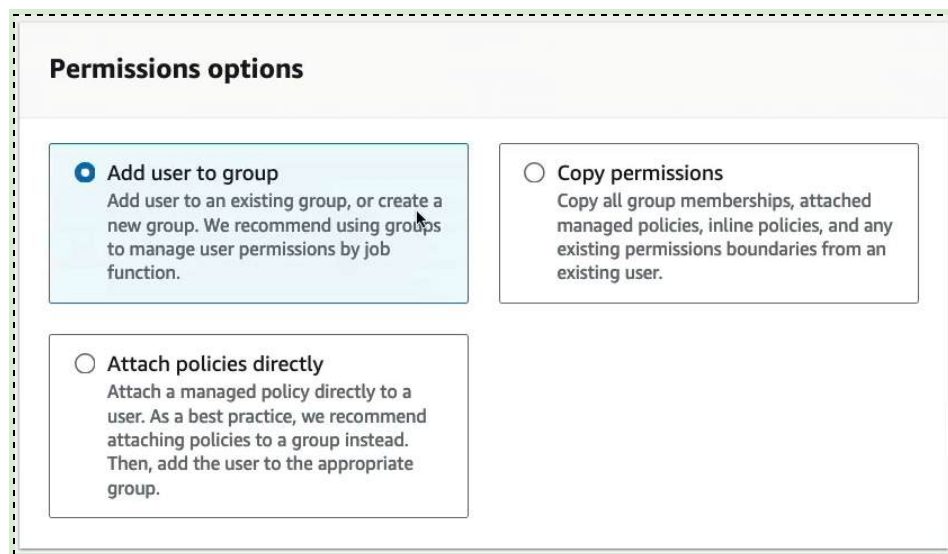


Users >> Stephane >> Permissions





"تقدر من تاني تحط ال permissions اللي انت عاوزها لل user stephane"



"قبل كدا دخلناه جروب وخذ ال permissions بتاعته اللي كانت Administration"

"permissions ودلوقتي هن Attach policies .. وليكن read only policy"



Permissions policies (1132)

Filter by Type

Search

All types ▼

< 1 2 3 4 5 6 7 ... 57 > ⚙

☐	Policy name ↗	Type ▼	Attached e... ▼
☐	AccessAnalyzerSer...	AWS managed	0
☐	AdministratorAccess	AWS managed - job f...	1
☐	AdministratorAcce...	AWS managed	0
☐	AdministratorAcce...	AWS managed	0
☐	AlexaForBusinessD...	AWS managed	0

Permissions policies (1/1132)

Filter by Type

Search X

All types ▼ 1 match

< 1 > ⚙

☒	Policy name ↗	Type ▼	Attached e... ▼
☒	IAMReadOnlyAccess	AWS managed	0

"بعد ما staphane خد ال readonly permission كدا يقدر ي list ال users مثلاً.. لو جيت ب stephane اللي واخد readonly permission تحاول ت create group مش هينفع."



☐	Policy name ↗	Type	Attached via ↗
☐	AdministratorAccess	AWS managed - job f...	Group admin
☐	AlexaForBusinessD...	AWS managed	Group developers
☐	IAMReadOnlyAccess	AWS managed	Directly

"بيكون واضحك انت واحد permissions اية ومن انهى groups بالظبط ولو directly ف
 دة اللى root مديهولك ال root مدي ل stephane وهو بيكرية .readonly permission".

IAM >> Policies >> AdministratorAccess

Identity and Access Management (IAM)

Dashboard

Access management

User groups

Users

Roles

Policies

Identity providers

Account settings

IAM > Policies

Policies (1130) [Info](#)

A policy is an object in AWS that defines permissions.

Filter by Type

1
2
3
4
5
6
7
...
57

	Policy name	Type	Use...	Des
☐	AccessAnalyzerSer...	AWS managed	None	Allc
☐	AdministratorAccess	AWS managed ...	Permis...	Pro
☐	AdministratorAcce...	AWS managed	None	Gra



Permissions defined in this policy [Info](#)

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it

Summary

JSON

Search

Allow (384 of 384 services)

Service ▲	Access level ▼	Resource
Access Analyzer	Full access	All resources
Account	Full access	All resources
Activate	Full access	All resources
Alexa for Business	Full access	All resources
AMP	Full access	All resources
Amplify	Full access	All resources

"لما تخش علي ال policies وتخش علي واحدة منهم وليكن AdministratorAccess هتلاقي كل ال services اللي ال policy دي تقدر تتحكم فيهم وكمان اية نوع ال access اللي واخدها هنا مثلا Full Access"



I'm Here

Permissions
Entities attached
Policy versions (1)
Access Advisor

Permissions defined in this policy [Info](#)

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it

Copy
Summary
JSON

```

1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": "*",
7       "Resource": "*"
8     }
9   ]
10 }

```

"لو دخلت على JSON هتلاقي محتوى ال JSON File خاص بال Policy ياغني لو Policy
تانية زي ما هنشوف تحت هتلاقي محتوى تاني خصوصا فال statements لان كل policy ب
rules مختلفة"

"هنا معني ال JSON File ده .. بن Allow any action على كل ال resources .. وده
فعلا ال access اللي بياخدو ال AdministratorAccess."

Policies (1130) [Info](#)

A policy is an object in AWS that defines permissions.

Refresh
Actions
Delete
Create policy

Filter by Type
iamread
All types
1 match

Policy name	Type	Used as	Description
IAMReadOnlyAccess	AWS managed	Permissi...	Provides read only e



"لو دخلت مثلا على policy تانية زي IAMReadOnly هتلاقى واحدة access على service واحدة"

Permissions defined in this policy [Info](#)

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it

SummaryJSON

Allow (1 of 384 services)

Show remaining 383 services

Service	Access level	Resource
IAM	Full: List Limited: Read	All resources

"ولو دخلت عال service دي هتوضحلك اية بالظبط اللي اقدر أ read only عليه.. اية هيا ال API Calls اللي مسموح بيها"



List (38 of 38)	
Action	Resource
GetAccountSummary	All resources
GetLoginProfile	All resources
ListAccessKeys	All resources
ListAccountAliases	All resources
ListAttachedGroupPolicies	All resources
ListAttachedRolePolicies	All resources
ListAttachedUserPolicies	All resources
ListCloudFrontPublicKeys	All resources
ListEntitiesForPolicy	All resources
ListGroupPolicies	All resources
ListGroups	All resources

Permissions defined in this policy [Info](#)

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it

Copy

Summary

JSON

```

1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7         "iam:GenerateCredentialReport",
8         "iam:GenerateServiceLastAccessedDetails",
9         "iam:Get*",
10        "iam:List*",
11        "iam:SimulateCustomPolicy",
12        "iam:SimulatePrincipalPolicy"
13      ],
14      "Resource": "*"
15    }
16  ]
17 }
```

"هنا فال read only policy محتوي ال JSON File اختلف عن اللى policy اللى فاتت

وهكذا."



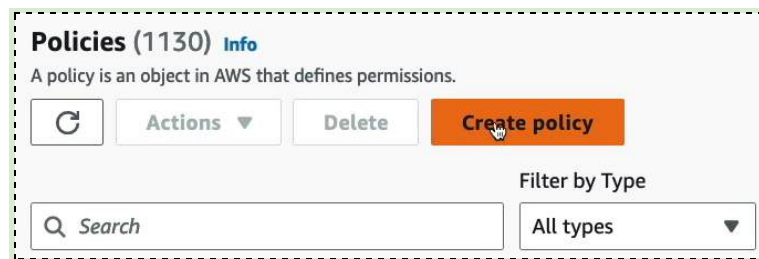
I'm Here

"Get" دي wild card .. مثلا Get users , Get Groups وهكذا نفس الكلام على "list"

يبقي الخلاصة : دة الترتيب 💡

IAM >> Policies >> One Of The Policies >> services can access by the policy , json file content

IAM >> Policies >> Create Policy



"يمكن انت ت Create ال policy الخاصة بيك .. هتلاقى عندك visual editor or json editor"



Policy editor Visual JSON Actions ▼

▼ **Select a service**
Specify what actions can be performed on specific resources in a service.

Service

[+ Add more permissions](#)

Policy editor Visual JSON Actions ▼

▼ **IAM**
Set permissions for IAM

Specify what actions can be performed on specific resources in IAM.

▼ **Actions allowed**
Specify actions from the service to be allowed.

Effect
☒ Allow ☐ Deny

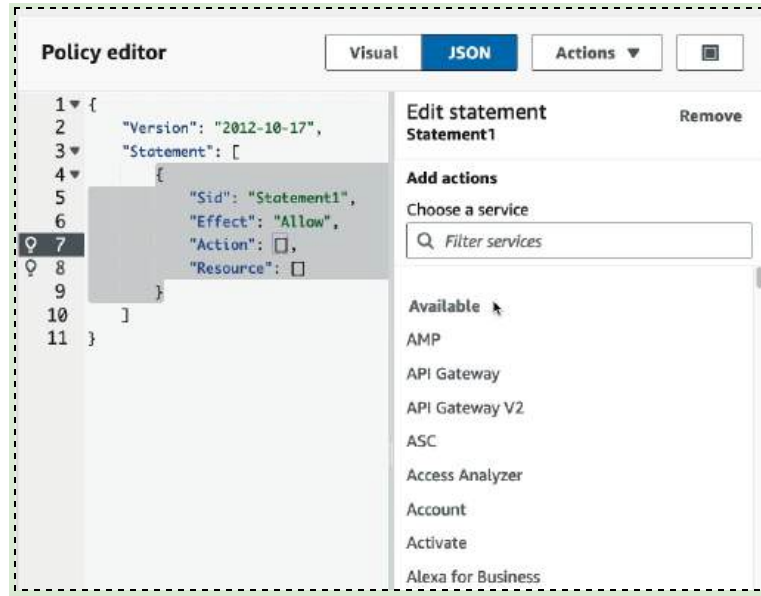
Manual actions | [Add actions](#)

☐ All IAM actions (iam:*)

"وبتختار بقا ال service ثم ال actions وهكذا

وبالنسبة لل JSON .. نفس الكلام"





"وبعدها بقا تحط لل policy name & description وهكذا"

➔ IAM – Password Policy

● Passwords:

IAM >> Account Setting >> Edit Password Policy

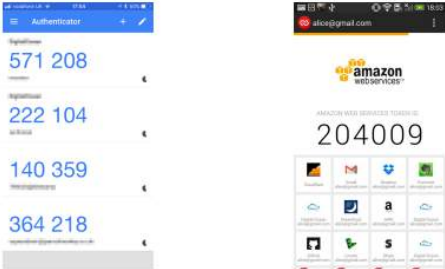
- تقدر تحدد لل users ال passwords بتاعتهم لازم يكون شكلها عامل ازاي
- تقدر تحدد المدة اللي لازم يغيرو فيها ال passwords دي
- تقدر تمنع انهم يعيدو استخدام passwords قديمة

● Multi Factor Authentication - MFA

- عبارة عن password محفوظ جوا device معاك



- تقدر تطبق ال Multi Factor Authentication عن طريق تطبيقات زي Google Authenticator او Authy .. او من خلال Physical Devices

Virtual MFA device	Universal 2nd Factor (U2F) Security Key
	
Google Authenticator (phone only)	YubiKey by Yubico (3 rd party)
Support for multiple tokens on a single device.	Support for multiple root and IAM users using a single security key

Hardware Key Fob MFA Device	Hardware Key Fob MFA Device for AWS GovCloud (US)
	
Provided by Gemalto (3 rd party)	Provided by SurePassID (3 rd party)

➡ IAM MFA Hands ON

- Passwords Policy:

IAM>> Access Management >> Account Setting >> Password



I'm Here

Policy >> Edit

تقدر تستخدم ال IAM Default بالشكل دة

Password policy

☒ **IAM default**
Default password requirements for IAM users.

☐ **Custom**
Use a customized password policy.

Password minimum length
8 characters

Password strength

Include a minimum of three of the following mix of character types:

- Uppercase
- Lowercase
- Numbers
- Non-alphanumeric characters (! @ # \$ % ^ & * () _ + - = [] { } | ')

Other requirements

- Never expire password
- Must not be identical to your AWS account name or email address

أوت Custom انت ال Password Policy الخاصة بيك بحيث تستخدمها بعدين.



I'm Here

Password policy

☐ IAM default

Default password requirements for IAM users.

☒ Custom

Use a customized password policy.

Password minimum length.

Enforce a minimum length of characters.

8

characters

Needs to be between 6 and 128.

Password strength

- ☐ Require at least one uppercase letter from the Latin alphabet (A-Z)
- ☐ Require at least one lowercase letter from the Latin alphabet (a-z)
- ☐ Require at least one number
- ☐ Require at least one non-alphanumeric character (!@#\$%^&*()_+-=[]{}|')

Other requirements

- ☐ Turn on password expiration
- ☐ Password expiration requires administrator reset
- ☐ Allow users to change their own password
- ☐ Prevent password reuse

● Assign MFA:

Account Name >> Security Credentials

My security credentials (root user) [Info](#)

The root user has access to all AWS resources in this account, and we recommend following [best practices](#). To learn more about the types of AWS credentials and how they're used, see [AWS Security Credentials](#) in AWS General Reference.



MFA not activated for root user

The root user for this account does not have multi-factor authentication (MFA) activated. Activate MFA to improve security for this account.

[Assign MFA](#)

من خلال Assign MFA بت set ال Device name , Device Type



I'm Here

ودي ال Authenticating Apps لو مش معاك device تقدر تستخدمها

Android	Twilio Authy Authenticator , Duo Mobile , LastPass Authenticator , Microsoft Authenticator , Google Authenticator , Symantec VIP
iOS	Twilio Authy Authenticator , Duo Mobile , LastPass Authenticator , Microsoft Authenticator , Google Authenticator , Symantec VIP

لو قفلت ال root acc وجيت تفتح تاني هيطلب منك MFA Code هتجيبه من خلال التطبيق او الجهاز ايا كان يايني .. لو التطبيق او الجهاز ضاعو هتبقى مشكلة كبيرة.

➔ AWS Access Keys, CLI and SDK:

- اتعاملنا مع AWS من خلال ال Management Console من خلال المتصفح يايني.. وعندك طريقتين كمان هما ال CLI وال SDK والاتنين علشان نشغل منهم لازمنا Access Keys

Access keys				
Use access keys to make secure REST or HTTP Query protocol requests to AWS service APIs. For your protection, you should never share your secret keys with anyone. As a best practice, we recommend frequent key rotation. Learn more				
Create access key				
Access key ID	Created	Last used	Status	
AKIAASK4E37PV4TU3RD6C	2020-05-25 15:13 UTC+0100	N/A	Active	Make inactive ✕

ال Access Key بتاعك انت بس ميخرجش بره. 💡



- **SDK (Software Development Kit):**

- عبارة عن مجموعة من ال libraries & tools التي developers يقدرو يستخدموهم
وهما بيكتبو كود لل Applications بتاعتهم بحيث يقدرو يخلو ال Applications تـ
interact مع AWS Services

- **DevOps and CI/CD:** SDKs are used in continuous integration and continuous deployment (CI/CD) pipelines to automate the deployment and management of applications on AWS

مثال على استخدام SDK هو ال CLI التي هنتستخدمه في اننا ن Access AWS كمان 💡

شوية .. دة مبني على AWS SDK For Python واسمو boto3

➡ AWS CLI Hands On:

Users >> Stephane >> Security Credentials >> Access Keys >>
Create Access Key



Access key best practices & alternatives [Info](#)

Avoid using long-term credentials like access keys to improve your security. Consider the following use cases and alternatives.

Use case

☐ **Command Line Interface (CLI)**

You plan to use this access key to enable the AWS CLI to access your AWS account.

☐ **Local code**

You plan to use this access key to enable application code in a local development environment to access your AWS account.

☐ **Application running on an AWS compute service**

You plan to use this access key to enable application code running on an AWS compute service like Amazon EC2, Amazon ECS, or AWS Lambda to access your AWS account.

☐ **Third-party service**

You plan to use this access key to enable access for a third-party application or service that monitors or manages your AWS resources.

☐ **Application running outside AWS**

You plan to use this access key to authenticate workloads running in your data center or other infrastructure outside of AWS that needs to access your AWS resources.

Create access key



Success

This is the **only** time that the secret access keys can be viewed or downloaded. You cannot recover them later. However, you can create new access keys at any time.

Download .csv file

Access key ID	Secret access key
AKIA4OHDCKKLTZNZBRUT	D32MzVn1lqqrPPYxvWAtvxO3ft6l/Dyan6PF Hide

Close

بعد ما وصلت لل Access Key وليكن بعد ما اخترت CLI بتروح بقا على ال terminal

علشان تربط الاتنين ببعض



```
~ ➤ aws configure
AWS Access Key ID [None]: AKIA40HDCKKLZNZBRUT
AWS Secret Access Key [None]: D32MzVn1IqqrPPYxvWAtvxOMlt03ft6I/Dyan6PF
Default region name [None]: eu-west-1
Default output format [None]:
~ ➤ aws iam list-users
```

```
{
  "Users": [
    {
      "Path": "/",
      "UserName": "stephane",
      "UserId": "AIDA40HDCKKLSE4HHDCOE",
      "Arn": "arn:aws:iam::855174697623:user/stephane",
      "CreateDate": "2020-05-27T16:28:16+00:00",
      "PasswordLastUsed": "2020-05-27T16:33:03+00:00"
    }
  ]
}
(END)
```

بت Configure .. وبالنسبة لل Region بتختار الاقرب ليك.

من خلال ال commands تقدر تعمل بقا كل اللي كنت بتعملو على aws management console قبل كدا

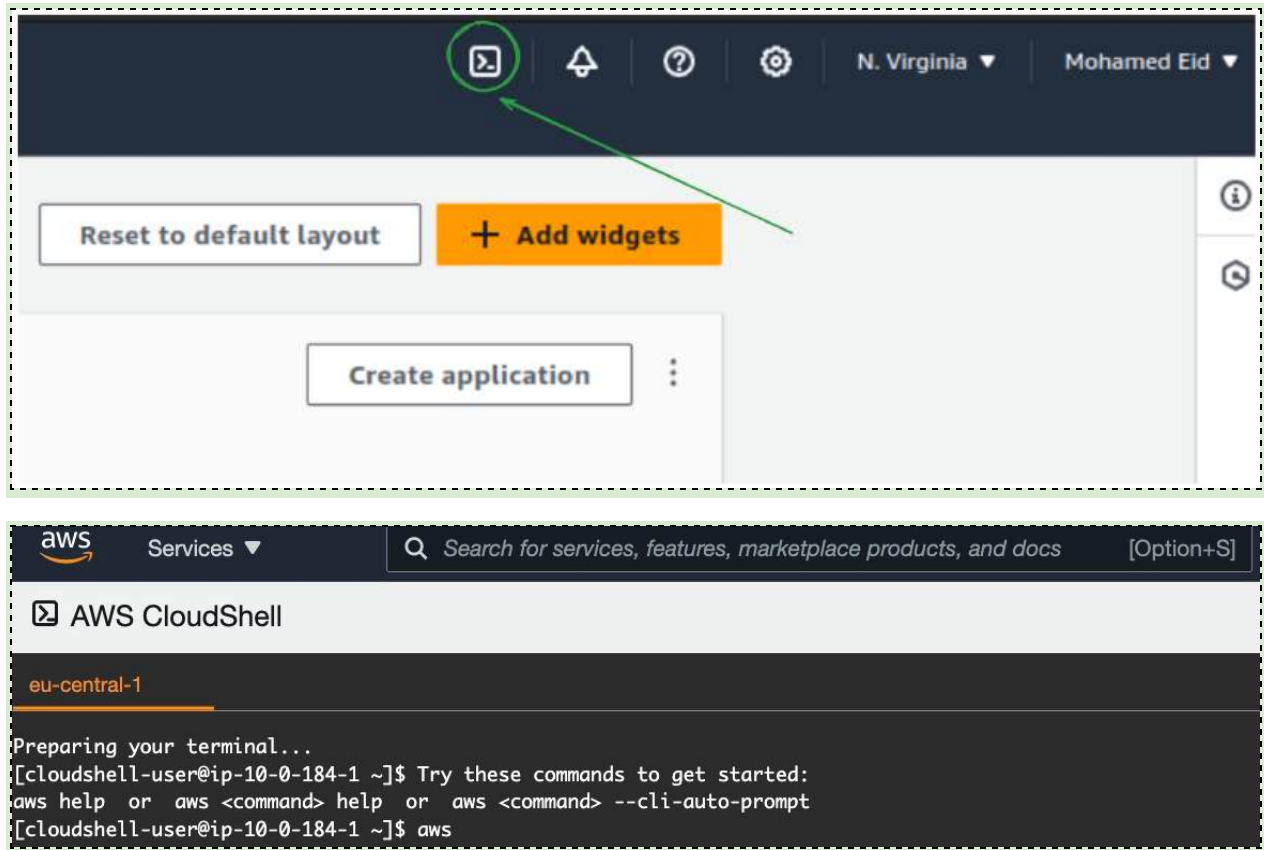
لو شيلت ال user اللى هو stephane من جروب ال admins يعانني مبقاش واخذ اي

permissions وجيت فى ال terminal نفذت نفس ال command اللى هو list the

users هتلاقي انو مدكش نتيجة ودة حصل مباشرة.



➔ AWS CloudShell:



- Terminal عادي خالص علي AWS وبيكون متاح في بعض ال regions تقدر تشوفهم من هنا
- repository عادي ياعني لو انشأت file هتلاقه موجود متخزن وتقدر تعمله download وتعمل upload وهكذا اكنك فاتح terminal عادي.
- ال terminal اللي على جهازي افضل اكيد.



➔ IAM Roles for AWS Services:

- Role ياعني دور لك ترجمه حرفيه .. ياعني هديك permission تقوم ب دور معين وغالبا بيكون الموضوع دة لمدة قصيره ..
- لو عملت access على EC2 instance وجيت ت list ال user بالشكل دة aws iam list-users مش هتلاقيها واخده ال permission دة ف بنروح ن attach role ليها تكون ليها policy allow read only وبالتالي تقدر ت list users .. ودي امثله اكثر :

● Cross-Account Access

- شركة A عاوزه تدي Permission للأكونت بتاع شركة B إنو يقدر يستخدم الداتا اللي على ال S3 Bucket بتاع شركة A ف دة بيتم عن طريق ان شركة A تكرت role وتديها لاكونت شركة B .. ال role بتبقي عباره عن policies ف بتحدد فى ال policy انه ي permission مسموح بيها.

● AWS Lambda Functions

- Lambda function ودي service عاوزين نديها role دور ياعني تمكنها من انها يكون ليها ال permissions اللي تخليها قادره ت access DynamoDB tables and S3 buckets علشان ت process الداتا مثلا
- يبقي كدا عندنا service اديناها permission مكنها انها ت perform operations على some resources

● EC2 Instances and S3 Access

- عاوزين نخلي ال service اللي اسمها EC2 Instance قادرة انها ت retrieve configurations files & upload logs ل S3 Bucket ف بنديها role علشان تقدر تعمل دة



● Most Common Roles:

- EC2 Instance Roles
 - Lambda Function Roles
 - Roles for CloudFormation
-

➔ IAM Roles Hands ON:

Access Management >> Roles >> Create Role

Trusted entity type

☒ **AWS service**

Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☐ **AWS account**

Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ **Web identity**

Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.

☐ **SAML 2.0 federation**

Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

☐ **Custom trust policy**

Create a custom trust policy to enable others to perform actions in this account.

زي ماننت شايف فيہ كذا نوع ولكن هنتغل حاليا على AWS Service



Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case
EC2

Choose a use case for the specified service.
Use case

- ☒ **EC2**
Allows EC2 instances to call AWS services on your behalf.
- ☐ **EC2 Role for AWS Systems Manager**
Allows EC2 instances to call AWS services like CloudWatch and Systems Manager on your behalf.
- ☐ **EC2 Spot Fleet Role**
Allows EC2 Spot Fleet to request and terminate Spot Instances on your behalf.
- ☐ **EC2 - Spot Fleet Auto Scaling**
Allows Auto Scaling to access and update EC2 spot fleets on your behalf.
- ☐ **EC2 - Spot Fleet Tagging**
Allows EC2 to launch spot instances and attach tags to the launched instances on your behalf.
- ☐ **EC2 - Spot Instances**
Allows EC2 Spot Instances to launch and manage spot instances on your behalf.
- ☐ **EC2 - Spot Fleet**
Allows EC2 Spot Fleet to launch and manage spot fleet instances on your behalf.
- ☐ **EC2 - Scheduled Instances**
Allows EC2 Scheduled Instances to manage instances on your behalf.

بعدها بتختار ال Service اللى عاوز تديها ال role

Permissions policies (1/882) [Info](#)
Choose one or more policies to attach to your new role.

Filter by Type
IAMRe All types 1 match

☒	Policy name	Type	Description
☒	IAMReadOnlyAccess	AWS managed	Provides read only access to IAM via the ...

بعدها بت attach ال policy وليكن هندي ال EC2 IAMReadOnly .. كدا قادرة ت read

اي حاجة فى IAM

بعدها بنحط ال role name , description وبكدا انت كريت ال role



Roles (3) [Info](#)

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

[Refresh](#) [Delete](#) [Create role](#)

☐	Role name	Trusted entities	Last activity
☐	AWSServiceRoleForSupport	AWS Service: support (Service-Linker)	-
☐	AWSServiceRoleForTrustedAdvisor	AWS Service: trustedadvisor (Service-Linker)	-
☐	DemoRoleForEC2	AWS Service: ec2	-

Permissions policies (1) [Info](#)

You can attach up to 10 managed policies.

[Refresh](#) [Simulate](#) [Remove](#) [Add permissions](#)

Filter by Type All types

☐	Policy name	Type	Attached entities
☐	IAMReadOnlyAccess	AWS managed	1

ولو ضغطت عليها هتلاقي ال permission اللي واخده اللي هو IAMReadOnly

➔ IAM Security Tools, Hands On:

- IAM Credentials Report (account-level)**

عباره عن report بتعمله download من خلال ال root بيكون فيه معلومات عن كل ال users

Access Reports >> Credential Report



Credentials report of IAM users in this account [Info](#)

The credentials report lists all your IAM users in this account and the status of their various credentials. After a report is created, it is stored for up to four hours.

Credentials report

[Download credentials report](#)

No report created in the past 4 hours. A new report will be created.

● IAM Credentials Report Content

الاسم

user نوع ال

امتي اتركيت ال user

امتي اخر مرة ال password اتيغير

هل فيه MFA ولا لا

Password next rotation : امتي ال password هي expired و امتي المفروض هيتغير

	A	B	C	D	E
1	user	arn	user_creation_time	password_enabled	password_last_used
2	<root_account>	arn:aws:iam::855174697623:root	2020-05-27T16:01:07+00:00	not_supported	2020-05-27T16:56:56+00:00
3	stephane	arn:aws:iam::855174697623:user/stephane	2020-05-27T16:28:16+00:00	TRUE	2020-05-27T16:33:03+00:00
4					

	F	G	H	I	J
1	password_last_changed	password_next_rotation	mfa_active	access_key_1_active	access_key_1_last_rotated
2	not_supported	not_supported	TRUE	FALSE	N/A
3	2020-05-27T16:28:17+00:00	N/A	FALSE	TRUE	2020-05-27T17:25:55+00:00

● IAM Access Advisor (user-level)

بيوضحلك اية ال services اللى ال user دة استخدمها و امتي استخدمها و كمان لما استخدمها

كان واخذ عليها permission اية .. زي تحت كذا واخذ admin access



I'm Here

Access Management >> Users >> Access Advisor

[Permissions](#) | [Groups \(1\)](#) | [Tags \(2\)](#) | [Security credentials](#) | **[Access Advisor](#)**

Access Advisor shows the services that this user can access and when those services were last accessed. Review this data to remove unused permissions. [Learn More](#)

Allowed services (367)

IAM reports activity for services and management actions. [Learn more](#) about action last accessed information. To see actions, choose the appropriate service name from the list.

Filter by services access history
No Filter

< 1 2 3 4 5 6 7 ... 37 >

Service	Policies granting permissions	Last accessed
AWS Organizations	AdministratorAccess	Today
AWS Identity and Access Management	AdministratorAccess	Today
AWS Health APIs and Notifications	AdministratorAccess	Today
AWS User Notifications	AdministratorAccess	Today
Amazon EC2	AdministratorAccess	Today
AWS Resource Explorer	AdministratorAccess	Today

➡ IAM Summary:

- **Users:** mapped to a physical user, has a password for AWS Console
- **Groups:** contains users only
- **Policies:** JSON document that outlines permissions for users or groups
- **Roles:** for EC2 instances or AWS services
- **Security:** MFA + Password Policy
- **AWS CLI:** manage your AWS services using the command-line



I'm Here

- **AWS SDK:** manage your AWS services using a programming language
- **Access Keys:** access AWS using the CLI or SDK
- **Audit:** IAM Credential Reports & IAM Access Advisor

CH-05 - EC2 Fundamentals

➡ AWS Budget Setup:

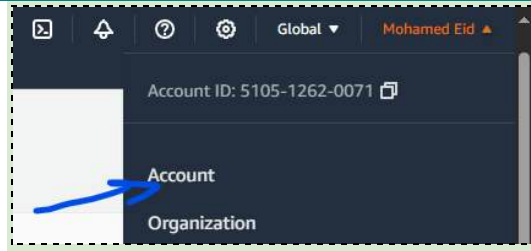
Your Name >> Billing And Cost Management

The screenshot shows the 'Billing and Cost Management home' dashboard. The 'Cost summary' section displays 'Month-to-date cost' and 'Total forecasted cost for current month', both with 'Access denied' errors. The 'Cost monitor' section shows 'Budgets status' and 'Cost anomalies status (MTD)', both with 'Setup required' messages. A 'View bill' link is visible at the bottom of the 'Cost summary' section.

- هتلاقي انك معندكش Access عال Billing & Cost لانك فاتح من user عادي رغم انك واخد Administrator Access .. ف علشان نحل المشكلة دي لازم نفعل ال access دة من ال root user



Open the root user >> Account



IAM user and role access to Billing information [Info](#)

Edit

IAM user/role access to billing information

Deactivated

IAM user and role access to Billing information [Info](#)

Edit

IAM user/role access to billing information

Activated

- وبعد activate ان ال user يقدر ي access ال billing & cost تقدر عادي تتحكم فيهم من خلال ال regular user

Billing and Cost Management home [Info](#)

Reset layout

Cost summary [Info](#)

Month-to-date cost

\$0.00

↓ 0% compared to last month for same period

Total forecasted cost for current month

Access denied

Last month's cost for same time period

\$0.00

Dec 1 – 15, 2023

Last month's total cost

\$0.00

[View bill](#)

Cost monitor [Info](#)

Budgets status

Setup required

No budget created

Cost anomalies status (MTD)

Setup required

No monitor created



I'm Here

Billing And Cost Management >> Billing and Payments >> Bills >> Charges By services

☒ Elastic Compute Cloud		USD 0.00
☒ No Region		(USD 44.57)
☒ EU (Ireland)		USD 43.77
☒ Amazon Elastic Compute Cloud NatGateway		USD 39.43
\$0.048 per GB Data Processed by NAT Gateways	77,449 GB	USD 3.72
\$0.048 per NAT Gateway Hour	744 Hrs	USD 35.71
☒ EBS		USD 0.62
\$0.05 per GB-Month of snapshot data stored - EU (Ireland)	3,518 GB-Mo	USD 0.18
\$0.11 per GB-month of General Purpose SSD (gp2) provisioned storage	4 GB-Mo	USD 0.44
☒ Elastic IP Addresses		USD 3.72
\$0.00 per Elastic IP address not attached to a running instance for the	1 Hrs	USD 0.00
\$0.005 per Elastic IP address not attached to a running instance per h	743 Hrs	USD 3.72

- من خلالها تقدر تشوف كل ال Active Services وكل Service انت اشتغلت عليها
كلفتك فلوس قد اية

Billing And Cost Management >> Cost Analysis >> Free tier



AWS Free Tier (43) Info				
Find service name < 1 2 3 4 5 > ⚙				
Service ▾	AWS Free Tier usage limit ▾	Current usage ▾	Forecasted usage ▾	MTD actual usage ▾
AWS Lambda	400000.0 seconds are always free per month as part of AWS Free Usage Tier (Global-Lambda-GB-Second)	400,000 seconds	826,667 seconds	
Amazon Simple Queue Service	1000000.0 Requests are always free per month as part of AWS Free Usage Tier (Global-Requests)	1,000,000 Requests	2,066,667 Requests	
AWS X-Ray	100000.0 Traces are always free per month as part of AWS Free Usage Tier (Global-XRay-TracesStored)	100,000 Traces	206,667 Traces	
AmazonCloudWatch	10.0 Alarms are always free per month as part of AWS Free Usage Tier (Global-CW:AlarmMonitorUsage)	10 Alarms	21 Alarms	
AWS Lambda	1000000.0 Request are always free per month as part of AWS Free Usage Tier (Global-Request)	646,310 Request	1,335,707 Request	

- من خلالها تقدر تتابع استخدامك الحالي والمتوقع لو فاتح acc free tier

Billing And Cost Management >> Budget And Planning >> Budgets

Templates - new

Choose a template that best matches your use case.

☒ Zero spend budget

Create a budget that notifies you once your spending exceeds \$0.01 which is above the AWS Free Tier limits.

☐ Monthly cost budget

Create a monthly budget that notifies you if you exceed, or are forecasted to exceed, the budget amount.

☐ Daily Savings Plans coverage budget

Create a coverage budget for your Savings Plans that notifies you when you fall below the defined target.

☐ Daily reservation utilization budget

Create a utilization budget for your reservations that notifies you when you fall below the defined target.



I'm Here

✓ Your budget **My Zero-Spend Budget** has been created successfully. [Submit feedback](#)

[Billing and Cost Management](#) > [Budgets](#) > Overview

Overview Info

Budgets (1) Info

[Download CSV](#)

Actions ▾

Create budget

Type - Show all budgets ▾

< 1 > ⚙

☐	Name	Thresholds	Budget	Amount u...	Forecaste...	Current vs. budgeted
☐	My Zero-Spend Budget	OK	\$1.00	-	-	0.0

- من خلالها تقدر تكرير Budget setup ي notificate you لو تعديت \$0.01 ك free tier account .. وتكتب ال email بتاعك اللي هيبيعتلك عليه ال alert دة.

☐ Zero spend budget
Create a budget that notifies you once your spending exceeds \$0.01 which is above the AWS Free Tier limits.

☒ Monthly cost budget
Create a monthly budget that notifies you if you exceed, or are forecasted to exceed, the budget amount.

☐ Daily Savings Plans coverage budget
Create a coverage budget for your Savings Plans that notifies you when you fall below the defined target.

☐ Daily reservation utilization budget
Create a utilization budget for your reservations that notifies you when you fall below the defined target.

Monthly cost budget - Template

Budget name
Provide a descriptive name for this budget.

Names must be between 1-100 characters.

Enter your budgeted amount (\$)
Last month's cost: \$94.85

Email recipients
Specify the email recipients you want to notify when the threshold has exceeded.

- بالنسبة لل monthly بتحددله انت عاوز تدفع كام هنا \$10 خلال الشهر.



❗ You will be notified when 1) your **actual spend** reaches 85% 2) your **actual spend** reaches 100% 3) if your **forecasted spend** is expected to reach 100%.

💡 هيبعتلك mail لو صرفت 85% من ال \$10 ولو وصلت 100% او لو المتوقع توصل 100%

Overview Info

Budgets (2) Info

Download CSV Actions Create budget

Find a budget Type - Show all budgets < 1 > ⚙

☐	Name	Thresholds	Budget	Amount u...	Forecaste...	Current vs. budgeted
☐	My Monthly Cost Budget (\$10)	OK	\$10.00	-	-	0.0
☐	My Zero-Spend Budget	Exceeded (1)	\$1.00	\$29.83	\$86.40	2982.6

- لما بتعدي ال budget بتتحول للشكل ده

➡ EC2 Basics:

- EC2 = Elastic Compute Cloud = Infrastructure as a Service

● عبارة عن service من خلالها تقدر :

- تأجر EC2 << virtual machines
- تخزين Data علي EBS << Virtual Drives
- توزيع ال load عال machines من خلال ELB
- ت Scale ال Services من خلال auto-scaling group << ASG



● EC2 sizing & configuration options

- Operating System (OS): Linux, Windows or Mac OS
- How much compute power & cores (CPU)
- How much random-access memory (RAM)
- How much storage space
- Network-attached (EBS & EFS)
- hardware (EC2 Instance Store)
- Network card: speed of the card, Public IP address
- Firewall rules: security group
- Bootstrap script (configure at first launch): EC2 User Data

- **User Data** : عبارته عن mechanism في AWS من خلاله ب pass configurations or commands لل EC2 Instance وهيا بت launch في الغالب بتكون في شكل shell script
- **Bootstrapping**: عبارة عن process معناها انك ت Run Commands لما ال machine ت Start .. ال Script دة بي run مرة واحدة بس لما ال machine ت start

يبقى ال **User data** بحط فيه ال script بتاعي اللي بيتنفذ من خلال ال **bootstrapping** لما ال machine ت start



User data **Info**

```
#!/bin/bash
# Use this for your user data (script from top to bottom)
# install httpd (Linux 2 version)
yum update -y
yum install -y httpd
systemctl start httpd
systemctl enable httpd
echo "<h1>Hello World from $(hostname -f)</h1>" > /var/www/html/index.html
```

● ال Script دة بيكون عبارته عن :

- Install updates , softwares
- Download something from the internet or any thing you can think of

كل ما كان ال Script اللي هيتنفذ اثناء ال boot time اكبر كل ما ال instance هتاخذ

وقت علي ما تـ launch

ال Script دة بيتنفذ من خلال ال root user يا عني في ال script هتشتغل as sudo

- عندك انواع كتير من ال EC2 Instances دول امثله ليهم.

- ال t2.micro اللي هتشتغل عليه وفي ال free tier بيتيحلك 750 hour / month

I'm Here



➔ Create An EC2 Instance with EC2 User Data to have a website Hands On:

هيكريت Instance زي ما عملنا في ال cloud practitioner و install web server عليها من خلال ال user data

لو عملت Stop ال Instance وجيت شغلتها تاني هتلاقي ال IPV4 اتغير ف خلي بالك.

➔ EC2 Instances Types:

- هتلاقي كل الانواع وكل اللي محتاج تعرفه عنهم [هنا](#)

PAGE CONTENT

[General Purpose](#)

[Compute Optimized](#)

[Memory Optimized](#)

[Accelerated Computing](#)

[Storage Optimized](#)

[HPC Optimized](#)

[Instance Features](#)

[Measuring Instance Performance](#)

M7g M7i M7i-flex M7a Mac M6g M6i M6in M6a M5 M5n M5zn M5a

M4 T4g T3 T3a T2

[Amazon EC2 M5 instances](#) are the latest generation of General Purpose Instances powered by Intel Xeon® Platinum 8175M or 8259CL processors. These instances provide a balance of compute, memory, and network resources, and is a good choice for many applications.

Features:

- Up to 3.1 GHz Intel Xeon Scalable processor (Skylake 8175M or Cascade Lake 8259CL) with new Intel Advanced Vector Extension (AVX-512) instruction set
- New larger instance size, m5.24xlarge, offering 96 vCPUs and 384 GiB of memory
- Up to 25 Gbps network bandwidth using Enhanced Networking
- Requires HVM AMIs that include drivers for ENA and NVMe
- Powered by the [AWS Nitro System](#), a combination of dedicated hardware and lightweight hypervisor
- Instance storage offered via EBS or NVMe SSDs that are physically attached to the host server
- With M5d instances, local NVMe-based SSDs are physically connected to the host server and provide block-level storage that is coupled to the lifetime of the M5 instance
- New 8xlarge and 16xlarge sizes now available.

Instance Size	vCPU	Memory (GiB)	Instance Storage (GB)	Network Bandwidth (Gbps)***	EBS Bandwidth (Mbps)
m5.large	2	8	EBS-Only	Up to 10	Up to 4,750
m5.xlarge	4	16	EBS-Only	Up to 10	Up to 4,750

● m5.2xlarge

- m : تشير إلى ال Class



- 5 : تيشر إلى ال generation .. هتلاقي AWS بتطور ال hardware ومع كل تطوير بتقدم generations جديدة ف هتلاقي m5 , m6 m7
- 2xlarge : بتيشر إلى ال size هتلاقي عندك 4x , 8x , 12x وهكذا
- Size اكبر ياغني CPU , Memory اكبر وهكذا. 💡

● هتلاقي ان AWS مقسمالك الأنواع دي على حسب غرضك:

- **General Purpose** : مناسب لل work loads زي ال load balancers وال code repository وفيه balance بين ال CPU , Memory , Networking
- النوع اللي هتستخدمه اللي هو T2 من ال General purpose 💡
- **Compute Optimized** : بستخدمها لل Applications اللي بتحتاج High performance processing زي ال media transcoding
- **Memory Optimized** : بستخدمها لل Applications اللي بتحتاج High performance ال Memory
- **Storage Optimized** : بستخدمها لل Applications اللي بتحتاج High performance ال Storage
- **Accelerated Computing** : بستخدمها لو محتاج اعمل machine learning او complex calculations او graphic workloads



● مقارنة بسيطة:

Instance	vCPU	Mem (GiB)	Storage	Network Performance	EBS Bandwidth (Mbps)
t2.micro	1	1	EBS-Only	Low to Moderate	
t2.xlarge	4	16	EBS-Only	Moderate	
c5d.4xlarge	16	32	1 x 400 NVMe SSD	Up to 10 Gbps	4,750
r5.16xlarge	64	512	EBS Only	20 Gbps	13,600
m5.8xlarge	32	128	EBS Only	10 Gbps	6,800

t2.micro is part of the AWS free tier (up to 750 hours per month)

- ودة [Website](#) بيقارن بين كل الانواع والاسعار

➔ Security Group & Classic Ports Overview

- أكنو Virtual Firewall بيكون على ال EC2 instances بتاعتك وبي control ال inbound and outbound traffic من خلال rules انت بتحددها بتخليك تتحكم ف ال protocols,ports and source/destination ip وهكذا
- هتلاقي ان **by default** كل ال **outbound traffic are allowed** يعايني تقدر تكلم اي حد بره
- هتلاقي ان **by default** كل ال **inbound traffic are denied** وف الحالة دي بتحدد ازاي هتتعامل مع ال inbound roles من خلال ال rules اللي بتحطها وليكن مثلاً:

● Example:

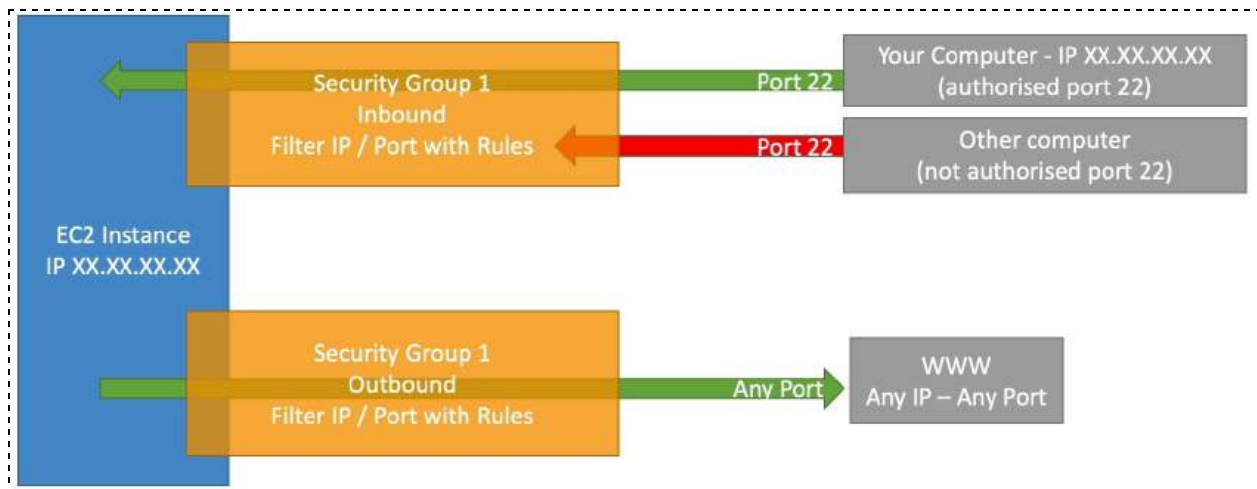
- حاطط rules بتسمح ب http traffic اللي جاي علي port 80
- حاطط rules بتسمح ب https traffic اللي جاي علي port 443



- حائط rules بتسمح ب ssh traffic اللي جاي علي port 22 من ip معين فقط وفي الغالب بيكون root user ip
- ممكن ت define اكثر من Security Group ل 1 instance
- ال Security Group بتشتغل على ال EC2 instance level مش على ال subnet level , اللي بيشتغل على ال subnet هو ال network access control list

● Control the inbound & outbound traffic by:

- IPV4 - IPV6
- Ports
- Inbound & Outbound Network



💡 يفضل فصل ال SSH فى Security group لوحدها.

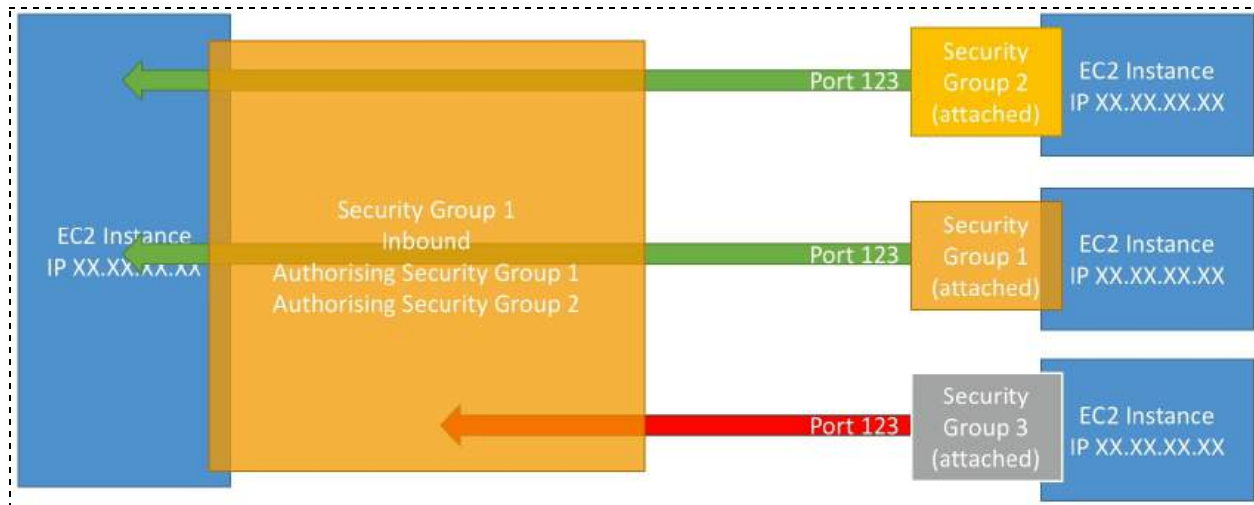
💡 لو بت Access ال App بتاعك وفيه time out او wait كثير ف 100% السبب ال

Security group

ولو Connection Refused Error فغالبا دي مشكلة Application or launching



● Important Example:



- فى ال Security Group اللى عالشمال انا سامح فيه بال inbound traffic من ال Security Group اللى علي اليمين اللى هما 1.2 بالتالي اي instances جواهم هيبقي مسموح بال traffic اللى جاي منها على ال instance اللى عالشمال

💡 انك تسمح بال inbound traffic اللى جاي من ال Security Group يغني عن انك تسمح بكل instance جواه مش هتفضل تكتب كل ip كل instance انت بتشوف مين ال security group اللى لامهم وتسمح بيه وخلاص .. ودة مناسب فال auto scaling علشان لو فيه instances بتزيد او بتنقص بردو اكيد مش هتفضل تعدل ال inbound rules عال Security Group اللى عالشمال

💡 انك ت Authorized Security Group عند Security Group ثاني بيتن من خلال ال ID بتاعه.

● Classic Ports:

- 22 = SSH (Secure Shell) - log into a Linux instance



- **3389 = RDP** (Remote Desktop Protocol) – log into a Windows instance
 - **21 = FTP** (File Transfer Protocol) – upload files into a file share
 - **22 = SFTP** (Secure File Transfer Protocol) – upload files using SSH
 - **80 = HTTP** – access unsecured websites
 - **443 = HTTPS** – access secured websites
-

➡ Security Groups Hands On:

💡 في حالة ال Security Group لو فتحت ال inbound traffic اللى جاي علي port 80

تلقائي ال outbound من port 80 هيكون allowed

طبعا ده علي عكس ال iptables مثلا لازم تحط للاتنين .rules

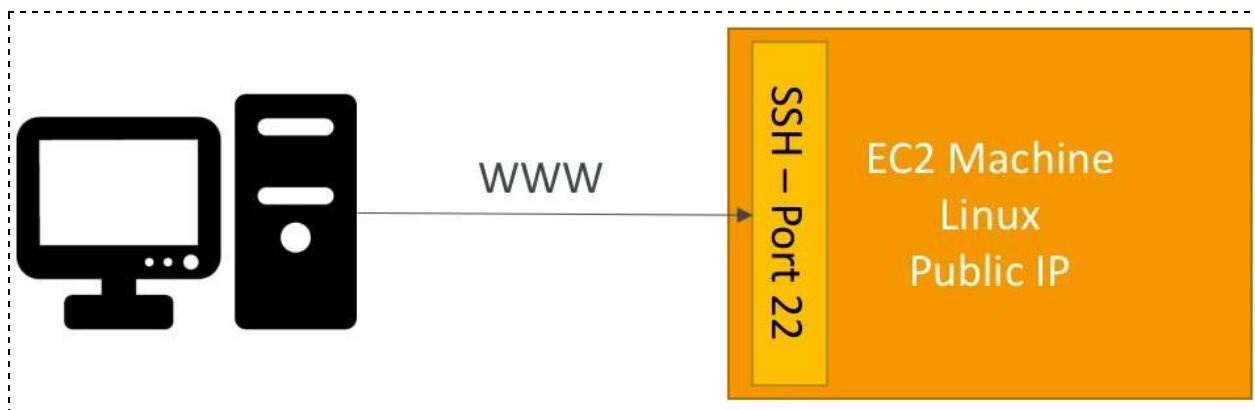


→ SSH Overview:

	SSH	Putty	EC2 Instance Connect
Mac	✓		✓
Linux	✓		✓
Windows < 10		✓	✓
Windows >= 10	✓	✓	✓

- اقل من 10 windows مينفعش معاها ssh
- **Putty** ل windows فى العموم
- EC2 Instance Connect هنتشرح قدام

→ How To SSH Using Linux Or Mac



- بتفتح ال Port 22 فى ال security group اللى بيحوي ال EC2 instance بتاعتك وبت ssh عال ipv4 public بتاع ال instance .. ولكن انت اكيد مش بتكون فاتح ال ssh لاي حد ي access ال instance ف لازم يا تكون محدد ال ip اللى هي access او عن طريق key

Ssh -i key.pem ec2-user@ipv4

- **Ec2-user** دة ال user اللى معمول عال instance واللى لما تخش هتخش بيه
- **Ipv4** دة بتاع ال instance

```
~/aws-course ➤ ssh -i EC2Tutorial.pem ec2-user@3.250.26.200
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@          WARNING: UNPROTECTED PRIVATE KEY FILE!          @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
Permissions 0644 for 'EC2Tutorial.pem' are too open.
It is required that your private key files are NOT accessible by others.
This private key will be ignored.
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@          WARNING: UNPROTECTED PRIVATE KEY FILE!          @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
Permissions 0644 for 'EC2Tutorial.pem' are too open.
It is required that your private key files are NOT accessible by others.
This private key will be ignored.
Load key "EC2Tutorial.pem": bad permissions
Received disconnect from 3.250.26.200 port 22:2: Too many authentication failures
Disconnected from 3.250.26.200 port 22
✖ ➤ ~/aws-course ➤ chmod 0400
```

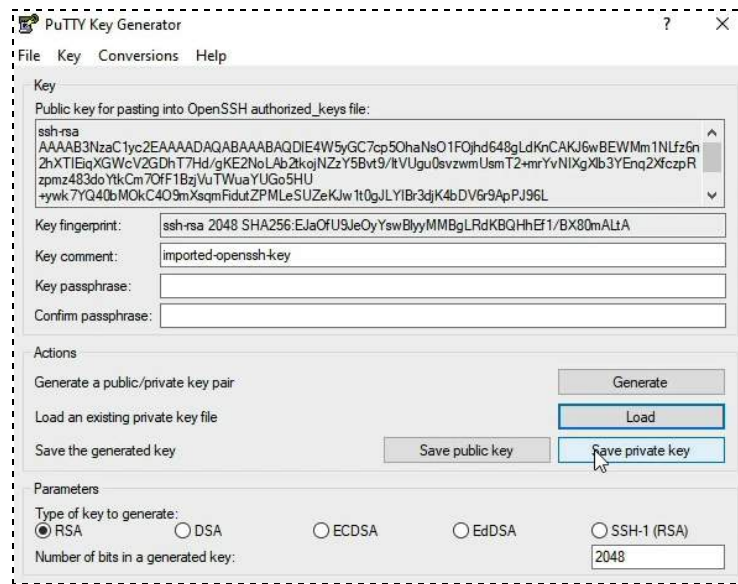
- ال error دة الحل بتاعه ف انك تدي الملف read permission لل owner فقط.



➔How To SSH Using Windows7.8:

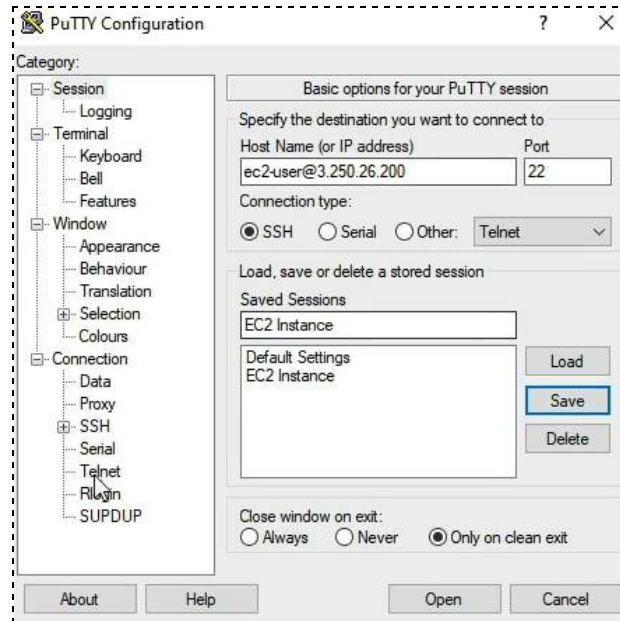
(1) هتحميل من جوجل putty

(2) بعد ما تسطبه هتلاقي عندك ليه جزاين puttygen خاص بال key وال putty ذات نفسه

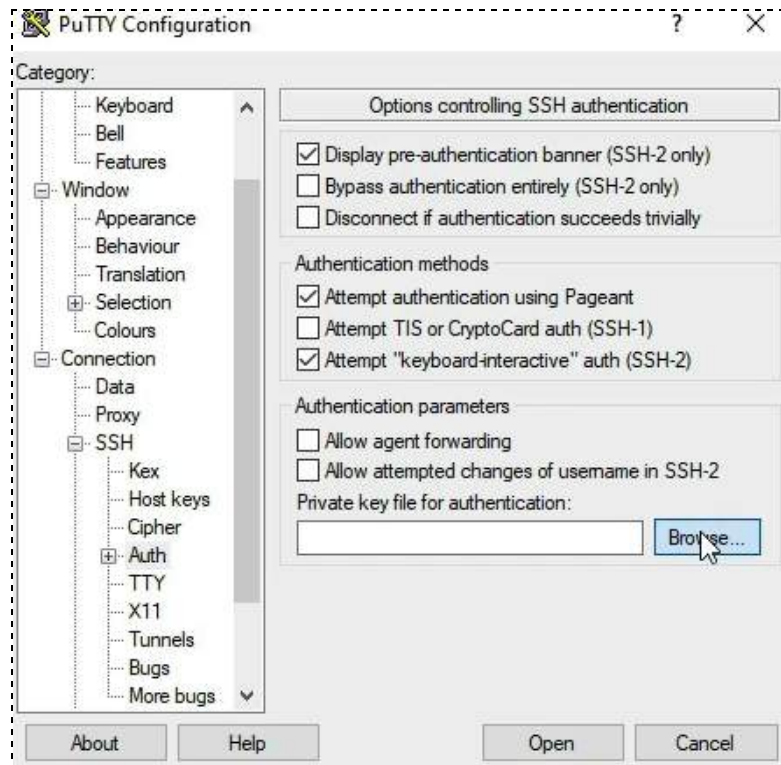


- هت load ال key.pem وتحوله ل key.ppk او لو حملته من الاول بصيغة ppk ف تمام ياعني .. وت save private key
- (3) هتملي الداتا دي :





4) Ssh ثم auth وبعدئذ هت load ال private key اللي هو .ppk



5) بت save اللي عملته دة فى الصفحة الاولى بحيث متعلمش الخطوات دي كل ما تحتاج

تخش عال instance



➔How To SSH Using Windows10:

- ممكن من خلال putty عادي

```
Ssh -i key.pem ec2-user@ipv4
```

- لو جالك error هيبقي نفس ال error اللي زي اللي فوق علشان ال owner ال
وهتغيرو من خلال :

Right click on the file >> properties >> security >> Advanced >>
Change >> choose your self.

```
PS C:\Users\stephanemaarek\Desktop> ssh -i .\EC2Tutorial.pem ec2-user@3.250.26.200
The authenticity of host '3.250.26.200 (3.250.26.200)' can't be established.
ECDSA key fingerprint is SHA256:INQMVOPTzV+fRkL+EeoRtTp+pWI4koar4F8a6QTmPgE.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.250.26.200' (ECDSA) to the list of known hosts.
Last login: Mon May 16 17:28:25 2022 from bl14-117-32.dsl.telepac.pt

 _ | _ | _ )
 _ | ( _ | /  Amazon Linux 2 AMI
 _ | \ _ | _ |

https://aws.amazon.com/amazon-linux-2/
[ec2-user@ip-172-31-33-135 ~]$
```

➔SSH Troubleshooting:

- ملف كويس اقرأه

➔EC2 Instance Connect (Easy way)



- هيثم الموضوع من خلال ال browser مجرد ما بتضغط connect ال session بتفتح عال browser

Instances (1/1) Info						
Search < 1 >						
☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm s
☒	My First Instance	i-034466697feb9ef80	Running	t2.micro	2/2 checks passed	No alarm

Connect to instance [Info](#)

Connect to your instance i-034466697feb9ef80 (My First Instance) using any of these options

EC2 Instance Connect

Session Manager

SSH client

EC2 Serial Console

Instance ID

i-034466697feb9ef80 (My First Instance)

Public IP address

3.250.26.200

User name

Connect using a custom user name, or use the default user name ec2-user for the AMI used to launch the instance.

Note: In most cases, the guessed user name is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI user name.

Cancel

Connect

Connect to instance | EC2 Man...
i-034466697feb9ef80 (My First...

<https://eu-west-1.console.aws.amazon.com/ec2/v2/connect/ec2-user/i-034466697feb9ef80>

Last login: Mon May 16 17:43:26 2022 from bl14-117-32.dsl.telepac.pt

```

  _ | ( _ | )
  _ | ( _ | ) Amazon Linux 2 AMI
  _ | \ _ | _ |
https://aws.amazon.com/amazon-linux-2/
[ec2-user@ip-172-31-33-135 ~]$
```





يمكن حتي لو فتحت ال ssh عال instance من anywhere/ipv4 متعرفش ت
connect فى الحاله دي افتح anywhere/ipv6

Any timeout (not just for SSH) is related to security groups or a
Firewall

لو عملت Stop لل Instance وجيت شغلتها تاني ممكن تلاقي ال IPV4 اتغير ف خلي
بالك.. ياعني مش هتعرف تخش لو محدد انك ت ssh بال ipv4 اللي اتغير

➡ EC2 Instance Roles Demo:

- اللي عمله انو access ال instance وحاول ي list ال user ولكن ال instance
مكنتش واخده ال role دي .. ف عملنا attach ل readonly role لل instance
علشان تقدر ت list ال users

لما تيجي ت attach ال role ممكن تاخذ وقت على ما تسمع عال instance ف
متستعجلش.

ال instance لما كريبتها ب linux جت بال CLI installed جاهز




```
[ec2-user@ip-172-31-17-250 ~]$ aws --version
aws-cli/1.18.147 Python/2.7.18 Linux/4.14.225-169.362.amzn2.x86_64 botocore/1.18.6
[ec2-user@ip-172-31-17-250 ~]$ aws iam list-users
Unable to locate credentials. You can configure credentials by running "aws configure".
[ec2-user@ip-172-31-17-250 ~]$ aws configure
AWS Access Key ID [None]:
AWS Secret Access Key [None]:
Default region name [None]:
Default output format [None]: ^C
[ec2-user@ip-172-31-17-250 ~]$
```

- متعلمش configure لل credential بالشكل دة اي حد لو دخل عال instances يقدر ي retrieve ال access key ويخرب الدنيا

➔ EC2 Instance Purchasing Options:

● On-Demand Instances :

- ودي pas as you go ف Windows , Linux بندفع بالثانية واي OS تاني بالساعة .. وده اعلى سعر ف ال Options يايني لو محتاج مثلا سيرفر اسبوع شهر شهرين ثلاثة.
- مفيش فيها upfront payment دفع مقدم يايني.
- 💡 **ال usage :** فى ال short-term usage ولل Apps اللى مش توقع ال load بتاعها.

● Reserved Instances & Convertible Reserved Instances (1&3 year):

● Reserved Instances

- وهنا AWS بتقلل ال Cost مقابل انك تلتزم ب Configuration ثابتة لل Instances بتاعتك زي ال region , type.
- فيها upfront payment دفع مقدم .. ممكن متدفعش وممكن تدفع جزاً ممكن تدفع كل الحساب وطبعاً كل ما دفعت اكثر كل ما هتأخذ خصم اكثر.



- بيبقي فيها تقريبا 70% discount عن ال on demand وتقدر تبيع ال instances دى فى ال market place لو مبقتش محتاجها
- سنة او 3 سنين مش سنة الي 3 سنين

💡 **ال usage :** فى ال steady state usage applications زي ال database

● Convertible Reserved Instances

- نوع من انواع ال Reserved Instances ميزته انك تقدر تغير ال
- EC2 instance type, instance family, OS, scope and tenancy
- بيبقي عليه discount بيوصل ل 66%



● Saving Plans (1&3 year):

- بيبقي فيها تقريبا 70% discount زي ال reserved
- بتلتزم فيها بالاستخدام مثلا (10 دولار/ساعة لمدة 1 أو 3 سنوات) ولو استخدامك زاد عن اللى انت محددة هتدفع بال on-demand
- Locked لل instance family & region ياعني لو حددت M5 على us-east-1 مش هتغيره



● Spot Instances:

- ودي سعرها بيبكون مخفض لنسبة قد تصل ل 90% من سعر ال On-Demand Instances ودي يعتبر اقل سعر ف ال options المتاحة. ف ال option دة انت بتستخدم ال unused EC2 Instances بسعر مخفض لكن ممكن تخسر ال instances بتاعك ف اي وقت لو سعر ال spot instance تخطي ال max price بتاعك انت ك customer ودة طبعاً مش مناسب لل critical services.

💡 **ال usage :** مش مناسب لل critical jobs ومناسب للآتي :

Batch jobs , Data analysis , Image processing , Any distributed , workloads , Workloads with a flexible start and end time



● Dedicated Hosts & Dedicated Instances:

- الاتنين متقاربين ال **hosts** بتاخد physical server ويكون ليك full control عال hardware اما ال **instances** يكون ليك control اقل ف بتختارها لما متكونش محتاج detailed hardware control
- علشان اللغطة من الاسم Dedicated Host يايني السيرفر ملكك .. Dedicated Instances يايني ال instances ملكك على hardware ملكك
- الدفع بيكون حاجة من الاتنين :
- **On-demand**
 - pay per second for active Dedicated Host
- **Reserved**
 - 1 or 3 years (No Upfront, Partial Upfront, All Upfront)

Comparison Summary

Feature	Dedicated Hosts	Dedicated Instances
Control	Full control over instance placement on specific physical servers	AWS manages instance placement
Hardware Visibility	Full visibility into physical hardware (sockets, cores)	No visibility into physical hardware
Licensing	Ideal for BYOL scenarios requiring specific hardware licensing	Less suitable for complex licensing
Isolation	Dedicated hardware for a single customer	Dedicated hardware for a single customer
Management	Requires more management effort	Simpler, less management overhead
Billing	Billed per host	Billed per instance
Use Cases	Regulatory compliance, detailed hardware control, BYOL	Physical isolation without hardware control



● Capacity Reservation:

- ال capacity هنا معناها عدد ال EC2 Instance من type معين في region او AZ معينه
- وهنا AWS بتعملك Reservation ل on-demand EC2 Instance ف Capacity Availability zone معينه لمدة بيتم الاتفاق عليها.
- مفيش time commitment التزام بالوقت .. ابدأ وكنسل في اي وقت براحتك
- مفيش discount ولكن ممكن تدمجها مع ال reserved instances و ال saving plans بحيث تاخذ خصم

💡 هتتاسب اكنك on-demand وال instance شغالة او مش شغالة هتتاسب عليها.

Usage : هتكون مناسبة لل short-term uninterrupted workloads في AZ معينه.

● Scheduled Instances:

- ودي من اسمها تقدر من خلالها انك تشغل ال Instances بتاعتك بجدولة ياعني خلال عدد ساعات معينه مثلا ف اليوم او الاسبوع او الشهر ودي بتكون لمدة سنة.

● Price Comparison - Example – m4.large – us-east-1

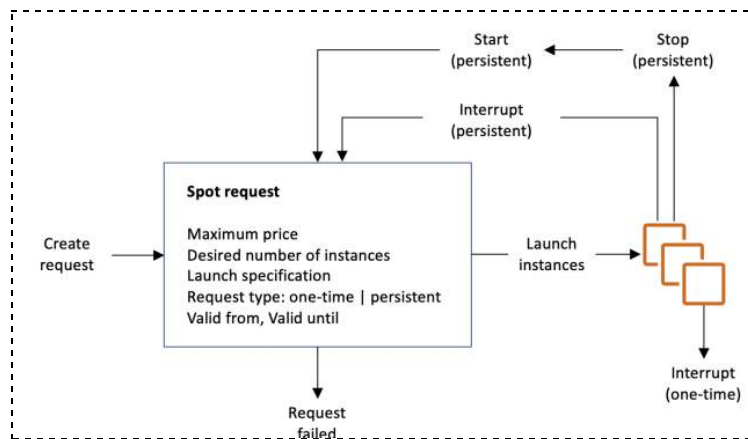
Price Type	Price (per hour)
On-Demand	\$0.10
Spot Instance (Spot Price)	\$0.038 - \$0.039 (up to 61% off)
Reserved Instance (1 year)	\$0.062 (No Upfront) - \$0.058 (All Upfront)
Reserved Instance (3 years)	\$0.043 (No Upfront) - \$0.037 (All Upfront)
EC2 Savings Plan (1 year)	\$0.062 (No Upfront) - \$0.058 (All Upfront)
Reserved Convertible Instance (1 year)	\$0.071 (No Upfront) - \$0.066 (All Upfront)
Dedicated Host	On-Demand Price
Dedicated Host Reservation	Up to 70% off
Capacity Reservations	On-Demand Price



➔ Spot Instances & Spot Fleet:

- لو السعر تخطي السعر اقصى سعر انت هتدفعو قدامك دقيقتين تقرر فيهم هل هتعمل stop او terminate ال instance .. لو هتعمل stop لما ال spot price ينزل عن ال max price تقدر تعملها start من تاني.

● How Spot Request Work :



- بتحدد كام عدد ال instances و ال max price اللي هتدفعو وال launch specifications

● How to terminate spot instances?

- عندك نوعين من ال request :
- One-time request
- بمجرد ما بتعمل ال request ال instances بت launch ولو حصلها interrupt or termination مش هتقوم تاني الا لما انت تعمل request جديد دة مناسب لل work loads اللى بتحتاج one single run او مش محتاجة continuous execution



- Example

You have a data processing job that you run periodically. You submit a one-time Spot Instance request to complete this job. Once the job is finished, or if the instance is terminated, no further action is taken automatically.

● Persistent request

- Persistent request يحاول دائما يحافظ على عدد ال instances لو اي instance حصلها interrupt او termination عطلول بي launch واحدة جديدة كبديل ودة مناسب لل work loads اللى بتحتاج continuous execution or high availability

- Example

You have a long-running data analysis application that needs to run continuously. You submit a persistent Spot Instance request for 3 instances. If any of these instances are interrupted, AWS will automatically attempt to launch new instances to maintain the count of 3.

يبقى فى النوع ده لو محتاج ت terminate ال instances خالص لازم الاول توقف ال persistent request لانك لو موقتوش هيحصل relaunch ال instances بحيث يحافظ على عددها اللى انت محددهوله.



- **Spot Fleet:**

- **Example** : لو عندك web application وعاوز تهندل ال variable traffic اللى بيجيك خلال اليوم وتحافظ على انك تدفع اقل تكلفة وانو يكون available علطول هتعمل الآتي :

- **Define Target Capacity :**

- بت define عدد ال instances اللى عاوزها وال cpu وهكذا

- **Choose the allocation strategy:**

- **Lowest price**

- ودي لو محتاج اقل cost

- **Diversified or Capacity Optimized**

- لو محتاج Availability عالية وأنت تقفل the risk of interruption

- **priceCapacityOptimized (recommended) :**

- ميكس بين الاتنين Available high capacity و lowest price

- **Choose instances types and AZs**

- هت list ال types وال AZs اللى ال application بتاعك can run on it

- **Mix Purchase Options**

- Optionally include a mix of Spot and On-Demand Instances to balance cost and availability.

- **Launch Spot Fleet**

➡ EC2 Instance launch types hands on:

Instances >> spot request >> request spot instances

ممکن من هنا او وانت بت launch ال instance عادي هتلاقي دي ..



▼ **Advanced details** [Info](#)

Purchasing option [Info](#)

☒ Request Spot Instances

[Customize](#)

Request Spot Instances at the Spot price, capped at the On-Demand price

CH-06 - EC2 - Solutions

Architect Associate Level



I'm Here

001 Private vs Public vs Elastic IP:

Instance: i-03977f3eb2e4e0d4d		
Details	Security	Networking
▼ Instance summary Info		
Instance ID i-03977f3eb2e4e0d4d	Public IPv4 address 3.129.218.15 open address	Private IPv4 addresses 172.31.11.234
Instance state Running	Public IPv4 DNS ec2-3-129-218-15.us-east-2.compute.amazonaws.com open address	Private IPv4 DNS ip-172-31-11-234.us-east-2.compute.internal

- ال instance اللي انت بتأجرها بيكون ليها public & private ip
- لما بت ssh عال instance بنستخدم ال public ip مينفعش ال private لاننا مش فى same network
- ال instance لو عملتها stop وبعدين start ال public ip دة بيتغير

002 Private vs Public vs Elastic IP Hands On

● Create Elastic IP:

Network & Security >> Elastic IPs >> Allocate Elastic IP Address

بيكون ليه تكلفه خاصه لواحدة بحوالي \$0.005 فى الساعه

Elastic IP addresses (1/1)			
Filter Elastic IP addresses		Public IPv4 address: 18.216.64.85 X	Clear filters
☒	Name	Allocated IPv4 add...	Type
☒	-	18.216.64.85	Public



I'm Here

● بعد ما كريت ال elastic ip .. بتربطة بال instance بتاعتك وبستفاد الآتي :

- لو عملت stop لل instance وبعدين start مش هيتغير ودة بيفيد في حاجات كتير منها

ال DNS record مثلاً

- بعد ما بتربطة .. تقدر ت ssh عليه .. بتلاقي ان ال elastic ip , public ipv4 بقو

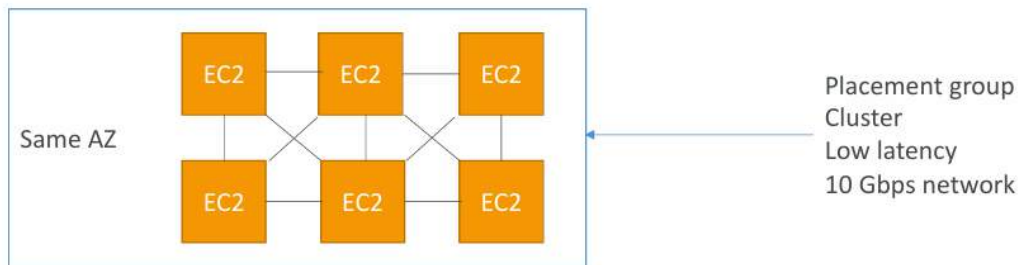
واحد .. هما نفس ال elastic

- تقدر بعد ما ربطته ب instance تشيله وتربطة بواحدة تانية



➡ 003 EC2 Placement Groups

● Cluster

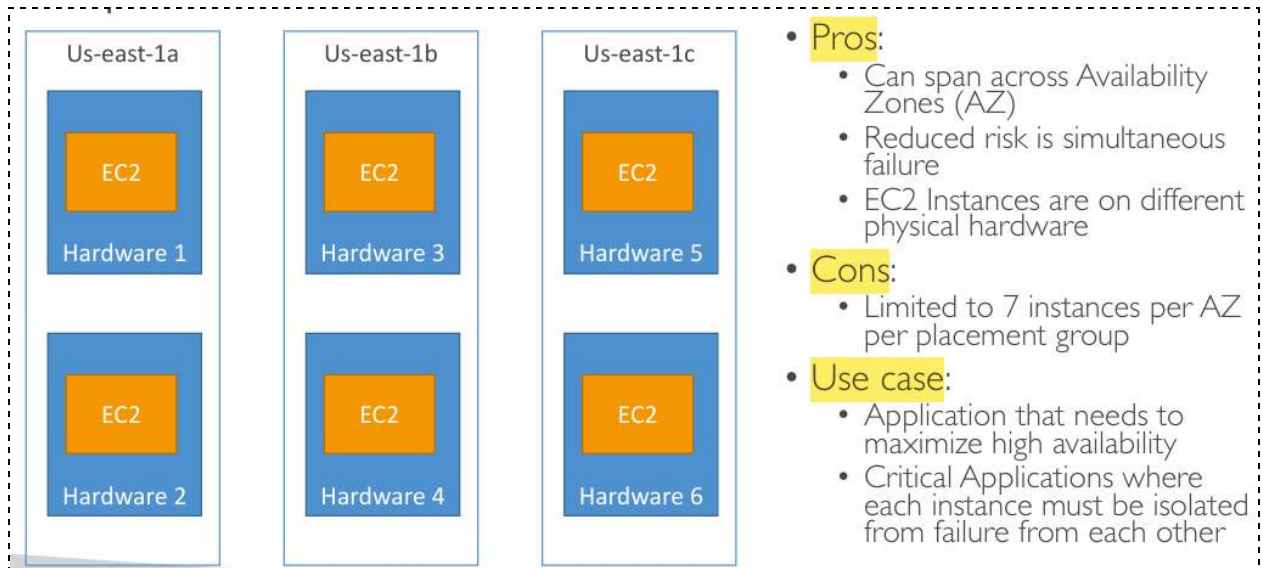


- **Pros:** Great network (10 Gbps bandwidth between instances with Enhanced Networking enabled - recommended)
- **Cons:** If the AZ fails, all instances fail at the same time
- **Use case:**
 - Big Data job that needs to complete fast
 - Application that needs extremely low latency and high network throughput

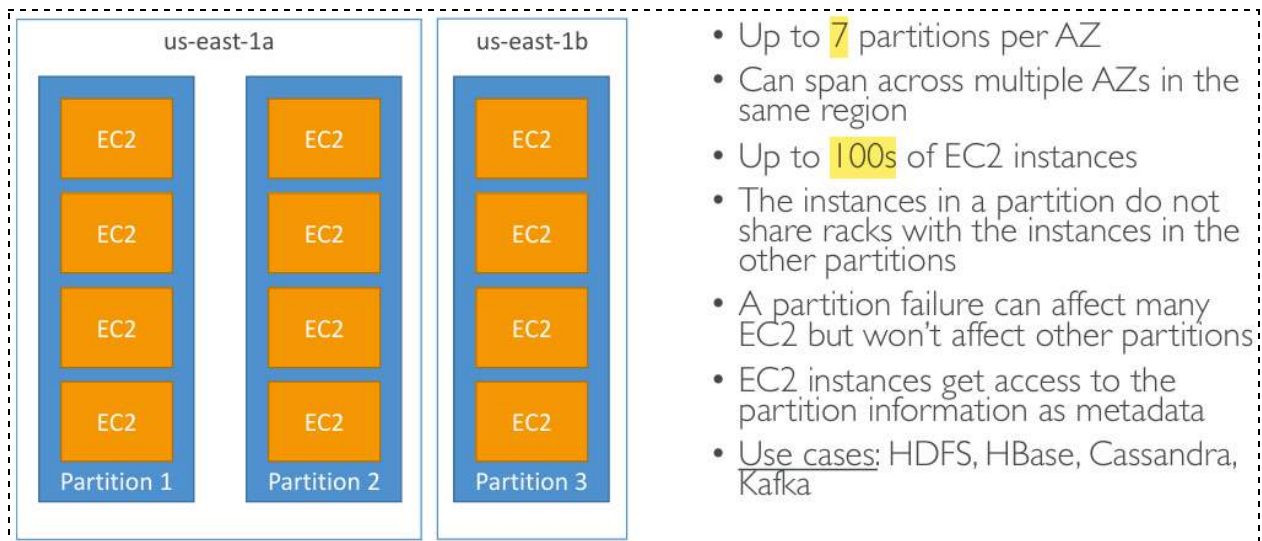


I'm Here

• Spread



• Partition



- highest networking performance >> **Cluster**
- maximum availability when there's an AZ failure >> **Spread**



➔004 EC2 Placement Groups - Hands On

Network & Security >> Placement group >> Create Placement group >> name,type

- بعد ما بتكرت ال placement group بتروح تكرت ال instances اللي هتحتها فيها

Instances >> Create Instance >> Advanced details >> Placement groups >> choose the placement group

➔005 Elastic Network Interfaces (ENI) -

Overview

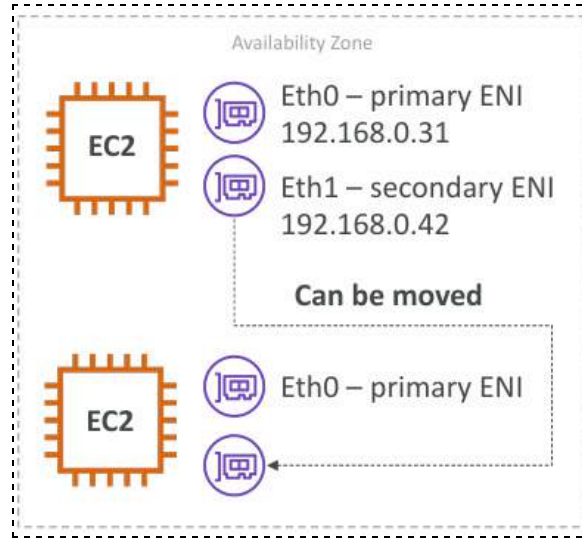
- Network Cards / Interface



Connects the computing device to the Outside world. Can be wireless (WiFi) or wired (use Ethernet/LAN cables).

- عن طريقها بنوصل لل internet سواء wireless زي اللاب كذا او لو كمبيوتر ممكن بكابل او ممكن ب وصلة تخليه يلقط واي فاي.. فى الحالة بتاعتنا ال instance بتوصل للانترنت من خلاله.





- **ENI** : عبارة عن virtual network card

● ENI Have :

- Primary private IPv4, one or more secondary IPv4
- One Elastic IP (IPv4) per private IPv4
- One Public IPv4
- One or more security groups
- A MAC address

- عبارة عن interface virtual بتكريرتها وتعملها attach لل instance بتاعتك
- انت اول مبيت launch ال EC2 Instance بتاعتك بتاخذ private ip جوا ال subnet اللى انت حددت انها تكون فيها
- جوا ال EC2 Instance هتلاقى option من خلاله تقدر تحدد هل هتدي ال EC2 Instance public ip ولا لا
- ال ENI زي مقولنا بيتعملها attach عال EC2 Instance ف بتكون جزا من ال EC2 Instance دي .. ف ساعتها ال EC2 Instance هيكون ليها 2 ip ال primary اللى هو ال private وال secondary بتاع ال ENI
- ممكن تشيلها من instance وت attach it لواحد تانية
- محكومة بال AZ اللى هيا فيها



006 Elastic Network Interfaces (ENI) - Hands

On

هنعمل 2 instances كل واحدة هتلاقي عندها network interface ببيكون ليها id ومتوضح ال network interface دي مربوطة بانهي instance id وهكذا

Network & Security >> Network Interface

Network interfaces (2) Info						Actions	Create network interface
Filter network interfaces						< 1 >	
☐	Name	Network interface ID	Subnet ID	VPC ID	Availability Zone		
☐	-	eni-0479b24bf2debda40	subnet-444bf62f	vpc-6047d20b	us-east-2a		
☐	-	eni-00f63c73c7d0187fc	subnet-444bf62f	vpc-6047d20b	us-east-2a		

Network interfaces (2) Info						Actions	Create network interface
Filter network interfaces						< 1 >	
Description	Instance ID	Status	Public IPv4 address	Primary private IPv4 address			
-	i-0575f9fde3525b143	In-use	3.143.214.169	172.31.7.169			
-	i-0145a88f58ba656db	In-use	18.189.185.151	172.31.13.28			

بمجرد ما تكريرت network interface هتحددها ومن action تقدر ت attach ال network interface دي لاي instance من الاثنين ف هتروح عليها هتلاقي عليها ال network interface الاصلية والجديدة



I'm Here

☒	-	i-0145a88f58ba656db	Running	t2.micro	Initializing	No alarms	+	us
☐	-	i-0575f9fde3525b143	Running	t2.micro	Initializing	No alarms	+	us
☐	-	i-03977f3eb2e4e0d4d	Terminated	t2.micro	-	No alarms	+	us

Network interfaces (2)

Filter network interfaces

Interface ID	Description	Public IPv4 address	Private IPv4 address	Private IPv4 DNS	IPv6
eni-00f63c73c7...	-	18.189.185.151	172.31.13.28	ip-172-31-13-28.us-...	-
eni-021963833f...	DemoENI	-	172.31.10.247	ip-172-31-10-247.us...	-

وتقدر بعد كذا تنقلها من واحدة للتانية

● Example: Initial Configuration

● Instance A:

- Attached ENI: eni-12345678
- Private IP: 10.0.0.5
- Elastic IP: 54.123.45.67

● Instance B:

- No ENI initially attached

● User Connection:

- Users connect to the application using the Elastic IP
54.123.45.67

● How It Works Internally

● Before Failover:

- Users access the application on Instance A
via Elastic IP 54.123.45.67.



- The traffic is routed through the Elastic IP to the private IP **10.0.0.5** of Instance A.

- **During Failover:**

- You detach the ENI from Instance A.
- The ENI retains its IP configuration.

- **After Failover:**

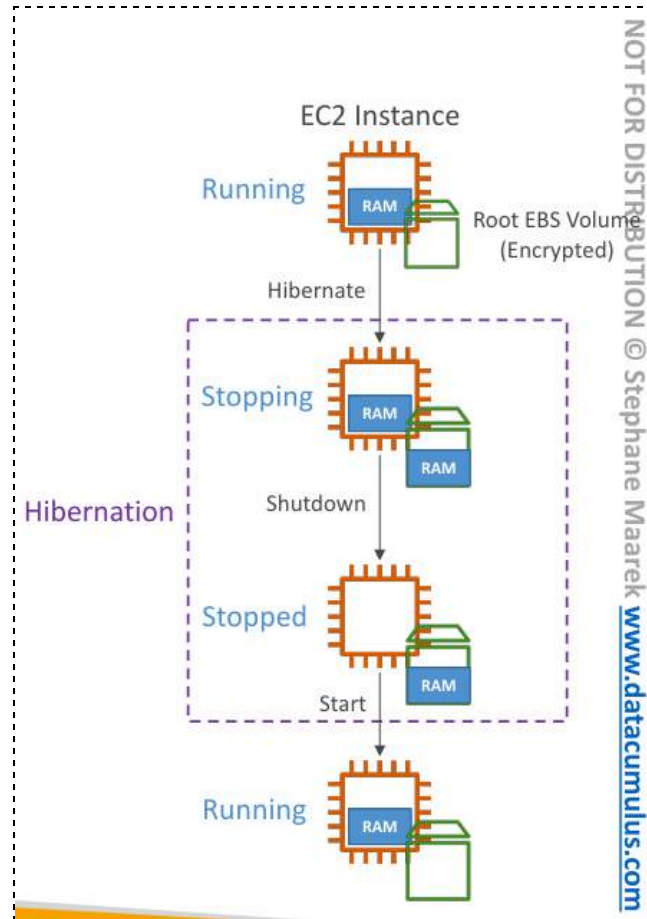
- You attach the ENI to Instance B.
- Instance B now uses the same IP configuration.
- Users access the application on Instance B via Elastic IP **54.123.45.67**.
- The traffic is routed through the Elastic IP to the private IP **10.0.0.5** of Instance B.

يبقي انت رابط ال ENI عال instance A لو بعمل صيانة مثلا لل app اللي عليها
وعاوز انقل علي instance B بنقل ال ENI على B ولما ال user ي access ال
Elastic ip هينقلو لل private ip الخاص ب instance b.



008 EC2 Hibernate

- **Hibernate** ال instance اكنك بتعمل لل computer بتاعك sleep
- فى الطبيعي لو انت قفلت الكمبيوتر بتاعك shutdown وفتحته من تاني بي boot from scratch وبيبتدي يشغل البرامج من تاني وبتفتح ال files من تاني وهكذا
- انما لو عملته Sleep الحوارات دي كلها اللى بتكون موجودة فى ال RAM بتتخزن فى ال SSD او فى ال HDD على حسب انت عامل ال OS بتاعك على مين فيهم ولما تشغله من تاني وبتفتح ال data دي بترجع من تاني من ال SSD or HDD لل RAM



- دة نفس اللى بيحصل عال instance بنفعل ليها ال Hibernate mode واحنا بنكريتها ف لما بنعمل ليها Hibernate بتخزن ال data اللى كانت فى ال RAM عندها



فى ال EBS root volume لحد ما تقوم من جديد وترجع تاني ال data دي لل RAM بتاعتها تاني.

● Use Cases:

- لو process بتاخذ وقت طويل ومش عاوزها تفصل او لو فيه services شغالة عال instance وبتاخذ وقت طويل على ما تفتح ومش عاوزها تفتح from scratch

● فيه شويه حاجات بتحكمك لو هتستعمل ال Hybernate :

- Supported Instance **Families** – C3, C4, C5, I3, M3, M4, R3, R4, T2, T3, ...
- **Instance RAM Size** – must be less than 150 GB.
- **Instance Size** – not supported for bare metal instances.
- **AMI** – Amazon Linux 2, Linux AMI, Ubuntu, RHEL, CentOS & Windows...
- **Root Volume** – must be EBS, encrypted, not instance store, and large
- Available for **On-Demand**, **Reserved** and **Spot Instances**

💡 An instance can NOT be hibernated more than **60 days**

➡ 009 EC2 Hibernate Hands On:

Create EC2 Instance >> Advanced Details >> Hibernate Behavior



Stop - Hibernate behavior [Info](#)

Select

Select

Enable

Disable

● الحاجات دي لازم تحصل :

لازم يكون ال root volume عليه مساحة كافيه تكفي ال RAM Content اللى هيتخزن عليه لحد ما ال instance ت start من جديد.. هنا عندنا ال instance 1 ال 8GiB Memory وال EBS root volume 8GiB ف كذا دة مناسب.

▼ **Storage (volumes)** [Info](#) Simple

EBS Volumes Hide details

▼ Volume 1 (AMI Root) (Custom)

Storage type Info EBS	Device name - required Info /dev/xvda	Snapshot Info snap-00022cfee7f3b9690
Size (GiB) Info 8	Volume type Info gp2	IOPS Info 100 / 3000
Delete on termination Info Yes	Encrypted Info Yes	KMS key Info (default) aws/ebs Key ID: 6e839668-f27c-4cd1-b6...

▼ **Instance type** [Info](#)

Instance type

t2.micro Free tier eligible [Compare instance types](#)

Family: t2 1 vCPU 1 GiB Memory
On-Demand Linux pricing: 0.0126 USD per Hour
On-Demand Windows pricing: 0.0172 USD per Hour

لازم يكون ال root volume encrypted



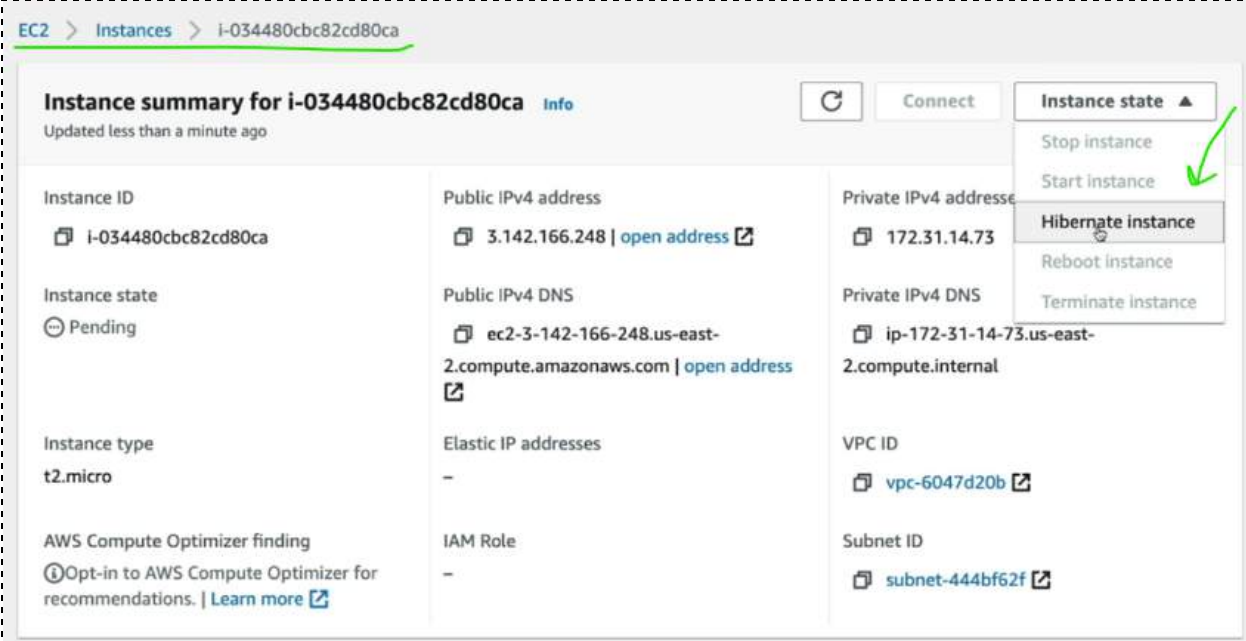
I'm Here

● TEST

علشان نشوف الدنيا دي شغالة ولا اية هنستخدم command اسمو uptime بيعرفنا ال instance دي بقالها قد اية شغالة.

```
[ec2-user@ip-172-31-14-73 ~]$ uptime
23:34:14 up 1 min, 1 user, load average: 0.30, 0.12, 0.04
[ec2-user@ip-172-31-14-73 ~]$ uptime
```

ظاهرنا دلوقتي بقالها شغالة لمدة دقيقة .. ف انت لو مش مفعل ال Hibernate وعملتها stop & start المفروض لو جيت تنفذ نفس ال command يظهر لك 0 ولكن بما انك مفعل ال Hibernate ف هتلاقي العداد دة متصفرش وكمل ثاني ودة معناه ان الداتا دي اللي كانت عال RAM انتقلت عال EBS root volume لحد ما ال instance تقوم من جديد ورجعته ثاني وكملت.



EC2 > Instances > i-034480cbc82cd80ca

Instance summary for i-034480cbc82cd80ca Info
Updated less than a minute ago

Instance ID i-034480cbc82cd80ca	Public IPv4 address 3.142.166.248 open address	Private IPv4 address 172.31.14.73
Instance state Pending	Public IPv4 DNS ec2-3-142-166-248.us-east-2.compute.amazonaws.com open address	Private IPv4 DNS ip-172-31-14-73.us-east-2.compute.internal
Instance type t2.micro	Elastic IP addresses -	VPC ID vpc-6047d20b
AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more	IAM Role -	Subnet ID subnet-444bf62f

```
[ec2-user@ip-172-31-14-73 ~]$ uptime
23:35:57 up 2 min, 1 user, load average: 0.23, 0.12, 0.05
[ec2-user@ip-172-31-14-73 ~]$
```



I'm Here

CH-07 - EC2 Instance Storage

➔001 EBS Overview

- عبارة عن Virtual hard drive بتعمله attach لل instance بتاعتك , اكنك عندك كمبيوتر ومعاك هارد خارجي بتوصله بيه وتحتفظ عليه الداتا بتاعتك.

● مميزاتها :

- **Persistent Storage**: الداتا هتفضل موجودة حتي لو عملت stop او restart لل instance بتاعتك .. لو مسحت ال instance هتلاقي ان ال EBS volume لسة موجودة ممكن بعد كدا تعملها attach ل instance تانية.
- **Different Types**: فيه انواع كتير منها على حسب الغرض من استخدامك محتاجة بالظبط (general-purpose, high-performance, or cost-effective storage).
- **Backup and Restore**: تقدر تاخذ (snapshots backup) لل EBS volumes دي بحيث تقدر ت restore ال data
- **Resize and Change**: تقدر ت change , resize ال EBS volumes دي من غير ما تفقد ال data بتاعتك

💡 متقدرش ت attach ال EBS Volume دي لاكثر من instance الا في حالة معينة

هنعرفها بعدين.

💡 ال EBS Volume محكومة بال AZ اللي هيا فيها بمعنى .. ال EBS Volume انت

بتكريتها وتعملها attach عال instance وال instance بتكون موجودة وليكن ف AZ1 ف

لو انت ف AZ2 تانية مش هتشوف ال EBS Volume اللي ف AZ1

I'm Here



يبقي عندك EBS root volume الاساسية واي واحدة تانية انت بتكريتها بتبقى EBS

volume اضافيه

● Delete On Termination:

▼ Configure storage Info Advanced

1x 8 GiB gp3 Root volume (Not encrypted)

Add new volume

- وانت بتكريت ال instance بتاعتك كنت بتخصص root volume بعدد giga معينه .. ف انت لو عملت delete لل EC2 instance بتاعتك ال root volume default behavior هيتسمح باللي فيه ودة ال default behavior

- علشان اغير ال default behavior ودميتمسحش ال root volume لو جيت اعمل delete لل instance بشيل ال **check** من هنا

Delete On Termination

- By default Root EBS is deleted
- By default Other Attached EBS is not deleted

Volume Type	Device	Snapshot	Size (GiB)	Volume Type	IOPS	Throughput (MB/s)	Delete on Termination
Root	/dev/xvda	snap-09f18f682fd23a1b1	8	General Purpose SSD (gp2)	100 / 3000	N/A	☒

- ف ال screen اللي فى الصفحة اللي فاتت هتلاحظ انك ممكن تـ add new volume .. يقصد EBS volume يعنى او انت تكريتها بعدين مش لازم وانت بت configure ال instance وبعد دة ت attach ال EBS volume دي لل instance عادى .. ف بكدا بقت ال instance بتاعتك عندها 2 volume .. الاول root volume والتانية ال EBS Volume اللي انت كريتته .. وزى مقولنا قبل كدا لو مسحت ال instance دي ال EBS Volume مش بتتمسح

- او وانت بتكريت ال EBS Volume تخش علي **Advanced** وبعدين تخلي **delete on termination = no** علشان ال EBS متتمسحش وكذلك لل root volume



I'm Here

تقدر تخلي بر دو ليه **delete on termination = no** علشان لما ال instance
تتمسح ال root volume دة يفضل موجود



▼ Configure storage [Info](#)

[Advanced](#)

1x GiB ▼ Root volume (Not encrypted)

1x GiB ▼ **EBS volume** (Not encrypted)

[Remove](#)

[i](#) Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage [X](#)

[Add new volume](#)

▼ Storage (volumes) [Info](#)

[Simple](#)

EBS Volumes

[Hide details](#)

▶ Volume 1 (AMI Root) (8 GiB, EBS, General purpose SSD (gp3))

▼ Volume 2 (Custom)

[Remove](#)

Storage type [Info](#)

EBS

Device name - *required* [Info](#)

/dev/sdb ▼

Snapshot [Info](#)

Select ▼

Size (GiB) [Info](#)

5

Volume type [Info](#)

gp2 ▼

IOPS [Info](#)

100 / 3000

Delete on termination [Info](#)

No ▼

Encrypted [Info](#)

Not encrypted ▼

KMS key [Info](#)

Select ▼

KMS keys are only applicable when encryption is set on this volume.



I'm Here

002 EBS Hands On

لو حددت ال instance من تحت هتعرف اية ال volumes اللى مرتبطة بيها سواء EBS root او EBS اضافيه

Instances (1/1) Info

Search

Name	Instance ID	Instance state	Instance type	Status check	Alarm st
My First Instance	i-034466697feb9ef80	Running	t2.micro	2/2 checks passed	No alarm

Instance: i-034466697feb9ef80 (My First Instance)

Details Security Networking **Storage** Status checks Monitoring Tags

Root device details

Root device name: /dev/xvda

Root device type: EBS

EBS optimization: disabled

Block devices

Filter block devices

Block devices

Filter block devices

Volume ID	Device name	Volume size (GiB)	Attachment status	Attachment time	Encryp
vol-0ec2ac1777f918db3	/dev/xvda	8	Attached	Mon May 16 2022 17:11:03...	No

لو ضغطت عال volume ده هيدخلك عالصفحة دي .. هتلاقي معلومات عنه ومرتبطة بانهي instance وهكذا ومن هنا تقدر ت create volume .. او من هنا

Elastic Block Store (EBS) >> Volumes



I'm Here

Volumes (1/1) Refresh Actions Create volume

Filter volumes

Volume ID: vol-0ec2ac1777f918db3 Clear filters

☒	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot
☒	-	vol-0ec2ac1777f918db3	gp2	8 GiB	100	-	snap-00022cf...

Volume ID: vol-0ec2ac1777f918db3

Details | Status checks | Monitoring | Tags

Details

Volume ID vol-0ec2ac1777f918db3	Size 8 GiB	Type gp2	Volume status ✓ Okay
Volume state ✓ In-use	IOPS 100	Throughput -	Encryption Not encrypted

بعد ما بتكرت ال EBS volume بتاخذ وقت على ما تكون Available ف لازم دي
تكون Available علشان تقدر تعملها attach على instance



I'm Here

Volumes (1/2)

Filter volumes

Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot	Created
-	vol-0ec2ac1777f918db3	gp2	8 GiB	100	-	snap-00022cf...	2022/05/16 17:11 GMT+1
-	vol-07547a181f5ae63fc	gp2	2 GiB	100	-	-	2022/05/16 18:59 GMT+1

Volume ID: vol-07547a181f5ae63fc

Details Status checks Monitoring Tags

Details

Volume ID: vol-07547a181f5ae63fc

Size: 2 GiB

Type: gp2

Volume status: ✔ Okay

Volume state: 🔄 Creating

IOPS: 100

Throughput: -

Encryption: Not encrypted

Details

Volume ID: vol-07547a181f5ae63fc

Size: 2 GiB

Type: gp2

Volume status: ✔ Okay

Volume state: ✔ Available

IOPS: 100

Throughput: -

Encryption: Not encrypted

بعد ما عملت attach لو رجعت لل instance ودخلت على ال storage هتلاقي ان ال EBS Volume ده Attached بالشكل ده

Block devices

Filter block devices

Volume ID	Device name	Volume size (GiB)	Attachment status	Attachment time	Encryption
vol-0ec2ac1777f918db3	/dev/xvda	8	✔ Attached	Mon May 16 2022 17:11:03...	No
vol-07547a181f5ae63fc	/dev/sdf	2	✔ Attached	Mon May 16 2022 18:59:56...	No

لو عملت terminate لل instance هتلاقي ان ال root volume اتمسح اما ال EBS volume by default لسة موجود .. ده بناءا على انك معملتتش check على



I'm Here

ال delete on termination وانت بتكريته زي ما وضعنا قبل كذا في ال CCP1

ودة بيكون متوضح على اليمين من نفس الاسكرين اللي فاتت

Instance: i-034466697feb9ef80 (My First Instance)

▼ Block devices

Filter block devices

Attachment status	Attachment time	Encrypted	KMS key ID	Delete on termination
Attached	Mon May 16 2022 17:11:03...	No	-	Yes
Attached	Mon May 16 2022 18:59:56...	No	-	No

003 EBS Snapshots



- وليكن انت محتاج تاخذ backup من AZ1 ل AZ2 .. زي مقولنا مينفعش ال EBS1

تعملها attach ل instance ف AZ2 تانية .. وبالتالي الحل هنا انك بتاخذ

snapshot لل EBS1 .. ووانت بتكريت volume لل instance فال AZ2 بتروح

عامل restore لل snapshot دي... ف كذا كل ال data اللي فال snapshot

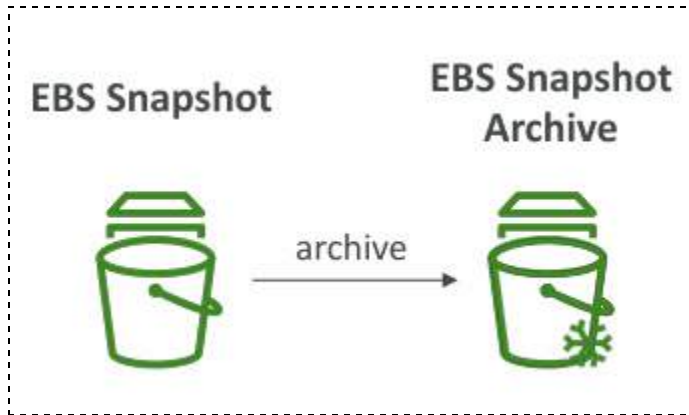
هتروح عال EBS2 volume.



I'm Here

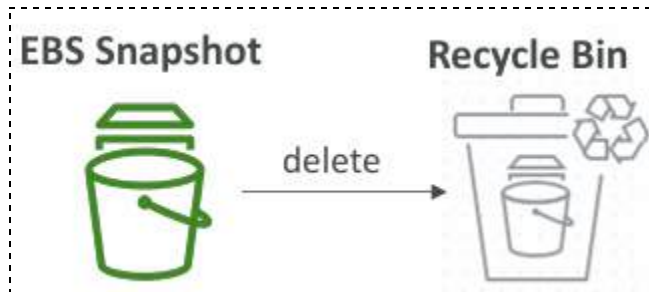
- **EBS Snapshots Features**

- **EBS Snapshot Archive**



- يمكن ت move ال snapshot دي ل "Archive tier" ودة بيكون ارخص بنسبة 75% ولكن بتستني حوالي 24:72 ساعة لحد ما ترجع ال snapshot دي.

- **Recycle Bin for EBS Snapshots**



- بت setup rules علشان لو مسحت snapshot بالغلط هتلاقىها فى ال recycle bin وبتفضل جوا لمدة سنة

- **Fast Snapshot Restore (FSR)**

- لو هتكرت من ال snapshot instance وعاوزها فى اسرع وقت ممكن ف دة انسب نوع ليك.



004 EBS Snapshots - Hands On

علشان تكریت snapshot

Elastic Block Store (EBS) >> Volumes

Volumes (1/1)

Filter volumes

Name	Volume ID	Type	Size
-	vol-066e0164889e6ed9f	gp2	2 GiB

Actions

- Modify volume
- Create snapshot
- Create snapshot lifecycle policy

Create volume

Snapshot

هتلاقبها هنا ..

Elastic Block Store (EBS) >> Snapshots

Snapshots (1)

Owned by me

Filter snapshots by attributes and tags

Name	Snapshot ID	Size	Description	Storage...	Snapshot status
-	snap-0c379c9b4dcbfdebf	2 GiB	DemoSnapshot	Standard	Completed

Recycle Bin

Actions

Create snapshot

علشان ت copy ال snapshot دي ل region تانية

وعلشان ت create volume منها.. وال volume دة ممكن تخليه فى اي AZ

Snapshots (1/1)

Owned by me

Filter snapshots by attributes and tags

Name	Snapshot ID	Size	Description
-	snap-0c379c9b4dcbfdebf	2 GiB	DemoSnapshot

Recycle Bin

Actions

- Create volume from snapshot
- Create image from snapshot
- Copy snapshot
- Modify permissions

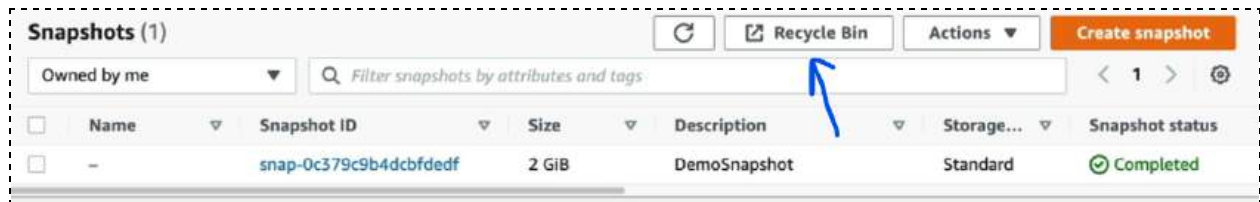
Create snapshot



I'm Here

جوا ال snapshots هتلاقي ال recycle bin اللي اتكلمنا عنها منها بتحدد عدد الايام

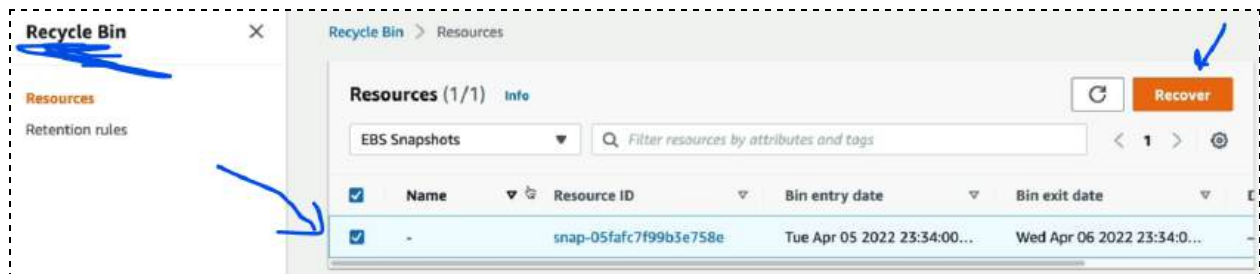
اللى تقدر فيها تسترجع ال snapshots المحسوخة



Snapshots (1)

Owned by me

☐	Name	Snapshot ID	Size	Description	Storage...	Snapshot status
☐	-	snap-0c379c9b4dcbfdedf	2 GiB	DemoSnapshot	Standard	Completed



Recycle Bin

Resources

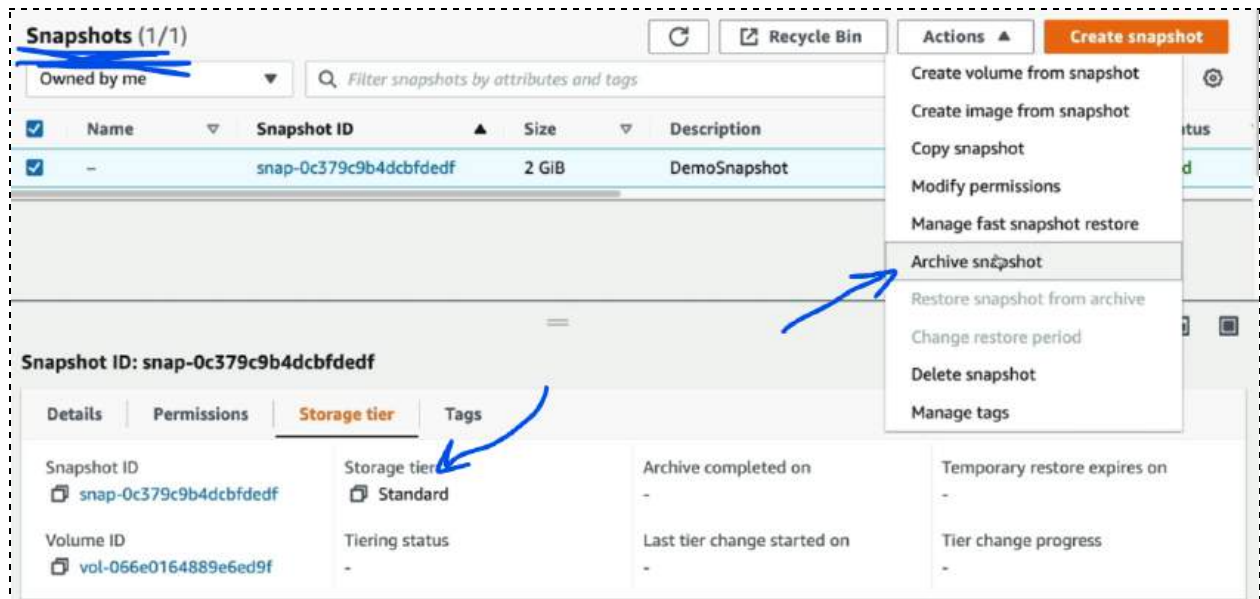
Retention rules

Resources (1/1)

EBS Snapshots

☒	Name	Resource ID	Bin entry date	Bin exit date
☒	-	snap-05fafc7f99b3e758e	Tue Apr 05 2022 23:34:00...	Wed Apr 06 2022 23:34:00...

من هنا تقدر ت archive ال snapshot زي ما شرحنا



Snapshots (1/1)

Owned by me

☒	Name	Snapshot ID	Size	Description
☒	-	snap-0c379c9b4dcbfdedf	2 GiB	DemoSnapshot

Snapshot ID: snap-0c379c9b4dcbfdedf

Details | Permissions | **Storage tier** | Tags

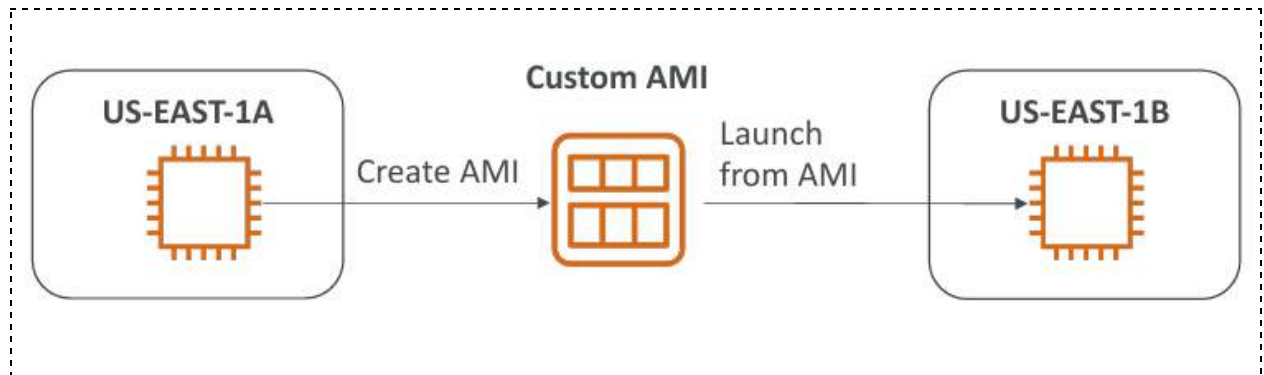
Snapshot ID snap-0c379c9b4dcbfdedf	Storage tier Standard	Archive completed on -	Temporary restore expires on -
Volume ID vol-066e0164889e6ed9f	Tiering status -	Last tier change started on -	Tier change progress -

- Create volume from snapshot
- Create image from snapshot
- Copy snapshot
- Modify permissions
- Manage fast snapshot restore
- Archive snapshot**
- Restore snapshot from archive
- Change restore period
- Delete snapshot
- Manage tags



I'm Here

➡ 005 AMI Overview (Amazon Machine Image)



- ودي عبارہ عن Pre Configured Instances يا عني عبارہ عن Instance Template جاهزة معمولوها كل ال configuration اللى كنا بنعملها ف اول محاضرة وبعملها Save as AMI image وفيما بعد لو عاوز اعمل نفس ال task بختار ال image دي علطول مش هفضل كل شوية اعمل نفس ال Configuration

● أنواع ال AMI :

● Public AMI:

- ناس عملتها ومشيراها Public وتقدر انت تستخدمها علطول

● Private AMI (Yours):

- بتاعتك انت اللى عاملها

● AWS Market Place AMI:

- ممكن يكون حد عامل Image معينه ويبييعها مثلاً.



006 AMI Hands On

The screenshot shows the AWS Management Console 'Instances' page. A context menu is open for the instance 'i-0a404d2f2e247ab5f', which is in a 'Running' state. The 'Create image' option is highlighted in the menu. The left sidebar shows the 'Instance summary' for the selected instance, including its ID and IPv6 address.

بمجرد ما بتعمل ال instance تقدر تعمل image بالشكل دة وهتلاقي ال image هنا
تقدر ت launch instance منها وال instance الجديدة هتلاقيها بالظبط زي القديمة
بكل ال configuration بتاعتها

Images >> AMIs

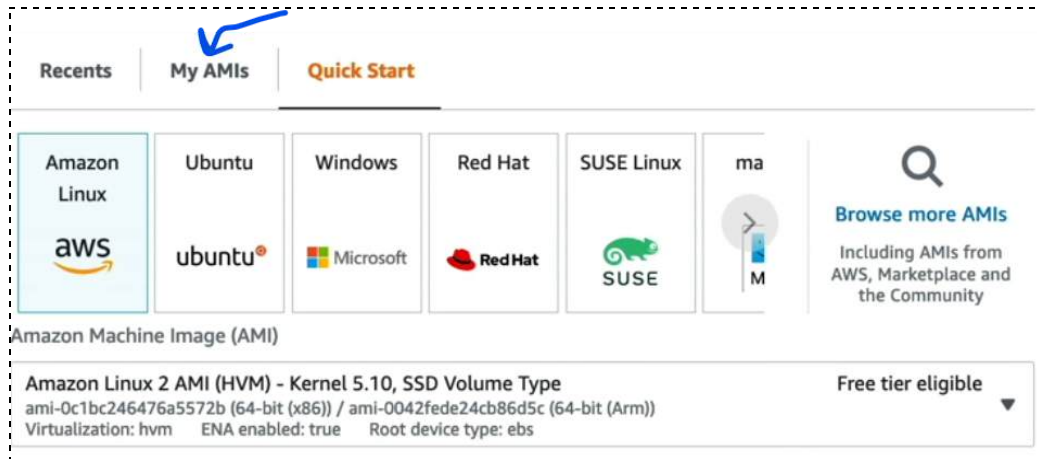
The screenshot shows the AWS Management Console 'Amazon Machine Images (AMIs)' page. A table lists the AMIs, including 'ami-019206cd8b19c8ffa' with the name 'DemoImage'.

Name	AMI ID	AMI name	Source	Owned by me
-	ami-019206cd8b19c8ffa	DemoImage	783768293452/DemoImage	783768293452

او تقدر من ت launch ال instance دي بال AMI دي وانت بتعمل ال instance
بالشكل العادي وبدل ما تختار AWS linux هتختار My AMI وتختار ال image



I'm Here



007 EC2 Instance Store

- احنا قولنا ان ال EBS volume بيبقي عبارة عن Virtual hard drive بتعمله attach لل instance بتاعتك.. هي كويسة ولكن ال performance بتاعها بيكون limited شوية .. ف لو محتاج حاجة ب Performance اعلي بيبقي ال EC2 Instance Store

● Cons:

- فيها عيب وهو انك لو عملت terminate لل instance ال instance store دي هتطير.. على عكس ال EBS Volume لما كنت بتمسح ال instance كانت بتفضل موجودة وكنت تقدر ت attach it لل instance تانية إلا بقا لو كنت عاملها delete on termination

● Use Case:

- Short term storage : Buffer, Cache, Temporary Data

يبقي لو محتاج Short long storage دي تمام .. لو محتاج Long term storage ال

EBS افضل.



➔008 EBS Volume Types

● GP2/GP3 (SSD) - General Purpose

- دة ssd volume تقدر تستخدمه لل general-purpose وتكلفته قليلة فيه balance بين ال price , performance
- ودة اللى غالبا هنستخدمه واحنا بنجرب لانه بيستخدم فى Virtual boot volumes, desktops, Development and test environments

● io1/io2 (SSD)

- دة ssd volume تقدر تستخدمه لو محتاج high-performance و low latency و high throughput
- high throughput : يعنى هينقل داتا بشكل اسرع

● st1 (HHD)

- دة HHD volume تقدر تستخدمه لو عاوز high throughput و low cost ومش فارقلك ال latency لان هنا ال latency بتكون عالية شوية

● sc1 (HHD)

- زيو زي اللى قبله دة ولكن هنا لو انت مش بت access الداتا بتاعتك كل شوية يعنى كل فترة بت access الداتا يبقى دة انسب نوع ليك واكيد سعرو اقل من اللى فوقه دة.

- بيتصنيف ال EBS Volumes علي حسب ال IOPS , throughput , size
- **IOPS** : Operation Per Second



- **General Purpose SSD :**

- 1 GiB - 16 TiB
- System boot volumes, Virtual desktops, Development and test environments

- **Gp3:**

- Baseline of 3,000 IOPS and throughput of 125 MiB/s
 - Can increase IOPS up to 16,000 and throughput up to 1000 MiB/s independently

- **Gp2:**

- Small gp2 volumes can burst IOPS to 3,000
 - Size of the volume and IOPS are linked, max IOPS is 16,000
 - 3 IOPS per GB, means at 5,334 GB we are at the max IOPS

اقدر اتحكم فى ال IOPS , throughput منفصلين فى Gp3 علي عكس ال Gp2 💡

- **Provisioned IOPS (PIOPS) SSD Use Cases**

- Critical business applications with sustained IOPS performance
- Or applications that need more than 16,000 IOPS
- Great for databases workloads (sensitive to storage perf and consistency)

- **io1 (4 GiB - 16 TiB)**

- Max PIOPS: 64,000 for Nitro EC2 instances & 32,000 for other
 - Can increase PIOPS independently from storage size



- **io2 Block Express (4 GiB – 64 TiB):**

- Sub-millisecond latency
- Max PIOPS: 256,000 with an IOPS:GiB ratio of 1,000:1

- **Supports EBS Multi-attach**

● **Hard Disk Drives (HDD)**

- Cannot be a boot volume
- 125 GiB to 16 TiB

- **Throughput Optimized HDD (st1)**

- Big Data, Data Warehouses, Log Processing
- Max throughput 500 MiB/s – max IOPS 500

- **Cold HDD (sc1):**

- For data that is infrequently accessed
- Scenarios where lowest cost is important
- Max throughput 250 MiB/s – max IOPS 250

يبقي gp2,gp3 لو محتاج اقل من 16000 IOPS 💡

وio1 لحد 64000 واكثر من كذا لحد 250000 يبغي io2

مش محتاج تعرف كل ال details دي هتلاقي كل التفاصيل دي [هنا](#) 💡



● Solid state drive (SSD) volumes

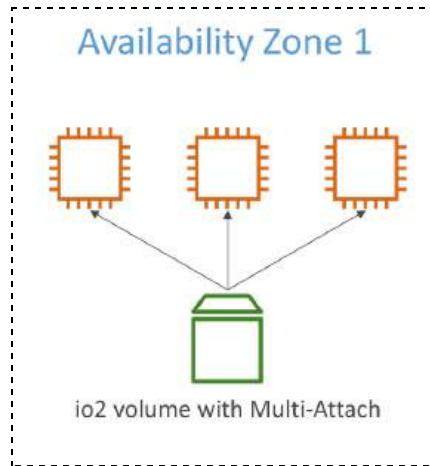
	General Purpose SSD volumes		Provisioned IOPS SSD volumes	
Volume type	gp3	gp2	io2 Block Express ³	io1
Durability	99.8% - 99.9% durability (0.1% - 0.2% annual failure rate)		99.999% durability (0.001% annual failure rate)	99.8% - 99.9% durability (0.1% - 0.2% annual failure rate)
Use cases	- Transactional workloads - Virtual desktops - Medium-sized, single-instance databases - Low-latency interactive applications - Boot volumes - Development and test environments		Workloads that require: - Sub-millisecond latency - Sustained IOPS performance - More than 64,000 IOPS or 1,000 MiB/s of throughput	- Workloads that require sustained IOPS performance or more than 16,000 IOPS - I/O-intensive database workloads
Volume size	1 GiB - 16 TiB		4 GiB - 64 TiB ⁴	4 GiB - 16 TiB
Max IOPS per volume	16,000 (64 KiB I/O)	16,000 (16 KiB I/O)	256,000 (16 KiB I/O) ⁵	64,000 (16 KiB I/O)
Max throughput per volume	1,000 MiB/s	250 MiB/s ¹	4,000 MiB/s	1,000 MiB/s ²
Amazon EBS Multi-attach	Not supported		Supported	
NVMe reservations	Not supported		Supported	Not supported
Boot volume	Supported			

● Hard disk drive (HDD) volumes

	Throughput Optimized HDD volumes	Cold HDD volumes
Volume type	st1	sc1
Durability	99.8% - 99.9% durability (0.1% - 0.2% annual failure rate)	
Use cases	- Big data - Data warehouses - Log processing	- Throughput-oriented storage for data that is infrequently accessed - Scenarios where the lowest storage cost is important
Volume size	125 GiB - 16 TiB	
Max IOPS per volume (1 MiB I/O)	500	250
Max throughput per volume	500 MiB/s	250 MiB/s
Amazon EBS Multi-attach	Not supported	
Boot volume	Not supported	



➡ 009 EBS Multi-Attach



EBS Multi-Attach is a feature of Amazon EBS that **allows you to attach a single Provisioned IOPS SSD (io1 or io2) volume to multiple EC2 instances in the same Availability Zone**. Here's an overview of EBS Multi-Attach:

- **Key points:**

متاحة فقط للنوع io1/io2 💡

- تقدر تربط ال EBS Volume دة بعدد 16 Nitro-based EC2 instances فى نفس الوقت
- لازم ال instances تكون فى نفس ال AZ
- تستخدم فى clustered Linux applications ودي applications بتشتغل across multiple servers
- Requires a cluster-aware file system (not standard file systems like ext4 or XFS).

- **Benefits:**

- High availability : لو instance حصلها fail الباقي بيشتغل
- بتمكنك ت build clustered applications



➔010 EBS Encryption

- لو عملت EBS Volume encrypted هتحصل على الآتي :

- كل ال data اللي في ال EBS Volume دي بتكون encrypted
- ال data اللي بتروح وتيجي بين ال EBS Volume وال instances بتكون encrypted
- كل ال snapshots وال instances اللي هتتعمل من ال snapshots دي بتكون encrypted
- ال encryption بييسبب latency طفيف جدا مش هتחס بيه

- EBS Encryption leverages keys from KMS (AES-256)

- **KMS** stands for AWS Key Management Service.
- **AES** stands for Advanced Encryption Standard, a widely used encryption algorithm, 256 refers to the key size in bits.
- يبقي ال encryption بيتم من خلال AWS KMS والنوع ال default اللي بيتم استخدامه هو (AES-256)

- Encryption: encrypt an unencrypted EBS volume

- Create an EBS snapshot of the volume
- Encrypt the EBS snapshot (using copy)
- Create new EBS volume from the snapshot (the volume will also be encrypted)



- Now you can attach the encrypted volume to the original instance

- يعني كان عندنا unencrypted EBS volume وكنا علوزين نخليه encrypted

Snapshots (1/1)

Owned by me

☒	Name	Snapshot ID	Size	Description
☒	-	snap-065047d562375dad5	1 GiB	-

Actions

- Create volume from snapshot
- Create image from snapshot
- Copy snapshot
- Modify permissions

New snapshot settings

Description
A description for the snapshot copy.
[Copied snap-065047d562375dad5 from eu-west-1]
255 characters maximum.

Destination Region
The Region in which to create the snapshot copy.
eu-west-1

Encryption [Info](#)
Use Amazon EBS encryption as an encryption solution for your EBS resources.
☒ **Encrypt this snapshot**

KMS key [Info](#)
(default) aws/ebs

KMS key description
☒ Default key that protects my EBS volumes when no other key is defined

Snapshots (1/2)

Owned by me

☒	Name	Snapshot ID	Size	Description
☒	-	snap-0278ddf9af9548627	1 GiB	[Copied snap-065047d562375dad5 from eu-west-1]

Actions

- Create volume from snapshot
- Create image from snapshot
- Copy snapshot

- هتلاقيه already encrypted بما انه معمول من encrypted snapshot



Encryption [Info](#)

Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances.

☒ Encrypt this volume

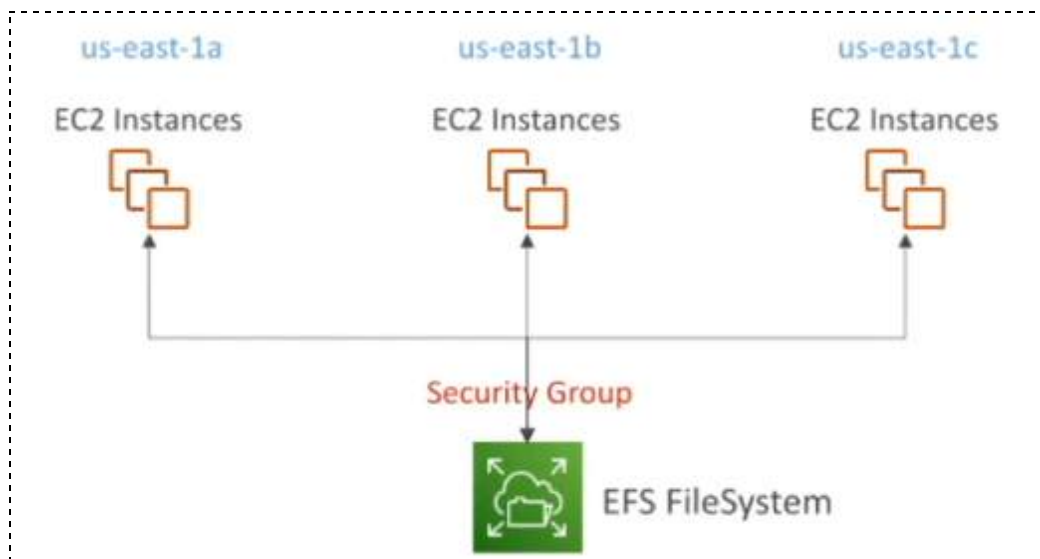
- أما لو بتعمل EBS Volume من جديد ف ممكن تخليه encrypted direct

Encryption [Info](#)

Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances.

☐ Encrypt this volume

011 Amazon EFS (Elastic File System)



- ال EBS كان storage أما هنا ال EFS عبارة عن File System

- بتدفع عال storage اللي بتستخدمها فقط



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● Key Points

- Use Cases: Content management, web serving, data sharing, Wordpress

- Uses NFSv4.1 (Network file system) protocol

- ودة استخدامه انو بيمكنك من انك تـ access files over the internet سواء هتـ read or write عليها

- Uses security group to control access to EFS

- اللى داخل واللى خارج من ال file system دة بنعرف نتحكم فيه

- Compatible with Linux based AMI (not Windows)

- شغالة مع Linux فقط

- Encryption at rest using KMS

- الداتا اللى عال file system دة بتكون encrypted باستخدام KMS اللى هيا AWS Key Management Service

- POSIX file system (~Linux) that has a standard file API

- بتشغل اكنها file system عادي زي بتاع linux دة معني انها بتتبع ال POSIX Standards اللى هيا بتتصرف بالطريقة العادية زي ما كنت فى لينكس بنفس طريقة ال commands وال file handling ياغني تقدر تستخدم معاها ls , mv , cp وهكذا



- File system scales automatically, pay-per-use, no capacity planning!

- "ال EFS بت grow , shrink .. يعني هت write data هتلاقيها بت grow هتمسح data هتلاقيها بت shrink .. يعني "pay for what you use"

● EFS – Performance & Storage Classes

- وانت بت Create ال EFS هتلاقي عندك:

● Performance Mode:

- **General Purpose** (default): latency-sensitive use cases (web server, CMS, etc...)
- ودة بيكون حاجة balanced كدا من جميع النواحي
- **Max I/O**: higher latency, throughput, highly parallel (big data, media processing)
- دة لو محتاج high throughput بيكون اغلي شوية أكيد .. هتفهم معنى ال throughput دلوقتي

● Throughput Mode:

- **Bursting** : 1 TB = 50MiB/s + burst of up to 100MiB/s
- عبارة عن Traffic عالي فجأة عال EFS ف هل عاوز ال EFS يهendl ال Bursting اللى حصل دة ولالا
- Bursting انفجار
- **Provisioned** : set your throughput regardless of storage size, ex: 1 GiB/s for 1 TB storage
- بتهيئها .. يعني بتحدد انت عاوز ال throughput بتاعك يكون قد اية وليكن مثلا 1 GB/sec



- **Elastic** : automatically scales throughput up or down based on your workloads

ودة بـ Scale Automatic مع ال work load اللي جاي وبيعمل throughput up and down ودة بيكون مفضل فى ال unpredictable workloads

يبقى مثلا لو عندك بيتبعت وليكن GB 1 عال file system ودة وفجأة بقو GB 20 فدة  bursting (انفجار) .. اما لو انت عارف ال throughput rate ف خليك provision .. اما لو عاوز الموضوع dynamic ف خليك ف elastic وكلو بحسابه 😊.

● Storage Tier:

- بن manage بيها ال lifecycle بتاعه ال file system ودة .. من خلالها تقدر تحدد هنعمل اية فى ال file system ودة بعد فتره معينه .. فى عندك فيها 3 option

● Frequently Accessed:

- مُصمم لل active data workloads ال data اللي بيحصلها access باستمرار

● Infrequently Accessed(EFS IA): cost to retrieve files, lower price to store.

- مناسبة لو ال data بتاعتنا مش بيحصلها access كثير .. بتكون هنا التكلفة اقل ولكن فيه تكلفه على استرجاع الملفات

● Archive: rarely accessed data (few times each year), 50% cheaper

- "لو ال data بيحصلها access اقل ك عدد مرات من ال 2 options اللي فوق وطبعاً ارخص منهم"



Lifecycle management

Automatically save money as access patterns change by moving files into the Infrequent Access (IA) or Archive storage class. [Learn more](#)

Transition into Infrequent Access (IA)

Transition files to IA based on the time since they were last accessed in Standard storage.

30 day(s) since last access

Transition into Archive

Transition files to Archive based on the time since they were last accessed in Standard storage.

90 day(s) since last access

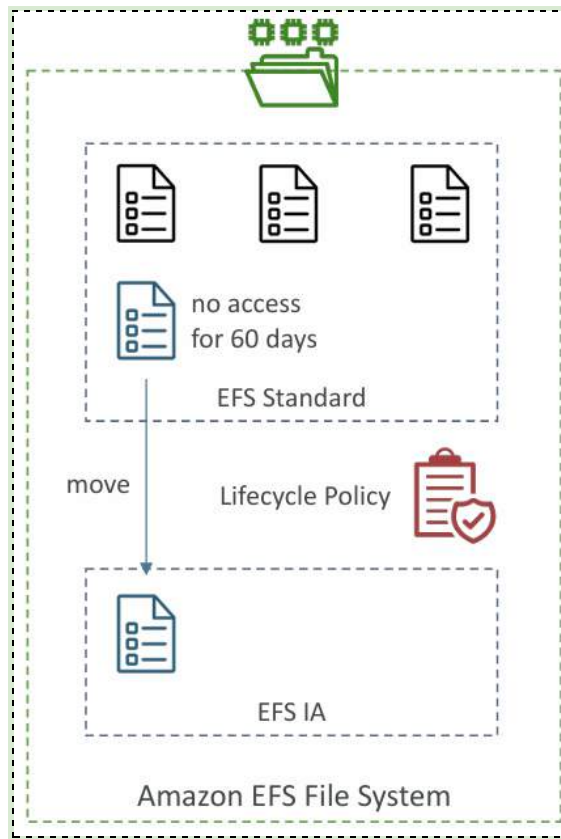
Transition into Standard

Transition files back to Standard storage based on when they are first accessed in IA or Archive storage.

None

علشان ت move ال data بتاعتك between ال storage tiers دي بيتم دة من خلال

انك بت set lifecycle polices تحققك دة.



- **Availability & Durability:**

- **Regional:** Multi-AZ, great for prod

- متاح ف اكثر من AZ .. يعني لو واحدة وقعت ال EFS file system دة يفضل شغال "اغلي .. مناسب لل production environment"



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- **One Zone:** One AZ, great for dev, backup enabled by default, compatible with IA (EFS One Zone-IA)

- يشتغل ف zone واحدة فقط .. ودة مناسب اكثر لل development or backup or "testing"

File system type

Choose to either store data across multiple Availability Zones or within a single Availability Zone. [Learn more](#)

☒ **Regional**
Offers the highest levels of availability and durability by storing file system data across multiple Availability Zones within an AWS Region.

☐ **One Zone**
Provides continuous availability to data within a single Availability Zone within an AWS Region.

012 Amazon EFS - Hands On

"Search على EFS وبتحدد ال setting اللى عاوزها بناء على اللى اتشرح فوق

وتختار security group ولو معملتش security group مخصص لل EFS اعملو

وارجع اختاره وبعدها تكرت ال instances وترتبط ال instances بال EFS."

Network settings [Info](#)

VPC - required [Info](#)

vpc-0799daa55c9fb6493 (default) [Info](#)

Subnet [Info](#)

No preference [Create new subnet](#)

subnet-0b033c07c2cfc2d1b
VPC: vpc-0799daa55c9fb6493 Owner: 510512620071 Availability Zone: us-east-1d
IP addresses available: 4091 CIDR: 172.31.32.0/20

subnet-0a3d455d7730cdc92
VPC: vpc-0799daa55c9fb6493 Owner: 510512620071 Availability Zone: us-east-1a
IP addresses available: 4091 CIDR: 172.31.0.0/20

subnet-017c65ab00faaf2af
VPC: vpc-0799daa55c9fb6493 Owner: 510512620071 Availability Zone: us-east-1c
IP addresses available: 4091 CIDR: 172.31.16.0/20

subnet-0d0cb74f90357682b
VPC: vpc-0799daa55c9fb6493 Owner: 510512620071 Availability Zone: us-east-1b
IP addresses available: 4091 CIDR: 172.31.80.0/20



"هكريت مثلا instance A مجرد اسم اول واحدة يعاني .. من ال network

setting هختارلها subnet طب انهى subnet ؟ هختار ال subnet اللى هيا

"us-east-1a وهكذا instance B هختارلها us-east-1a

"دي كدا وانت بتكرت ال EFS بتختار ال security group وال subnet وكدا"

Mount targets

A mount target provides an NFSv4 endpoint at which you can mount an Amazon EFS file system. We recommend creating one mount target per Availability Zone. [Learn more](#)

Availability zone	Subnet ID	IP address	Security groups	
us-east-1a	subnet-0a3d455d7730cdc92	Automatic	Choose security groups sg-0cb83134332e4b1c2 default	Remove
us-east-1b	subnet-0d0cb74f90357682b	Automatic	Choose security groups sg-0cb83134332e4b1c2 default	Remove
us-east-1c	subnet-017c65ab00faaf2af	Automatic	Choose security groups sg-0cb83134332e4b1c2 default	Remove
us-east-1d	subnet-0b033c07c2cfc2d1b	Automatic	Choose security groups sg-0cb83134332e4b1c2 default	Remove
us-east-1f	subnet-09a2ba7c68db5b25e	Automatic	Choose security groups sg-0cb83134332e4b1c2 default	Remove

بتختار نفس ال file system لل instances لان ممكن يكون عندك اكثر من file

system وبتختار نفس ال mount point



File systems

Hide details

EFS

FSx

Shared file system 1

Remove

File system Info

fs-03ad3ad91588ea5e7

Availability: Regional

Mount point Info

/mnt/efs/fs1

Add shared file system

Create new shared file system

4 remaining (Up to 5 file systems maximum)

Automatically create and attach security groups

To enable access to the file system, the required security groups will be automatically created and attached to this instance and the selected file system. To manually manage the security groups, clear the checkbox. [Learn more.](#)

Automatically mount shared file system by attaching required user data script

Automatically mount your file system by updating your user data to install efs-utils. If you would like to manually mount your file system, clear the checkbox.

تلقائي بيتعمل security groups عال AZs اللى بتحتوي عال instances بتاعتك اللى مرتبطة ب file system واحد .. ياعني ال 3 instances اللى عملناهم دلوقتي كل واحدة كانت فى AZ مختلفة ف لو دخلت على ال EFS هتلاقي ان فيه بالاضافة لل security group اللى عملناه فيه by default security groups اللى اتعملت security groups دي هتلاقي انها فى ال inbound rules بتسمح

ب NFS protocol الى اتكلمنا عنه قبل كدا.

Availability zone	Mount target ID	Subnet ID	target state	IP address	Network interface ID	Security groups
eu-west-1a	fsmt-095ef9845742400c7	subnet-09b25e9baf4dae1d7	Available	172.31.24.238	eni-01d64b8447127d289	sg-0e9492811e288e9e0 (efs-sg-1), sg-01357c5128cce2851 (efs-sg-2), sg-0600b2c430ec51f45 (efs-demo)
eu-west-1b	fsmt-0887a085cec67f874	subnet-069b03c0b1a519b91	Available	172.31.45.40	eni-0a1559b3f69e200c1	sg-0e9492811e288e9e0 (efs-sg-1), sg-01357c5128cce2851 (efs-sg-2), sg-0600b2c430ec51f45 (efs-demo)

Security Groups (1/11) Info

Filter security groups

Name	Security group ID	Security group name	VPC ID	Description
-	sg-01357c5128cce2851	efs-sg-2	vpc-0a40b157dcd7c81f1	Created by t...

Details Inbound rules Outbound rules Tags

You can now check network connectivity with Reachability Analyzer

Run Reachability Analyzer

Inbound rules (1/1)

Filter security group rules

version	Type	Protocol	Port range	Source
	NFS	TCP	2049	sg-0521b4deede655e...

لو روحنا بقا دخلنا على اول instances وكريتنا file في ال mount point اللي هيا mnt/efs/fs1/ وسجلنا خروج ودخلنا عال instance التانية او التالته ودخلنا على نفس ال mount point هتلاقى نفس ال file هناك .. ف ال 3 مشتركين في نفس ال mount point على نفس ال file system رغم انهم في different AZs.

I'm Here

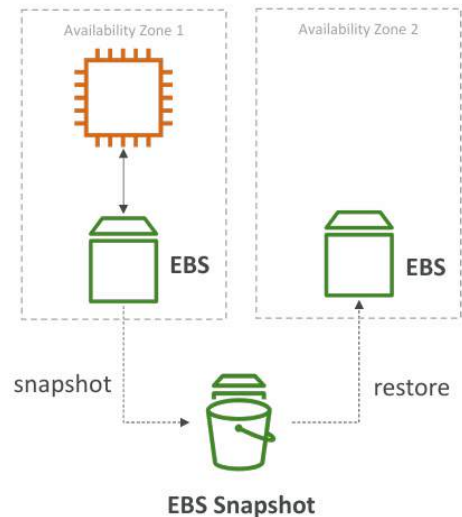


➡ 013 EBS vs EFS:

- إتكلّمنا ف كل دة قبل كدا دة بس ملخص سريع.

● EBS

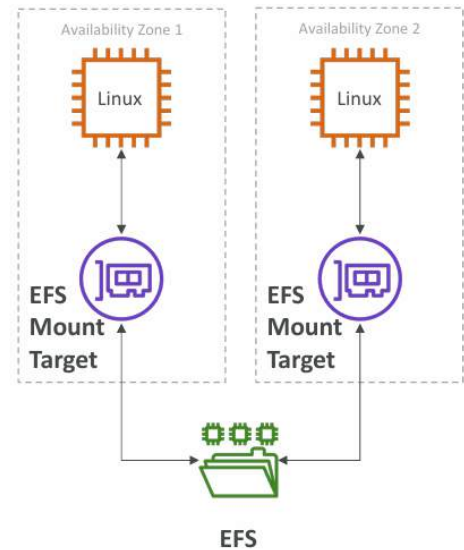
- EBS volumes...
 - one instance (except multi-attach io1/io2)
 - are locked at the Availability Zone (AZ) level
 - gp2: IO increases if the disk size increases
 - gp3 & io1: can increase IO independently
- To migrate an EBS volume across AZ
 - Take a snapshot
 - Restore the snapshot to another AZ
 - EBS backups use IO and you shouldn't run them while your application is handling a lot of traffic
- Root EBS Volumes of instances get terminated by default if the EC2 instance gets terminated. (you can disable that)



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● EFS

- Mounting 100s of instances across AZ
- EFS share website files (WordPress)
- Only for Linux Instances (POSIX)
- EFS has a higher price point than EBS
- Can leverage Storage Tiers for cost savings
- Remember: EFS vs EBS vs Instance Store



➡ 014 EBS & EFS - Section Cleanup

- هتمسح كل حاجة instances,volumes,snapshots ... كدا انت المفروض
بتمسح اول باول علشان التكلفة.



CH-08 - High Availability and Scalability ELB & ASG

➡001 High Availability and Scalability

- **Scalability:**

- قدرة ال system على انو يهندل ال increased load اللى جايله عن طريق adding resources وعندك منها نوعين :

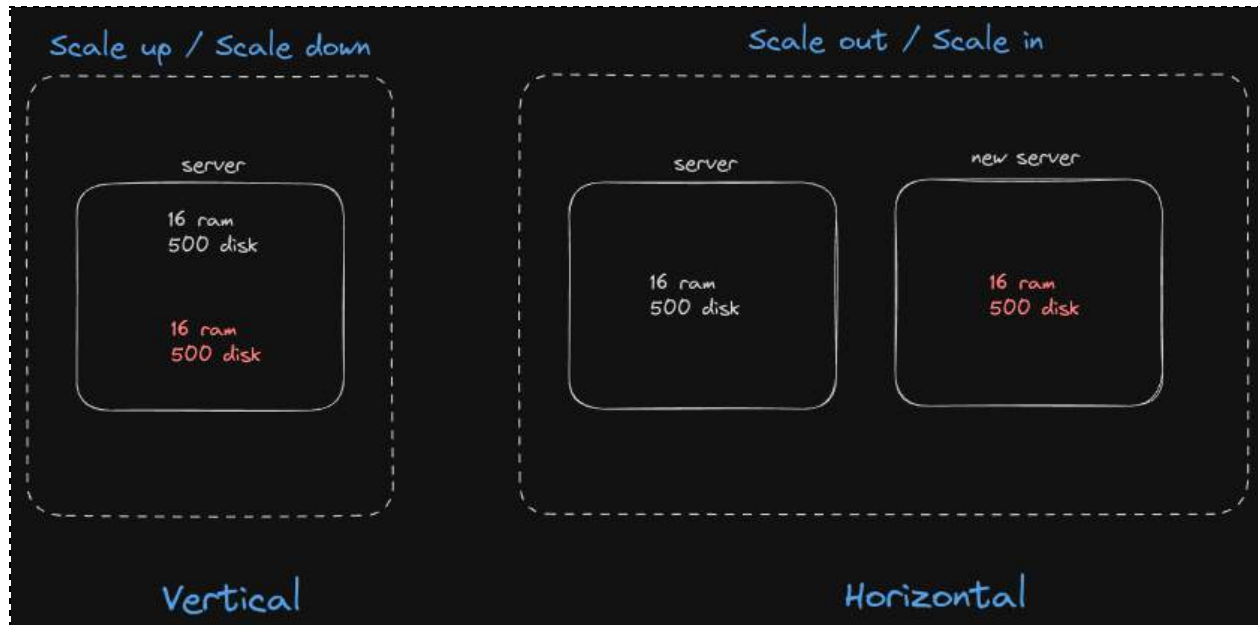


- **Vertical Scaling (Scaling Up/down):**

- اضافة power CPU, RAM على نفس ال Machine

- **Horizontal Scaling (Scaling Out/in)**

- اضافة Machines زيادة

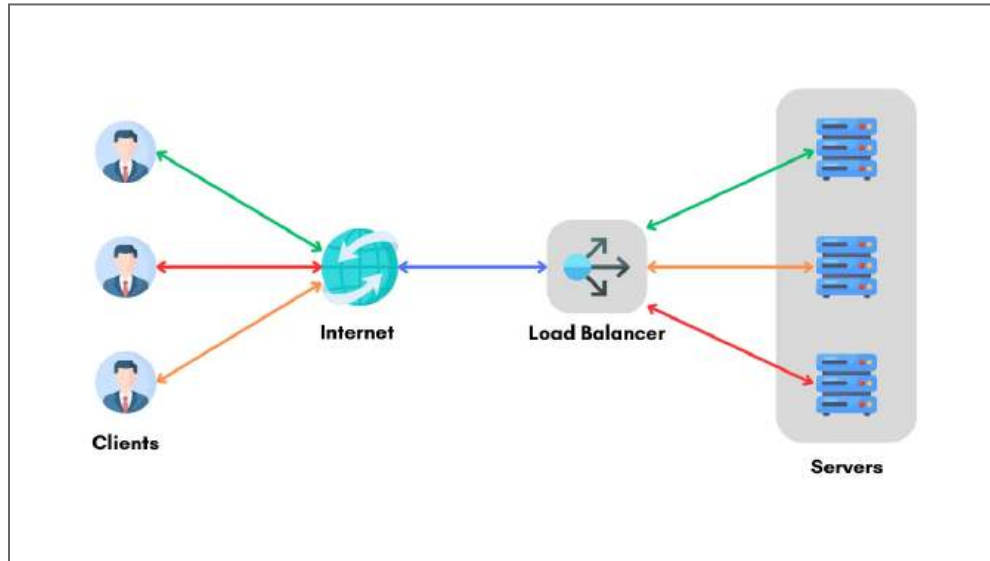


- **Availability:**

- قدرة ال system على انو يفضل up and running ودة بيتم عن طريق انو يكون ليه .redundancy, failover mechanisms, and reliable infrastructure.
موجود ف كذا AZs مثلا علشان لو instance وقعت التانيه تشتغل وهكذا.



002 Elastic Load Balancing (ELB) Overview



- Load Balancer : عبارہ عن server او مجموعة من ال servers وظيفتها انها توزع ال load او ال traffic اللى جايلها على مجموعة من ال back end servers بنسبيهم downstream instances .. موضوع ال load balancing دة AWS بتقدمه ك service اسمها Elastic Load Balancer



● Key Diff Between AWS LB & Nginx LB

Key Differences

1. Management and Maintenance:

- **NGINX:** Requires manual setup, configuration, and maintenance.
- **ELB:** Fully managed by AWS, reducing operational overhead.

2. Scalability:

- **NGINX:** Manual scaling and configuration required.
- **ELB:** Automatically scales to handle traffic fluctuations.

3. Availability and Redundancy:

- **NGINX:** Requires additional setup for high availability and failover.
- **ELB:** Built-in high availability and fault tolerance.

4. Customization and Control:

- **NGINX:** Greater control and customization options.
- **ELB:** Limited to AWS-provided configurations but sufficient for most use cases.

5. Cost:

- **NGINX:** Potentially lower costs but higher management overhead.
- **ELB:** Pay-as-you-go pricing, potentially higher costs but lower management overhead.

● Types of load balancer on AWS

- **Classic Load Balancer (v1 - old generation) – 2009 – CLB**
 - **Support Protocols:** HTTP, HTTPS, TCP, SSL
- **Application Load Balancer (v2 - new generation) – 2016 – ALB**
 - **Support Protocols:** HTTP, HTTPS, WebSocket
- **Network Load Balancer (v2 - new generation) – 2017 – NLB**
 - **Support Protocols:** TCP, TLS (secure TCP), UDP



- **Gateway Load Balancer – 2020 – GWLB**

- Operates at layer 3 (Network layer) – IP Protocol

- هنتكلم عن كل نوع بالتفصيل تحت.

Recommended انك تستخدم ال new generations بيبيقي فيهم features اكثر.

Some load balancers can be setup as **internal (private)** or **external (public)** ELBs

- **Load Balancer Security Groups**

- من ال user لل load balancer بتسمح في ال security group بال http/https

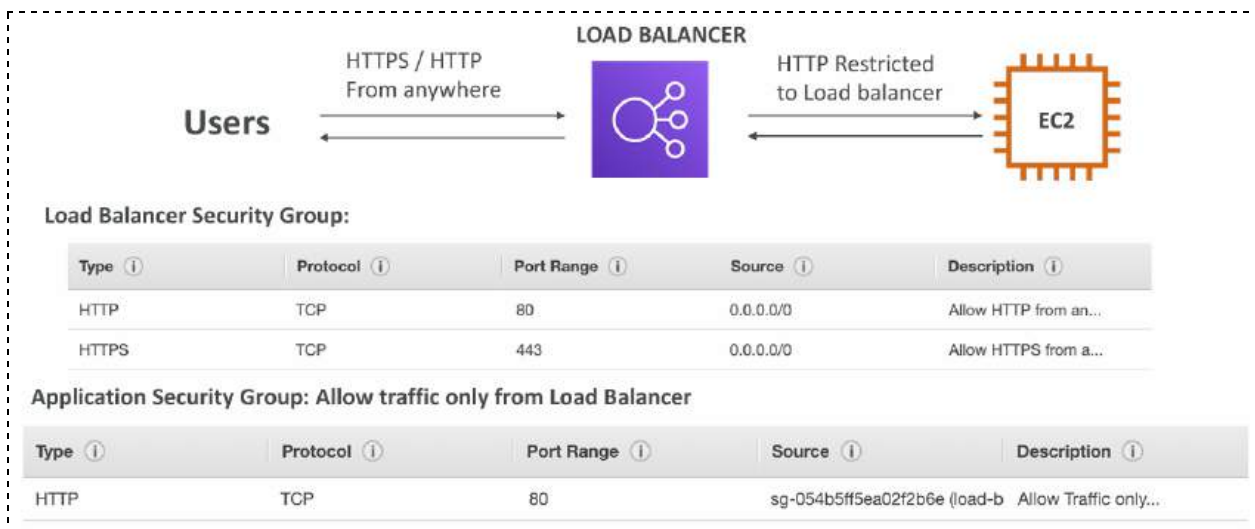
- requests اللي جايه من anywhere

- أما من ال load balancer لل instances ف بتسمح فقط بال requests اللي جايه

- عن طريق ال load balancer ودة بيحصل عن طريق انك بتحدد في ال security

- group بتاع ال instance ان ال source هو ال security group بتاع ال load

- balancer



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003 Note About the 1-Classic Load Balancer (CLB)

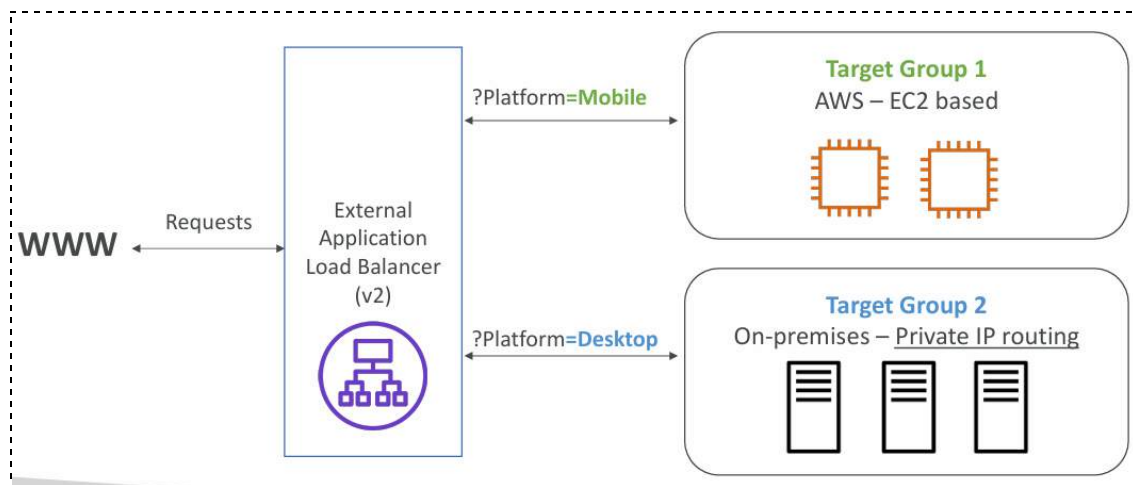
- The **Classic Load Balancer** is **deprecated** at AWS and will soon not be available in the AWS Console.
- The exam also has removed any references to it, so we will not cover it in depth in the course, nor the hands-on.

004 2-Application Load Balancer (ALB)

● Key Features

- بيشتغل ف layer 7 ويهندل ال traffic اللي جاي عليه بناء على rules specifications كتيره
- بيدعم ال SSL/TLS Termination

● Target Groups:



- عبارة عن Backend Component بتكون خلف ال load balancer ويمكن تكون اي حاجة من دول :



- **EC2 instances** (can be managed by an Auto Scaling Group) – HTTP
- **ECS tasks** (managed by ECS itself) – HTTP
- **Lambda functions** – HTTP request is translated into a JSON event
- **IP Addresses** – must be private IPs – must be private IPs

💡 ALB can route to **multiple target groups**

💡 Health checks are at the target **group level**

- **Listeners and Rules:**

- ال **Listener Rules** دي عبارة عن ان ازاي ال load balancer هيتعامل مع ال requests اللى جايه من ال user ويعملها routing على ال Target groups ودة بيعتمد علي بعض ال conditions :

- **Route based on URL PATH**

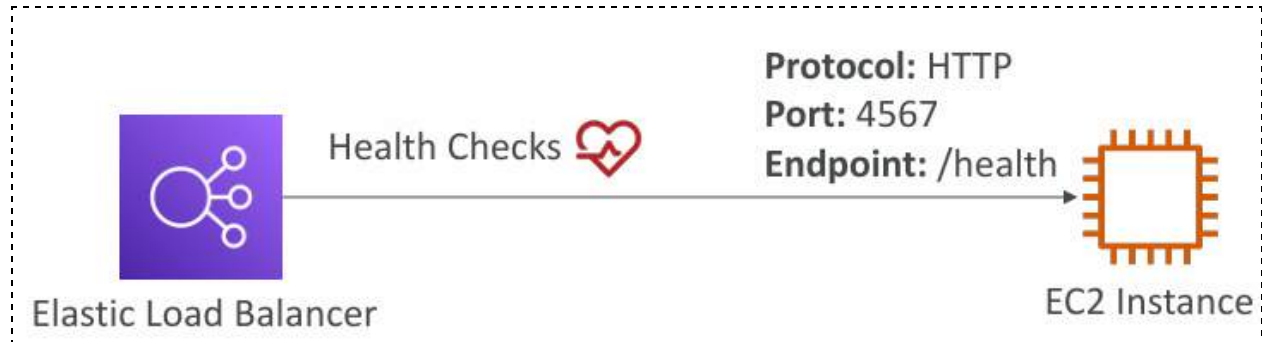
If the URL path is `/images/`, >> route traffic to **Target Group 1**.
 If the URL path is `/videos/`, >> route traffic to **Target Group 2**.

- **Route Based on Host Name**

If the hostname is `api.example.com`, >> route traffic to **Target Group 1**.
 If the hostname is `www.example.com`, >> route traffic to **Target Group 2**.



● Health Check



- عبارة عن process ال load balancer بيعملها بانتظام علشان يتأكد ان ال Targets قادرة انها تـ respond ال requests اللي هو باعتهاها ولا لا.
- سواء بتستخدم aws load balancer او load balancer software على سيرفر ك زي nginx فانت بتـ configure ال options الخاصة بال health check زي interval, timeout وكدا
- لما بتكرت Targets group بتقدر انك تعمل health check configuration .. والخطوة الجاية هيتوضح يا عني اية Health Check اكرر.

● Health Check Configuration:

- Configuration:
 - **Target Type:** EC2 instances
 - **Health Check Protocol:** TCP
 - **Protocol:** HTTP
 - **Path:** /healthcheck
 - **Port:** 80
 - **Interval:** 30 seconds
 - **Timeout:** 5 seconds
 - **Healthy Threshold:** 2



- Unhealthy Threshold: 2

- ف المثل دة ال load balancer هيبعت كل 30 ث request لـ

```
http://<target-ip>:80/healthcheck
```

- لو ال request دة اتعملو 200 ok respond ف خلال 5 ث ..

It is considered **healthy**.

- لو ال request دة اتعملو fails ف خلال 5 ث أو response with error ..

It is considered **unhealthy**.

- Unhealthy Threshold: لو ال Target حصله fail ف 2 health checks متتابعين ..

It will be marked as **unhealthy** and **stops receiving traffic**.

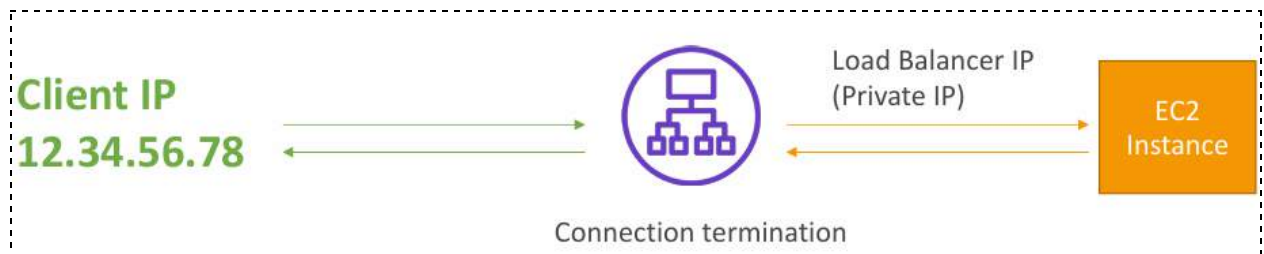
- Healthy Threshold: لو ال Target حصله success ف 2 health checks متتابعين .. وبلاش 1 دة مش مؤشر قوي ان ال instance تستحمل ولا لا

it is marked as **healthy** and **starts receiving traffic again**.

💡 if the health check found the instance not working it will be terminated then replaced

💡 Health Checks support the TCP, HTTP and HTTPS Protocols.

● 'X-Forwarded-For' Header (How it works?):



1. ال client لما بيعمل request بيروح لل << Application Load Balancer

ALP دة بيشتغل ف 7 layer وبيهندل ال HTTP , HTTPS Traffic

2. ال ALP بيضيف لل request ال X-Forwarded-For Header

3. ال ALP بي forward ال request بعد ما انضافلة ال Header لل backend targets

```
GET / HTTP/1.1
```

```
Host: www.example.com
```

```
X-Forwarded-For: 192.0.2.1
```

4. دا كذا شكل ال request اللي رايح لل backend targets بعد ما انضافلة ال

Header

💡 الفايده من الموضوع دة ان ال backend target بيـ

Can identify the original client's IP address for logging, analytics, and security purposes,
Helps in tracing the request path through multiple proxies.

➡ 005 Application Load Balancer (ALB) - Hands

On - Part 1

هتكريت 2 instances وال SG بتاعهم هتسمح فيه ان ال http connections

تيجي عن طريق any where وهتكريت ال load balancer وهتسمح ال

connection اللي داخلاه انها تكون http من any where واللى خارجه منه بردو

all all وبالتالي كذا قادر ال load balancer يستلم ويوزع عال ec2 instances

وتقدر تخش عليهم من خلال ال loadbalancer DNS او ال ip addresses بتاع

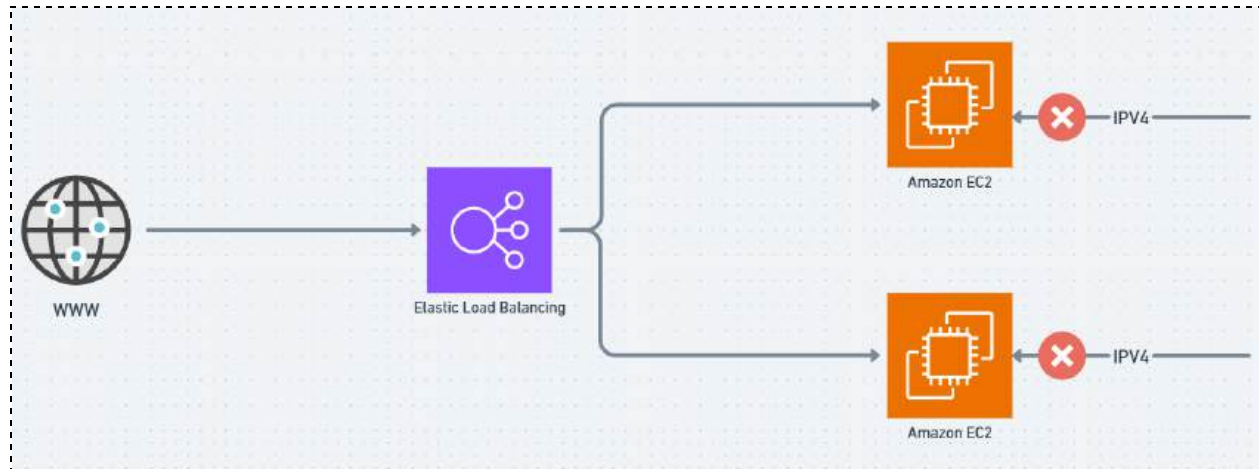
ال instances



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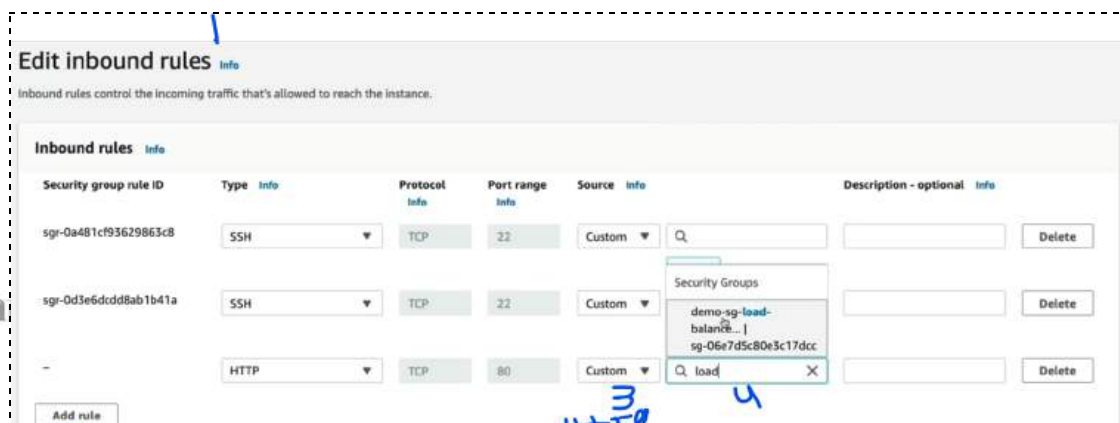
006 Application Load Balancer (ALB) - Hands On - Part 2

- **Tip1-** Enter the instances through ALP only



لما اشتغلنا في Part 1 كنا نقدر ن access ال EC2 Instances سواء من ال IPv4 بتاعهم او تخش عال Load Balancer DNS وهو بيوزك عليهم .. ف احنا عاوزين نقفل جزأيه ان حد يقدر يخش direct عال EC2 Instances عاوزين بس من خلال ال load balancer

ده هيتم عن طريق ال security group قولنا انها اكنها عباره عن firewall بتة security group ال control inbound , outbound traffic ف هنخش عال



بتاع ال instances ون allow فى ال inbound rules ال HTTP يجيلك فقط من ال load balancer بتاع ال Security group

دلوقتي لو جيت ت Access ال instances من خلال ال IPv4 هت timeout معاك.

● Tip2 - Modify the ALP Listeners & rules

Go to ALP >> Listeners >> HTTP-80 >> Listeners rules >> add rules

Add condition

Rule limits

Rule condition types

Route traffic based on the condition type of each request. Each rule can include one of each of the following conditions: host-header, path, http-request-method and source-ip. Each rule can include one or more of each of the following conditions: http-header and query-string.

Source IP

Include one of each

Host header

Path

HTTP request method

Source IP

Include one or more of each

HTTP header

Query string

is are not supported.

this rule.

Cancel

Confirm

تقدر تحط عدد ال rules اللى انت عاوزها وكل rule بال condition بتاعها .. وبعدين

بتحدد ال action اللى هيحصل بناء عال condition دة

هنا احنا هنعمل action معين يحصل لو ال user دخل عال path دة

Path

Define the path. For example: /item/*. Case sensitive.

is /error

Maximum 128 characters. Allowed characters are a-z, A-Z, 0-9; the following special characters: _-./~'"@:~; & (using &); and wildcards (* and ?).



I'm Here

Actions

Action types

☐ Forward to target groups
 ☐ Redirect to URL
 ☒ Return fixed response

Return fixed response [Info](#)

Use fixed-response actions to drop client requests and return a custom HTTP response. When a fixed-response action is taken, the action and the URL of the redirect target are recorded in the access logs.

Response code
The type of message you want to send.

404

2xx, 4xx, 5xx

Content type - optional
The format of your message.

text/plain

Response body - optional
Enter your response message.

Not found, custom error!

1024 character maximum

هتلاقى عندك حاجة اسمها priority معناها الأولوية .. فايدتها أننا لو عندنا rule 100
مثلا ال rule اللى الأولوية بتاعتها اعلى هي اللى هتتنفذ

Rule: DemoRule

Priority
Rule priority controls the evaluation order of a rule within the listener's set of rules. You can leave gaps in priority numbers.

5

1 - 50000

ودي كانت النتيجة



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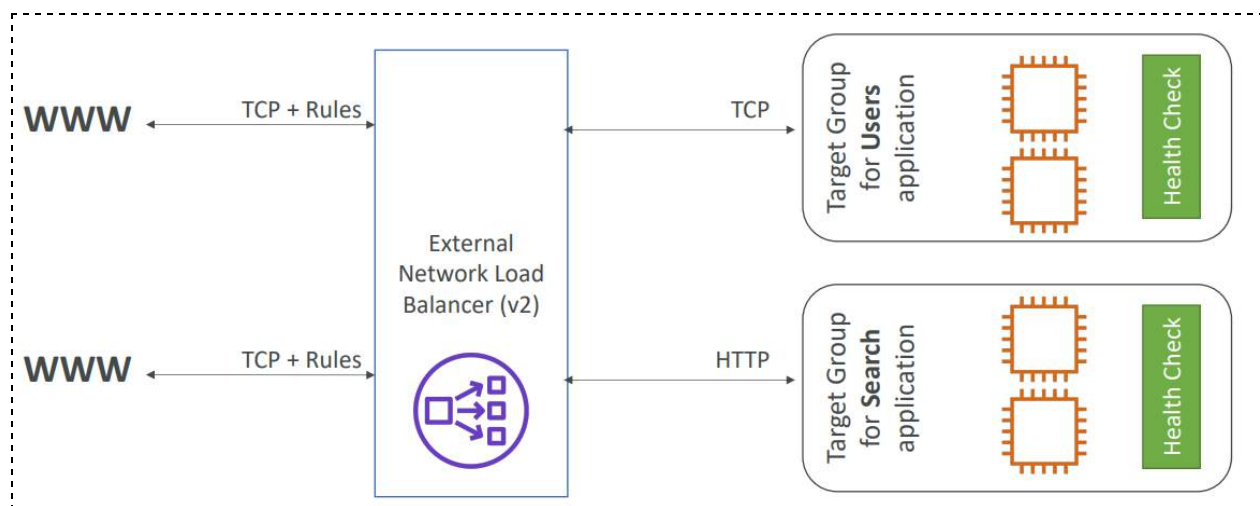
007 Network Load Balancer (NLB)

● Key Features

- بيشتغل فى Layer 4 وبي forward TCP & UDP traffic عال targets group
- بيهندل ال high volume of connections .. ودة بيخليه مناسب اكر لل real-time gaming, media streaming, or financial services
- ال latency قليلة هنا 100 ms مقارنة بال ALP اللى كان 400 ms

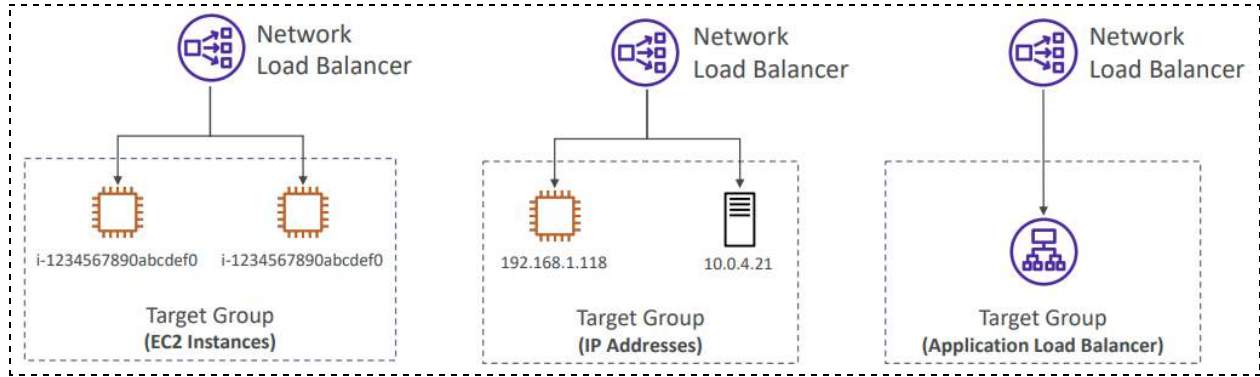
NLB has one static IP per AZ, and supports assigning Elastic IP
(helpful for whitelisting specific IP)

- ميزة ان ال NLB ليه Static IP بتخليك تستخدمها بشكل كويس ف مثلا لو عندك data base او اي backend services وعاوز تخليها تـ Accept ال Traffic اللى جاي عليها only from ال NLB ف بتروح فالـ firewall بتاعها تـ allow ال Traffic اللى جاي من الـ Static IP دة وبس كدا ضمانت ان مفيش اي Traffic هيجيلك غير منه فقط.. نفس الكلام وانت بتـ assign ال DNS عال Static IP .. مفيش تغيير ف ال IP ف دة كويس.
- مش متاح ف ال free tier



- **Network Load Balancer –Target Groups**

- نفس اللى قولناه فوق ممكن تكون EC2 instances او IP Addresses وهنا الزيادة ممكن يكون ALP



- **IP Addresses**

- فى حالة لو ال target group كان ip addresses لازم تكون Private IPs

- **NLB With ALP As a target group**

- ال scenario ده بيحصل لما اكون محتاج ال **NLB** يهتدل ال TCP/UDP traffic العالي اللى جايلى ويكون فيه low latency وال ALP يدبني اوبشنز كثيره علشان ي Handle HTTP/HTTPS traffic



008 Network Load Balancer (NLB) - Hands On

- Create >> NLP SG , Instances SG , Instances with user data , target group (wait for the instances to be healthy) , NLP

1. Instances SG >>

عندنا 2 instances وانت بتكريتهم هتروح الاول تكرت SG تسمح فيه بال http

inbound من NLP SG وتحط في ال user data الكود بتاعك اللي هيعرفك انت

بتاكس انه instance

Inbound rules

Security group rule ID	Type	Protocol	Port range	Source	Description - optional
sg-0225613f79833456d	HTTP	TCP	80	Custom	sg-08a015347b89ec1d7

Add rule

Cancel Preview changes Save rules

Outbound rules

Security group rule ID	Type	Protocol	Port range	Destination	Description - optional
sg-0b8cfe4638e17955	All traffic	All	All	Custom	0.0.0.0/0

Add rule

Rules with destination of 0.0.0.0/0 or ::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

2. NLP SG

دي ال inbound , outbound الخاصة بيه

Inbound rules

Security group rule ID	Type	Protocol	Port range	Source	Description - optional
sg-0f61eaa7040e8237	HTTP	TCP	80	Custom	0.0.0.0/0

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.



Outbound rules [Info](#)

Security group rule ID: sgr-0b5fa2d3d0ae166f1c9

Type: All traffic

Protocol: All

Port range: All

Destination: Custom

Description - optional

0.0.0.0/0

Add rule

Rules with destination of 0.0.0.0/0 or ::0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

3. NLP

هتختار ال instances target جروب اللي هتوزع عليه

Listeners and routing [Info](#)

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener TCP:80

Protocol: TCP

Port: 80

Default action: Forward to

Select a target group

1-65535

Create target group

Instances TG

Remove

زي مقولنا تستني ال target group توضحك ان ال instances healthy .. لان

في البداية بتكون unhealthy cause of bootstrapping

Targets [Monitoring](#) [Health checks](#) [Attributes](#) [Tags](#)

Registered targets (2)

Filter targets

	Instance ID	Name	Port	Zone	Health status	Health status details	Launch time
☐	i-0713330ed80be801	machine1	80	us-east-1b	Healthy		July 31, 2024, 23:40 (...)
☐	i-0c67f8d9646fb67ab	machine1	80	us-east-1b	Healthy		July 31, 2024, 23:40 (...)

لو عملت stop ل instance وروحت عال target group هتلاقيها بالشكل ده انها في

ال stopped state

Targets [Monitoring](#) [Health checks](#) [Attributes](#) [Tags](#)

Registered targets (2) [Info](#)

Anomaly mitigation: Not applicable

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

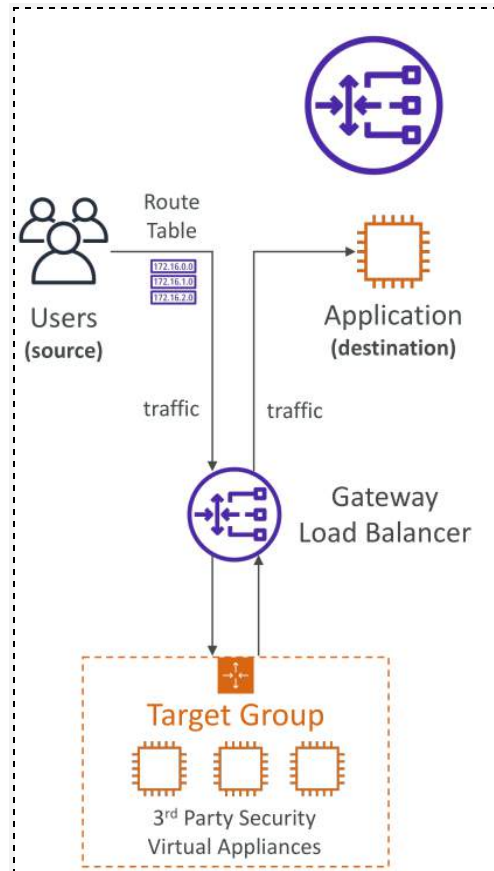
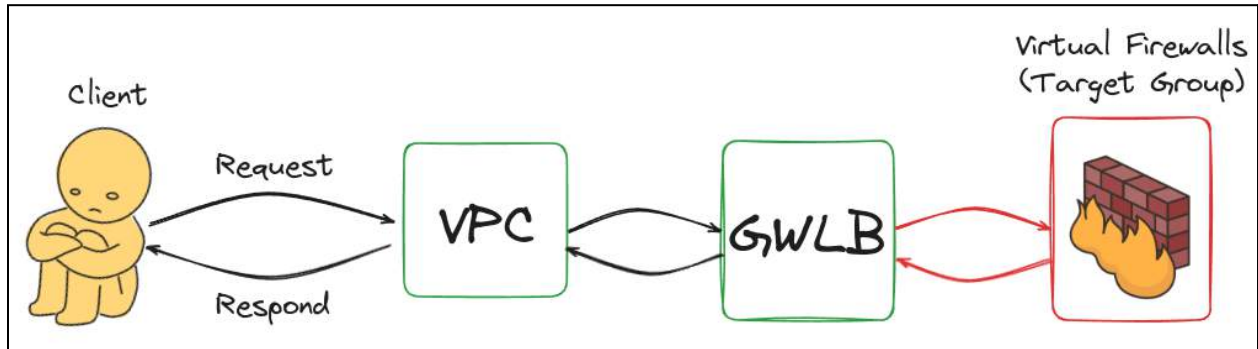
Filter targets

	Instance ID	Name	Port	Zone	Health status	Health status details	Launch...	Anomaly detection result
☐	i-Dc33c2852c982065d	Machine1	80	us-east-1b	Unused	Target is in the stopped state	August 1, ...	Normal
☐	i-01d07075fb54e4973	Machine1	80	us-east-1b	Healthy	-	August 1, ...	Normal



I'm Here

➔ 009 Gateway Load Balancer (GWLb)



● النوع ثالث من أنواع ال load balancers .. محتاجين نعرف عنها هنا طريقة عملها :

- سيكون فيه Traffic داخل عال AWS infrastructure

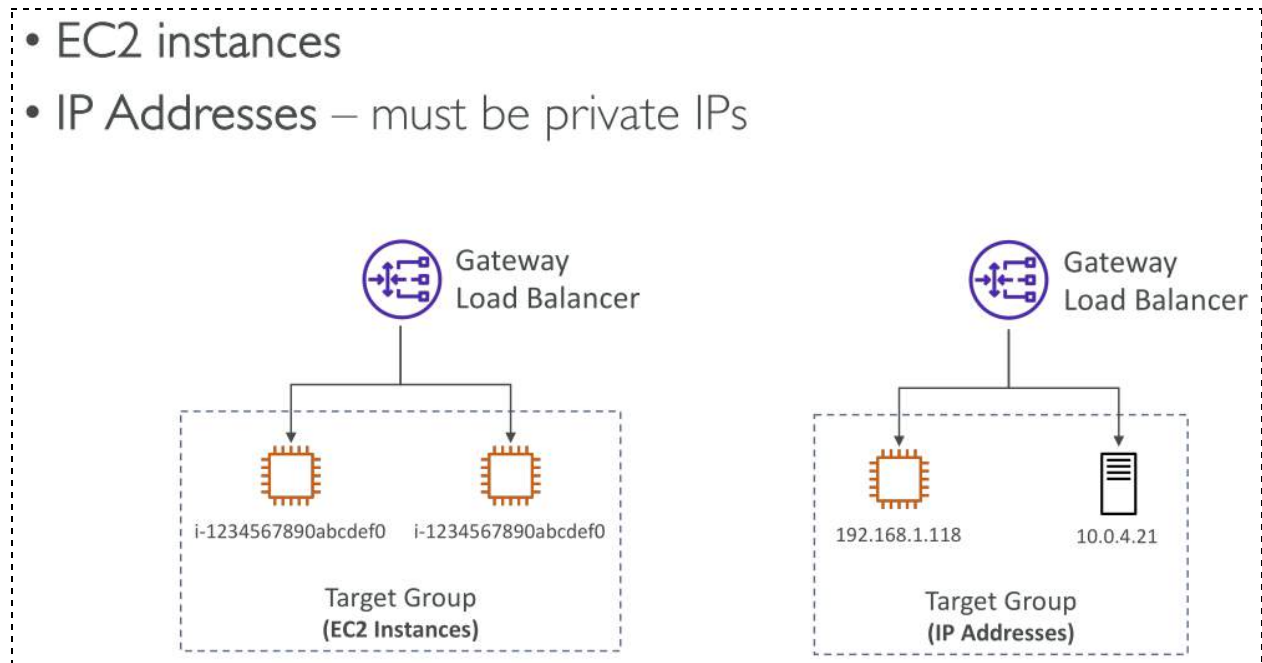
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- الـ GWLB بتستلم الـ Traffic دة وتعمله route علي firewall instance او اكثر من واحد بيكونو داخل target group
 - بيتتم فلترة الـ Traffic دة من خلال الـ firewall وبيشوف مين هيتعملو accept ومين reject
 - بعدها الـ firewall بي route الـ accepted Traffic للـ GWLB وبترجعه للـ AWS infrastructure
- هنعمل Hands on ولكن بعدين لان الموضوع معقد شوية.

• Gateway Load Balancer –Target Groups

- EC2 instances
- IP Addresses – must be private IPs

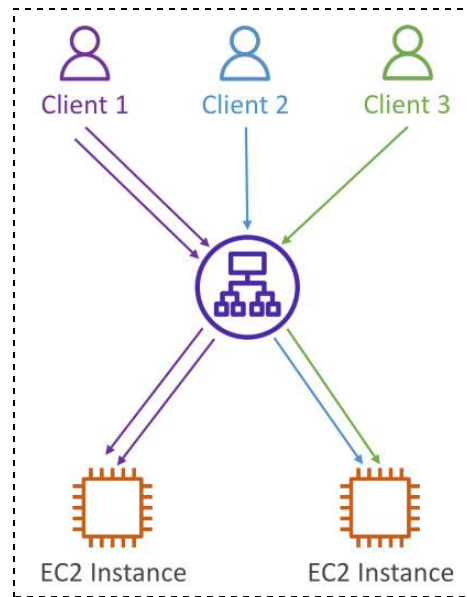


- نفس اللى قولناه فوق ممكن تكون EC2 instances او Private IP Addresses



➡ 010 Elastic Load Balancer - Sticky Sessions

- Overview:



- في الطبيعي ال load balancer سواء ال classic , network , application بيوزعو ال traffic عال instance الاولي ثم الثانية ثم الثالثة وهكذا .. في الحالة دي لو استخدمت ال sticky sessions اللي هيثم ان ال client لو بعث request 2 لل load balancer ف هيوديه على نفس ال instance مش هيووزعو

- Cycle Scenario:

- ال user بيبعث لل load balancer ال request ف بيحولو على اي server فال server بي generate session ID ويبعتها لل browser على هيئة cookie وال session ID دة بيكون unique لل user حتي ال load balancer من خلال ال session ID الموجود في ال cookie بيعرف هيوذي ال same requests على انهي server



• عندنا بقا نوعين من ال cookies :

1. Application-based Cookies

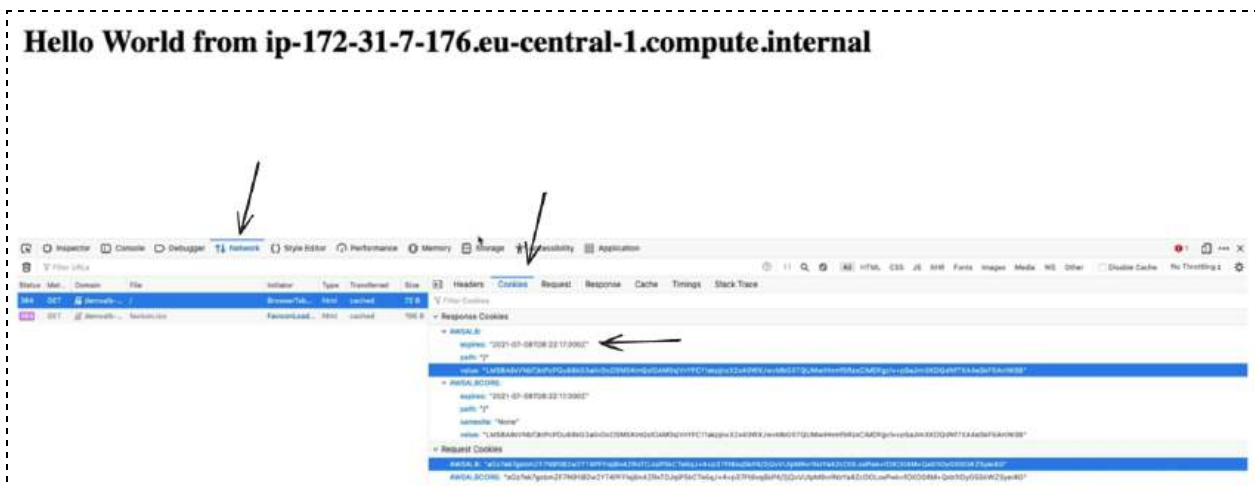
- ال user لما بيبيعت request لل server وال server بي generate cookie with session ID ال browser بيخزن ال cookie ده بال session id اللي جواه ويفضل الموضوع مستمر ان ال user ده بال session id بتاعة مرتبط ب specific server بيتحول عليه وده مش هيتغير .
- طب اية بيحصل لما بمسح cookies ال browser ؟ اللي بيحصل ان ال session دي خلاص اتقفلت ف هترجع بقا من ثاني تبعت request وسيرفر جديد تتحول عليه دا بمزاج ال load balancer بناء عال load وال server هي generate cookie & session id جداد وتبدا من جديد حتي تلاقي ان ال account عمل sign out من الموقع اللي بتستخدمه او ان ال themes بتاعة المتصفح اتمسحت او ال shop cart فضيت وهكذا

2. Duration-based Cookies

- لمدة معينة انت بتحددها وليكن مثلا لمدة 30 دقيقة ال user هيفضل يستخدم نفس ال cookie مش هتتجدد وهيفضل يتحول على نفس ال instance

هتلاقي حتي في ال responses , requests في جزأ ال network & cookies

في ال browser بتنتهي امتي وهكذا.

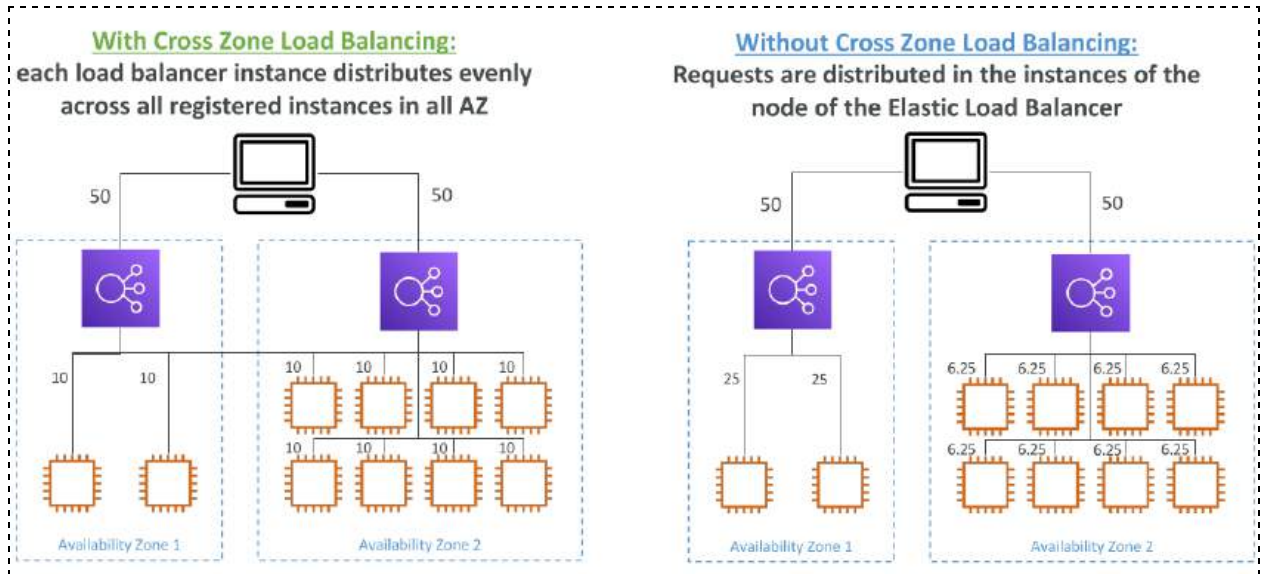


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011 Elastic Load Balancer - Cross Zone Load Balancing

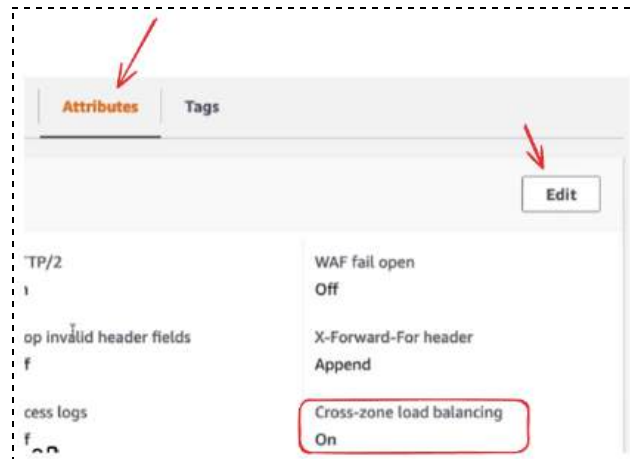
Balancing



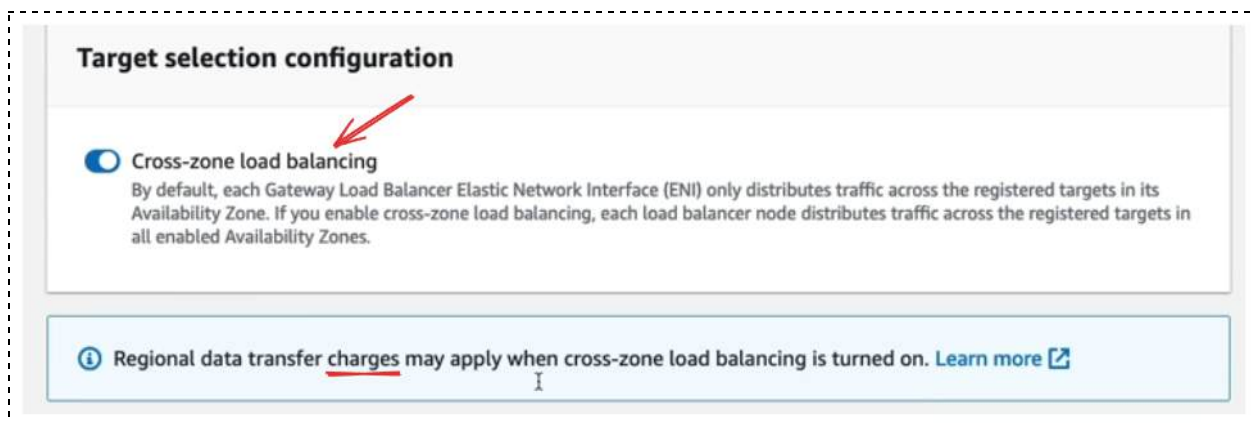
- ال Traffic اللى جاي لو انت مفعّل ال Cross zone LB فى الحالة دي هيتوزع عال 10 instance بالتساوي اما لو مش مفعّل ف ال traffic اللى جاي على AZ معينه هيتقسم عال instances اللى جواها فقط ومش هيخرج براها.
- بنستخدمه علشان الدنيا تمشي بالتساوي وعلشان مفيش instance تقع مننا.
- بتلاقيه **enabled by default** فى حالة ال **ALB**
- بتلاقيه **disable by default** فى حالة ال **classic , NLB**
- بالنسبة للدفع لو متفعل هتدفع فقط فى ال **NLB**

تقدر بعد ما تخش عال load balancer تعرف هو enable ولا disable من هنا



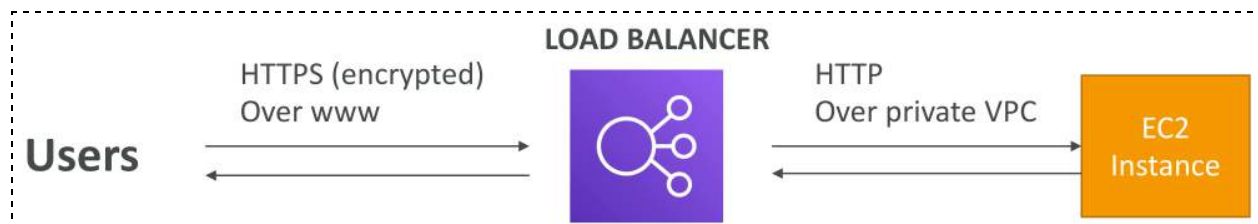


ومن edit تقدر تخليه enable or disable



012 Elastic Load Balancer - SSL Certificates

• Load Balancer SSL Certificate (Secure Sockets Layer):



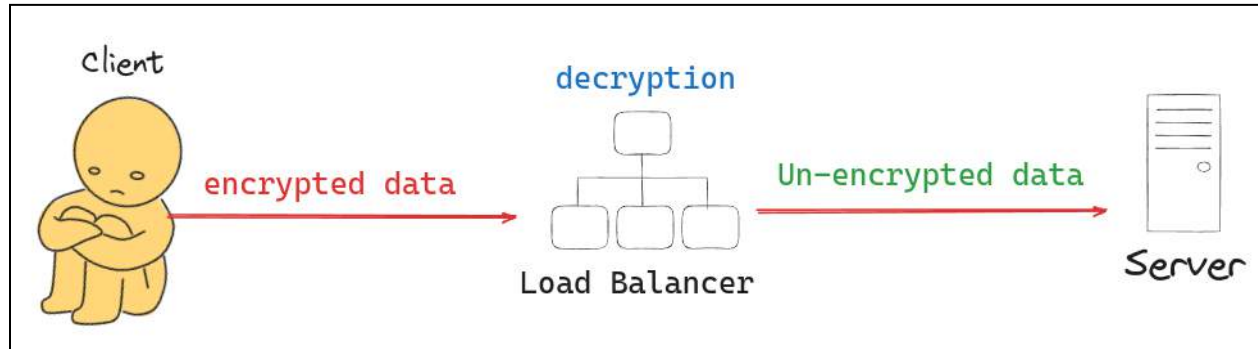
- بنستخدمها علشان نـ protect , encrypt الـ connection بين الـ user والـ load balancer



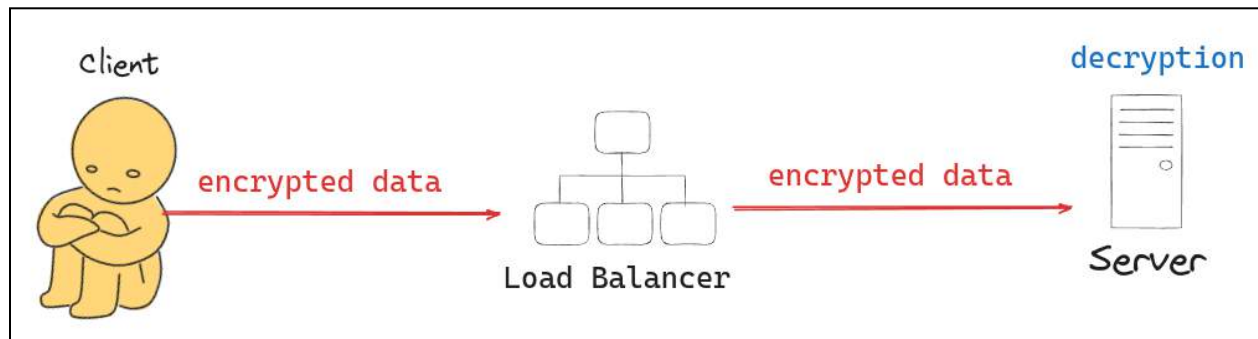
I'm Here

- الـ **SSL** : هو older protocol بيب secure الـ internet connection
- الـ **TLS** : الأحدث و more secure وحاليا مستخدم أكثر ولكن شائع اننا بنقول عليه SSL

• SSL Termination:



- الـ SSL/TLS traffic بييجي من الـ user معموله encrypt فالـ load balancer بيعملو decrypt وبيبعث الـ unencrypted data دي للـ backend servers ودة فايدته انو بيخفف الحمل عالـ backend servers انو ميعملش decrypt وكمات بتكون محتاج SSL Certificate واحدة بس عالـ load balancer مش لكل الـ back end server واحدة.



• SSL Passthrough:

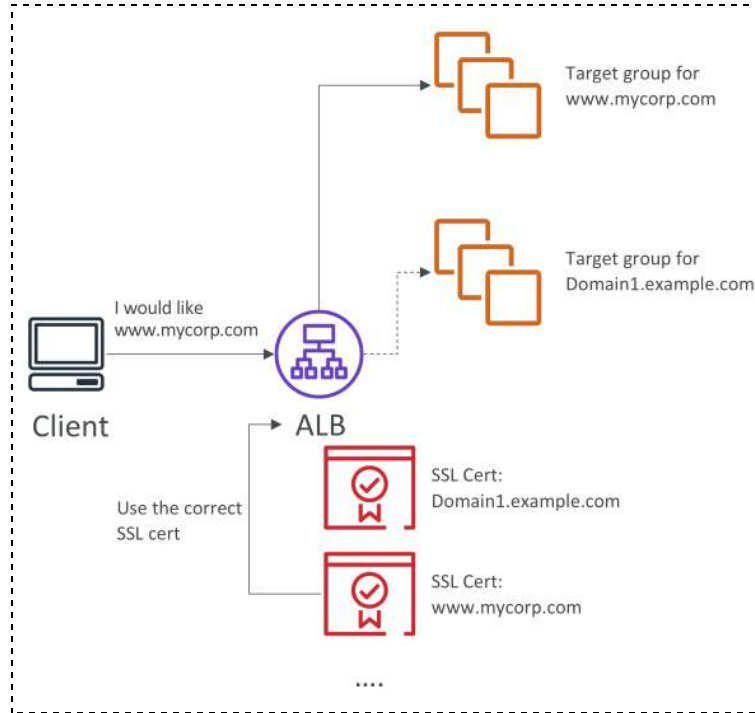
- الـ SSL/TLS traffic بييجي من الـ user معموله encrypt وبيفضل كدا لحد ما يوصل للـ backend servers .. ودة فايدته ان انت حققت End-to-End Encryption على طول المسار الـ data protected.



- **AWS Certificate Manager (ACM):**

- عبارة عن AWS Service بتـ simplify الـ provisioning, managing, and deploying SSL/TLS certificates for use
- او ممكن انت manual تـ upload الـ certificate details بتاعتك

- **Server Name Indication (SNI):**



- لو عندي server واحد فقط وعليه كذا website وليكن mycorp.com و Domain1.example.com وهكذا .. وكل website ليه certificate ف ازاى ال load balancer هيعرف ان ال certificate الفلانية تبع الموقع الفلاني؟

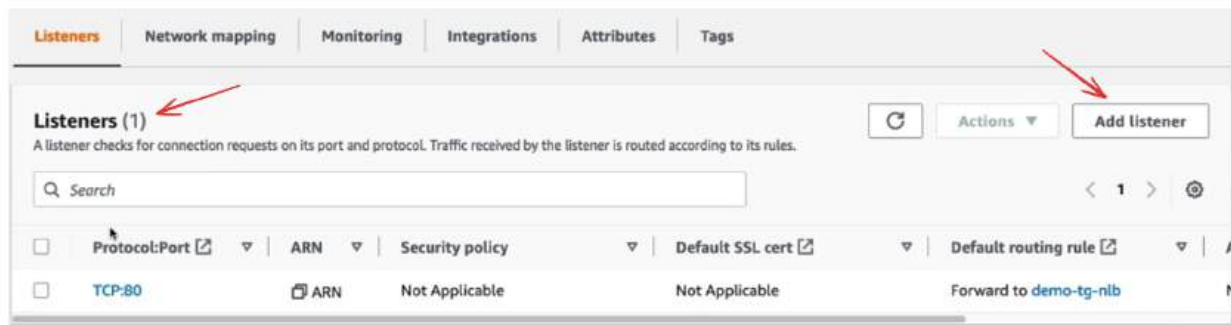
1. ال client بيبدأ SSL/TLS handshake اللى بيكون فيه ال host name اللى عاوز يوصله وضمن ال client hello message بردو بيكون فيه ال SNI Field
2. ال load balancer بيستلم ال client hello message ومن خلال ال SNI Field اللى فيها بيعرف هو يقصد أنه Certificate
3. بيروح ال load balancer عال domain names اللى معمولها configure عنده ويعرف ال certificate دي تبع انهى موقع (host name) ويوصل ال client بيه



4. بمجرد ما بيعرف ال load balancer بياخد بقا ال data دي سواء هطبق عليها مبدأ ال ssl termination or ssl passthrough وبيعتهها لل backend servers

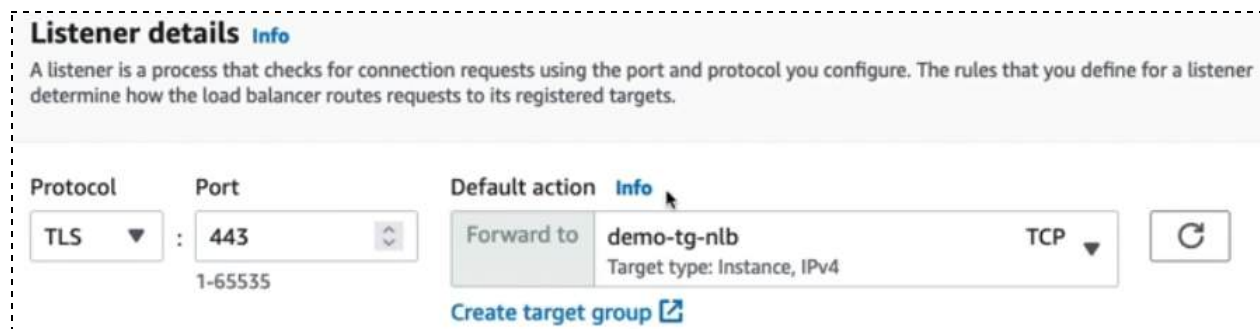
- **Only works for ALB & NLB** (newer generation), CloudFront, Does not work for CLB (older gen).

013 Elastic Load Balancer - SSL Certificates - Hands On



بتخس عال load balancer بتاعك سواء ALB , NLB وت Add listener

في حالة NLB هتختار TLS بدل ما كنا بنختار TCP



وفي حالة ال ALB هتختار HTTPS بدل HTTP



I'm Here

Listener details

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

Protocol	Port
HTTPS ▼	443
	1-65535

014 Elastic Load Balancer - Connection

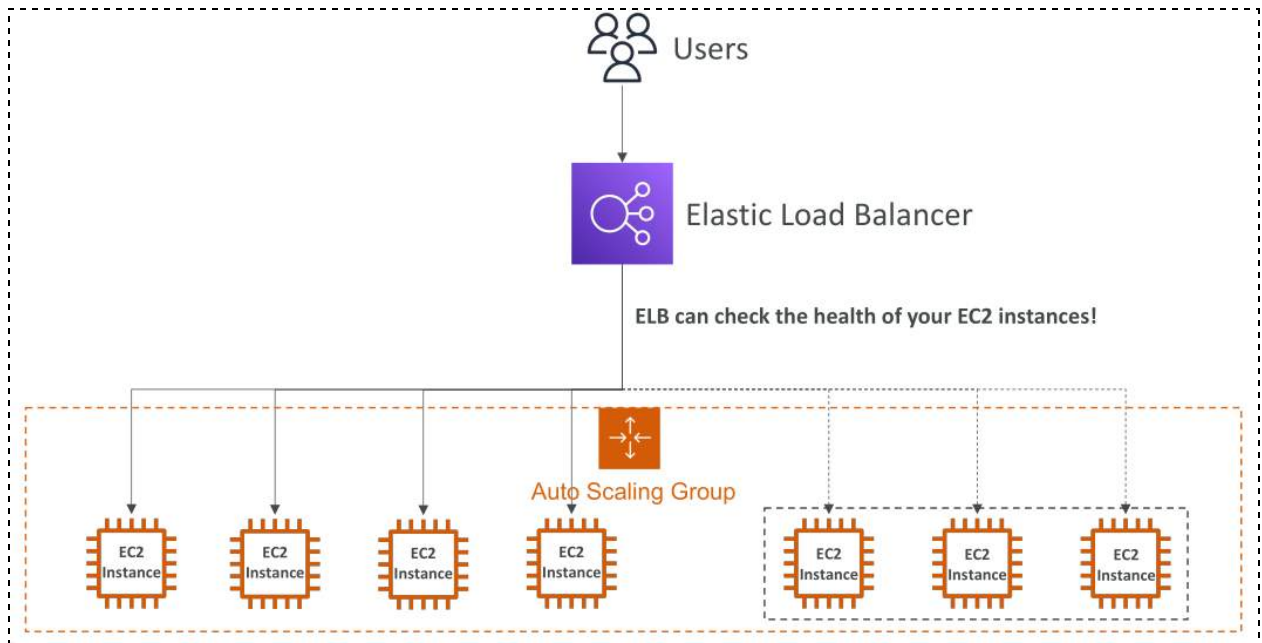
Draining

- محتاج مثلا تعمل update لل APP بتاعك وال old instances عاوز تغييرها مثلا ف لما تغييرها ال Load Balancer هيقف اي connections بين ال user & instances ولكن باستخدام ال Connection Draining ال connection حتي لو ال instance حصلها stop او حصل فيها اي مشكلة خلتها Unhealthy هيفضل ال connection مفتوح لمدة معينة انت بتحددها اسمها time out
- يبقي ال connection وليكن هيفضل لمدة 5 دقائق 300 ث شغال لحد ما ال instance تتبدل وال connections الجديدة تروح عليها
- ال Time out بتحدده في ال ELB Elastic Load Balancer ايا كان نوعه ويكون من 1 : 3600 ث .. علي حسب نوع ال requests بتاعتك بقا لو هيا بتاخذ وقت كبير بتكبر المدة لو بتاخذ وقت صغير بتصغر المدة على حسب ال app ونوعيه ال requests



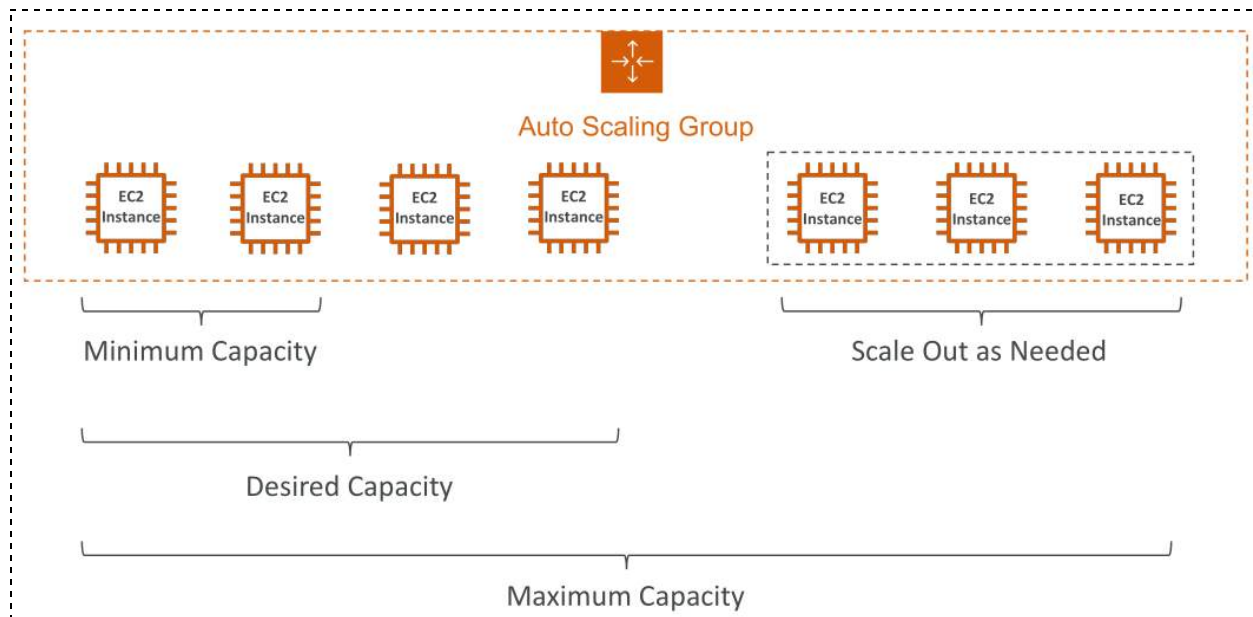
➔015 Auto Scaling Groups (ASG) Overview

- Overview:



- عبارہ عن **Free AWS feature** وظيفتها انها تحافظ عال application بتاعك انو يكون علطول Available من خلال انها بتزود وتنقص EC2 Instances علي حسب ال Traffic اللى بيبيعه ال load balancer عالجروب دة
- لو فيه traffic عالي بت Automatic scale out ولو فيه traffic قليل بترجع تاني تعمل Automatic scale in علشان تقلل التكلفة لان التكلفة اكيد زادت بما انك زودت instances





● Auto Scaling Group Capacity:

1. **Minimum Capacity** : أقل عدد Instances انا عاوزها تكون موجودة
 2. **Maximum Capacity** : أكبر عدد Instances انا عاوز اني اوصله
 3. **Desired Capacity** : العدد المتوسط اللي انا عاوزة
- ممكن أتحكم في عدد ال instances ازود او انقص بناء على ال cloud watch alarm ..
مثلا جالي alarm ان فيه استهلاك عالي فال CPU ف نزود EC2 Instances وهكذا

● Create Launch Template:

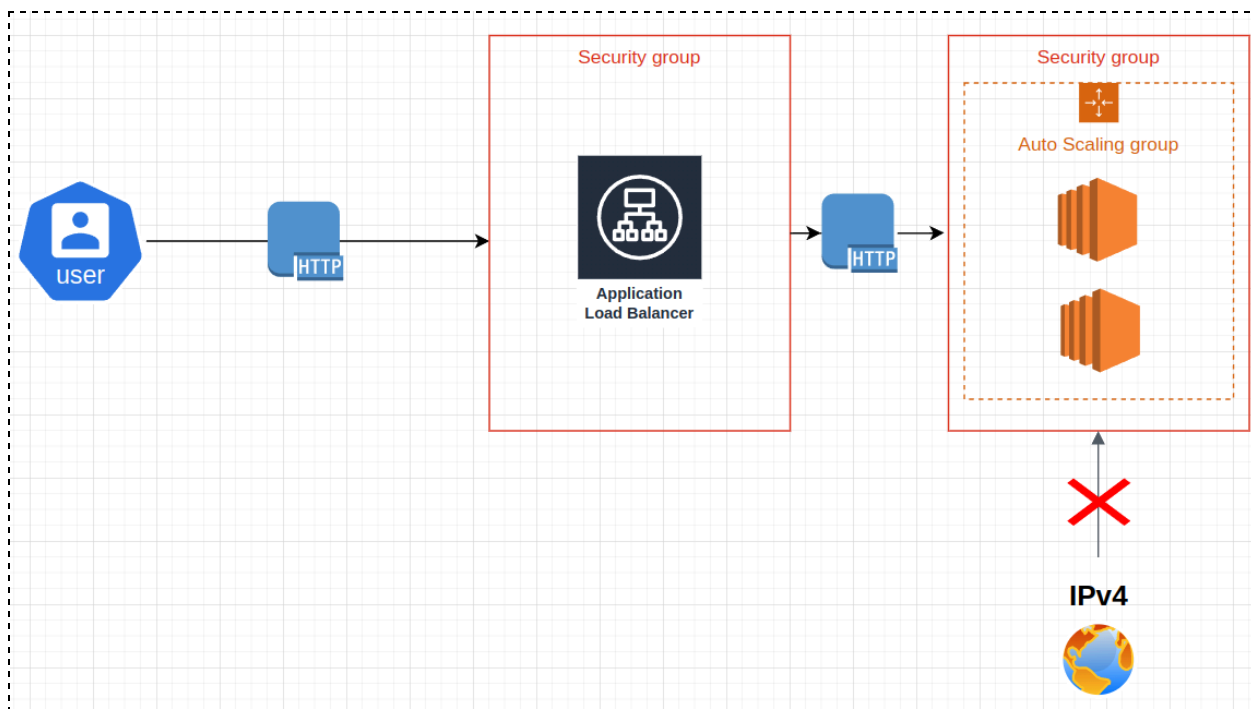


I'm Here

- شوية parameters بتعملها set بحيث اي EC2 Instance هتكريتها هتقوم بال parameters دي.. يا عين لما يحتاج يـ Scale out هياخد نفس ال Parameters دي ..
مثلا عملت set لل Subnet = 1A معني كذا ان اي EC2 Instance هتقوم من ال Zone 1A = وهكذا



016 Auto Scaling Groups Hands On



أول حاجة هتعملها ال launch template علشان اي instance تقوم تقوم بنفس ال configurations

ASG >> Launch Template

Launch template [Info](#) [Switch to launch configuration](#)

Launch template
Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

Select a launch template

[Create a launch template](#) [↗](#)

هتختار مثلا 3 AZ علشان هنقوم حوالي 3 instances



I'm Here

Load balancing Info

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

☐ **No load balancer**
Traffic to your Auto Scaling group will not be fronted by a load balancer.

☒ **Attach to an existing load balancer**
Choose from your existing load balancers.

☐ **Attach to a new load balancer**
Quickly create a basic load balancer to attach to your Auto Scaling group.

Attach to an existing load balancer

Select the load balancers that you want to attach to your Auto Scaling group.

☒ **Choose from your load balancer target groups**
This option allows you to attach Application, Network, or Gateway Load Balancers.

☐ **Choose from Classic Load Balancers**

Existing load balancer target groups
Only instance target groups that belong to the same VPC as your Auto Scaling group are available for selection.

تزداد ال desired وليكن ل 2 بدل 1 تلاقي فال instance واحدة بتقوم تروح عال ASG
تلاقي نفس الكلام ان فيه واحدة بتقوم تروح عال target جروب بعدها تلاقيها بتوصل وبقت
unhealthy علشان ال bootstrapping ثم بقت healthy تيجي تأكس ال load
balancer بدل ما كان بيوجهك على واحدة بدأ يبديل عالنتين وهكذا

ظهرلي مشكلة وهي اني لما جيت اعمل ال load balancer حددت zone 2 بس ولكن ال
instance لما جت تقوم قامت ف 1c وانا كنت محدد 1a & 1b ف رحت عال load
balancer عدلتها .. ولقيتها بدل ما كانت واقفة في ال instances بدات تقوم وفي ال target
group كذلك



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Targets							
Registered targets (2) Info							
Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets. Filter targets							
☐	Instance ID	Name	Port	Zone	Health status	Health status details	Launch...
☐	i-0e653d10d1436eda1		80	us-east-1c	Unused	Target is in an Availability Zone that is not enabled for the load balancer	August 1, ...
☐	i-0af8ee8d7b4b06632		80	us-east-1a	Healthy	-	August 1, ...

Instances (2) Info							
Find Instance by attribute or tag (case-sensitive) All states							
☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
☐		i-0e653d10d1436eda1	Running	t2.micro	Initializing	View alarms +	us-east-1c
☐		i-0af8ee8d7b4b06632	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a

Targets							
Registered targets (2) Info							
Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets. Filter targets							
☐	Instance ID	Name	Port	Zone	Health status	Health status details	Launch...
☐	i-0e653d10d1436eda1		80	us-east-1c	Initial	Target registration is in progress	August 1, ...
☐	i-0af8ee8d7b4b06632		80	us-east-1a	Healthy	-	August 1, ...

ال ASG مربوط بال TG وال TG مربوط بال ALB 💡

➡ 017 Auto Scaling Groups - Scaling Policies

- **Dynamic Scaling Policy:**
- **Target Tracking Scaling:**

- مثلاً أحددو ان أخرك ف استهلاك ال CPU انو يكون 60% زيادة عن كذا زودلي EC2 instances .. وطبعاً هيلتزم بال maximum اللي احنا حددناه فوق



I'm Here

- **Simple/Step Scaling:**

- لو فيه cloud watch alarm معين نفذ كذا .. لو مثلا عندي alarm ان ال cpu usage بقى فوق ال 70% ف فى الحالة دي زودلي EC2 instances وليكن زود 2 ولو اقل من 70% قلل 1 وهكذا

- **Scheduled Scaling:**

- بنعمل دة لو فيه event ف وقت معين بيكون فيه traffic عالي .. رأس السنة مثلا عيد الأم يوم الجمعة كذا ياعني .. فالأوقات دي فيه traffic اعلى من الطبيعي ف زودلي EC2 instances ولما الايام دي تنتهي ارجع للطبيعي.

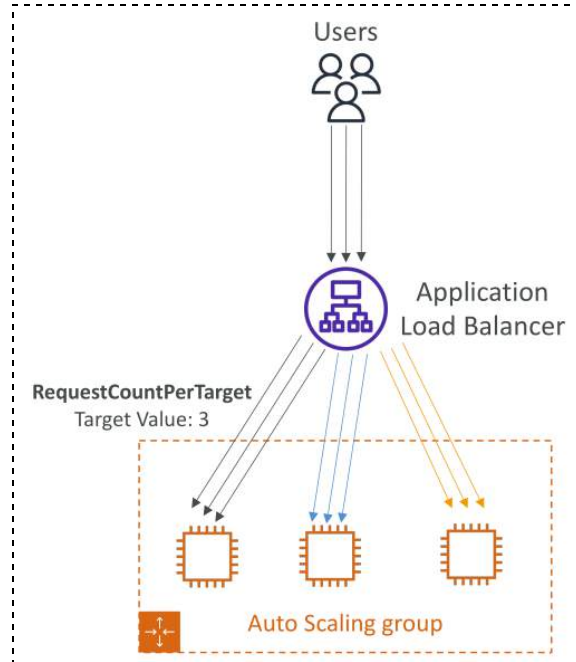
- **Good Metrics to Scale On in AWS:**

- حاجات ممكن ت Scale in/out بناء عليها:

- **CPU Utilization:**

- **Request Count Per Target:**

- بناء على عدد ال requests اللي جايه عال instances بتاعتي هنا مثلا 3



- **Average Network In / Out:**

- Measures the average network traffic in and out of your instances.

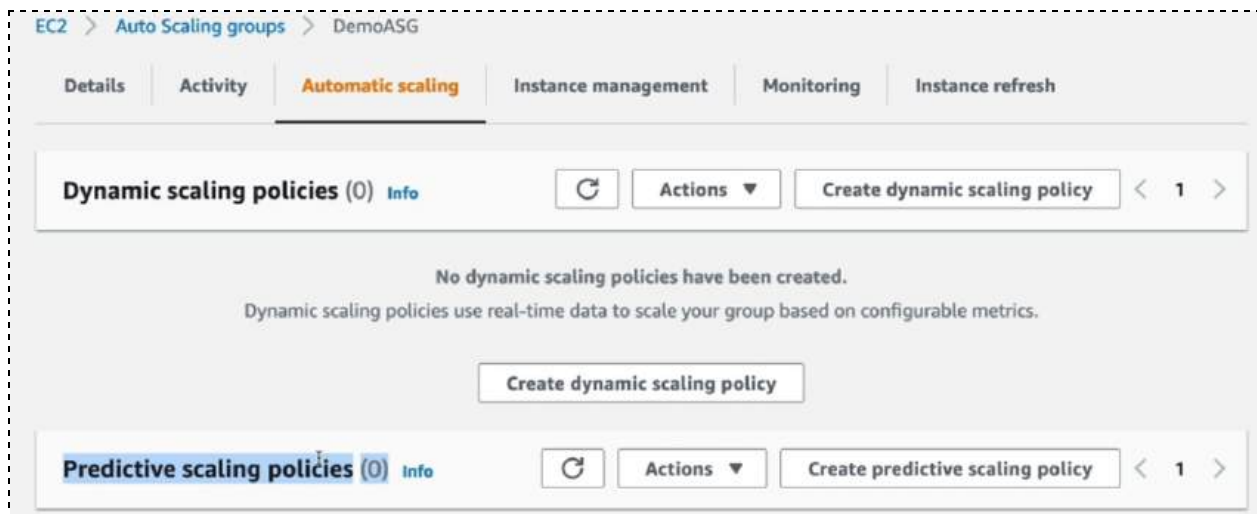
- **Custom Metrics:**

- اي specific metrics انت عاوزها وبتدخلها عال cloud watch وتستخدمها.

- **Scaling cooldown:**

- فتره فى خلالها ال ASG مبيقومش باي actions تخص ال scaling سواء in/out بحيث ي stabilize وميكونش فيه rapid scaling و by default بيكون 300s = 5 min لو معملتوش ال rapid scaling ده هيكلفك كثير excessive costs

- **Predictive Scaling:**



- عبارته عن feature موجودة فى ال ASG بتحلل ال traffic data اللى حصلت قبل كذا وبناء عليها بتعرف امتي ال traffic بيكون عالي وامتي اقل وكذا وبناءا عليه بت scale فى اي عطلة او مناسبة او كذا
- بنستخدمها لان من غيرها ال ASG ممكن متشتغلش بكفاءة ممكن يكون فيه مثلا lag time على ما ال instances تقوم اما لما بنستخدمها بت schedules scaling actions مقدما

- **Activity History Example :**

1. كنت محدد ف ال desired capacity = 1 ف هتلاقىه زود instance من 0 إلى 1



2. كنت محدد ف ال 50% target tracking = ف لما ال cpu زاد عن 50% زود 1 instance لان زي مقولنا معني ال target tracking (ان أخرجك ف استهلاك ال CPU انو يكون 50% زيادة عن كذا زودلي EC2 instances .. وطبعاً هيلتزم بال maximum اللي احنا حددناه فوق)
3. بعد ما زود instance وبقي عندنا 2 لقي ان فيه ضغط جاي تاني عال CPU وبيزيد عن 50% تاني فبالتالي زود واحدة كمان ووصلنا ل3 اللي هو ال maximum

Activity history (3)			
Status	Description	Cause	Start time
Successful	3 Launching a new EC2 instance: i-09f0a15800773c69f	At 2024-05-12T23:08:21Z a monitor alarm TargetTracking-CloudCode-ASG-AlarmHigh-cf4fd480-114d-4011-a374-8e2e709032fa in state ALARM triggered policy Target Tracking Policy changing the desired capacity from 2 to 3. At 2024-05-12T23:08:25Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 2 to 3.	2024 May 13, 01:08:27 AM +02:00
Successful	2 Launching a new EC2 instance: i-0bb0b0f24bcefe6a2	At 2024-05-12T23:06:21Z a monitor alarm TargetTracking-CloudCode-ASG-AlarmHigh-cf4fd480-114d-4011-a374-8e2e709032fa in state ALARM triggered policy Target Tracking Policy changing the desired capacity from 1 to 2. At 2024-05-12T23:06:27Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 1 to 2.	2024 May 13, 01:06:30 AM +02:00
Successful	1 Launching a new EC2 instance: i-0331d8fd086b43c89	At 2024-05-12T22:39:57Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 1. At 2024-05-12T22:39:58Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 1.	2024 May 13, 12:40:00 AM +02:00

018 Auto Scaling Groups - Scaling Policies

Hands On

ال setup بتاع ال predictive scaling بسيط بتحدد بس ال metrics و خلاص



I'm Here

Turn on scaling [Info](#)

☒ **Scale based on forecast**
 If switched off, predictive scaling will only forecast capacity and not take any scaling action. Only one predictive scaling policy can have scaling turned on at a given time.

Metrics and target utilization [Info](#)

Metrics
 There are two metrics for predictive scaling. One metric tells the policy about the load your application has been under. The other metric and the target utilization determine the overall average utilization to target.

CPU utilization
 Uses the total CPU to create the load forecast and the average CPU to define your target utilization...

Target utilization
 The average CPU utilization rate to target in the forecast period.

50 % per instance
 Must be greater than 0.

بالنسبة لل cloud watch alarm Dynamic scaling policy بتحدد بقا نوعها ولو فيه

وهكذا بردو setup بسيط

هنا هنكريت target tracking scaling بناءا عال 40% cpu

- خلي بالك بقا انك تغير ال desired , maximum capacity وكدا لان دلوقتي لما

تشتغل وال cpu يزيد عن 40% ف هي scale out ف لازم تكون سامح ب دة قبل كدا

في ال ASG

Create dynamic scaling policy

Policy type
 Target tracking scaling

Scaling policy name
 Target Tracking Policy

Metric type
 Average CPU utilization

Target value
 40

Instances need
 300 seconds warm up before including in metric

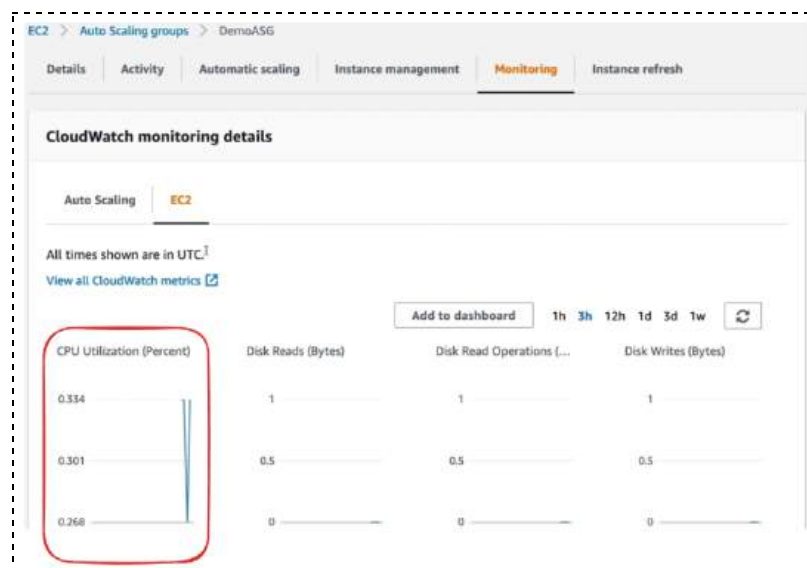
☐ Disable scale in to create only a scale-out policy



تقدر ت connect عال instance من خلال AWS عادي وبدها command ب دول بحيث
ت stress ال cpu انو يرفع عن 40%

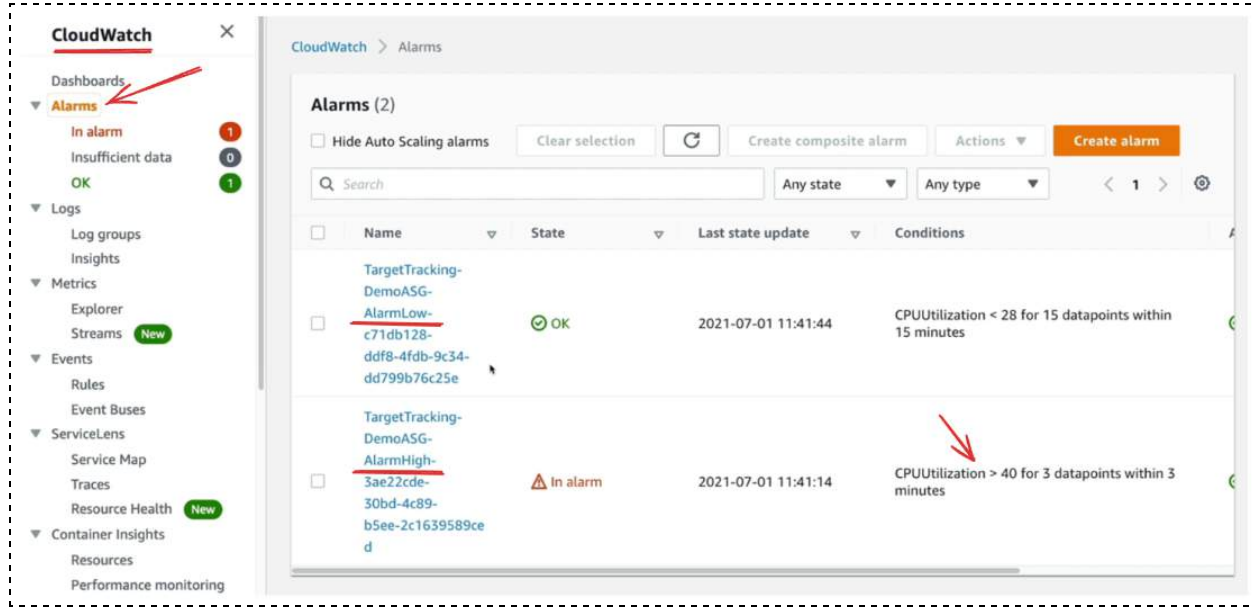
```
sudo amazon-linux-extras install epel -y  
sudo yum install stress -y  
Stress -c 4
```

زي مانت شايك في ال monitoring ال cpu utilization رفع .. وبالتالي هي scale out
ب 1 او 2 instances على حسب الضغط وعلى حسب ال maximum capacity



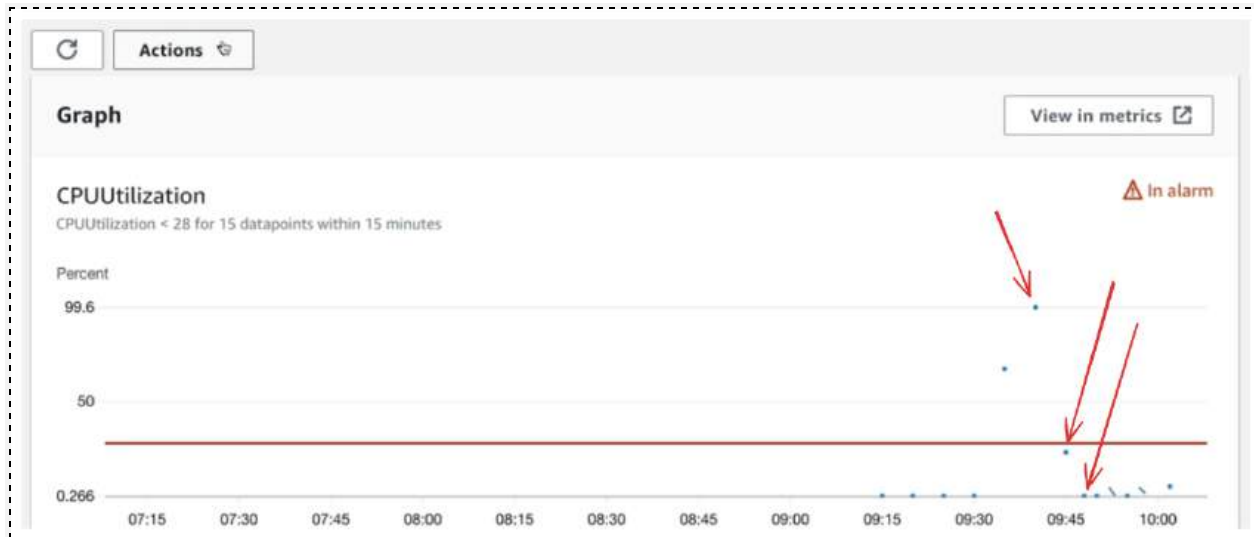
ولو روجت عال cloud watch alarm هتلاقي فعلا ان فيه alarm جايلك من target
policy اللي احنا كربتناها الأول لما بدانا خالص ان ال cpu اقل من 28 ف تمام هنكريت
instance واحدة ودي البداية والثاني alarm high بيوضحلك ان ال cpu زاد عن 40% ف
كدا هنخش ف instances اكثر





لما تيجي ترجع لل connection بينك وبين ال instance هتلاقيها معلقه علشان استهلاك ال cpu العالي ف روح عال instances dashboard واعملهم reboot يبقى كدا ال 0 هيرجع لل

هتلاقي بقا graph لل حصل في ال cloud watch .. طلع ثم نزل تاني لما عملنا reboot



ال ip بيتغير سواء v4 or v6 اما ال dns ثابت بتخش من خلاله عال instances



I'm Here

Question 3:

Elastic Load Balancers provide a

☐ static IPv4 we can use in our application

☒ static DNS name we can use in our application

☐ static IPv6 we can use in our application

Submit

Question 11:

For compliance purposes, you would like to expose a fixed static IP address to your end-users so that they can write firewall rules that will be stable and approved by regulators. What type of Elastic Load Balancer would you choose?

☐ Application Load Balancer with an Elastic IP attached to it

☒ Network Load Balancer

☐ Classic Load Balancer

Submit

مش هيجصل حاجة لانو على 3 = desired

Question 16:

You have an application hosted on a set of EC2 instances managed by an Auto Scaling Group that you configured both desired and maximum capacity to 3. Also, you have created a CloudWatch Alarm that is configured to scale out your ASG when CPU Utilization reaches 60%. Your application suddenly received huge traffic and is now running at 80% CPU Utilization. What will happen?

☐ Nothing

☐ The desired capacity will go up to 4 and the maximum capacity will stay at 3

☒ The desired capacity will go up to 4 and the maximum capacity will stay at 4

Submit



I'm Here

Question 20:

You have an ASG and a Network Load Balancer. The application on your ASG supports the HTTP protocol and is integrated with the Load Balancer health checks. You are currently using the TCP health checks. You would like to migrate to using HTTP health checks, what do you do?

☐ Migrate to an Application Load Balancer

☒ Migrate the health check to HTTP

Submit

Question 21:

You have a website hosted in EC2 instances in an Auto Scaling Group fronted by an Application Load Balancer. Currently, the website is served over HTTP, and you have been tasked to configure it to use HTTPS. You have created a certificate in ACM and attached it to the Application Load Balancer. What you can do to force users to access the website using HTTPS instead of HTTP?

☐ Send an email to all customers to use HTTPS instead of HTTP

☒ Configure the Application Load Balancer to redirect HTTP to HTTPS

☐ Configure the DNS record to redirect HTTP to HTTPS



I'm Here

CH-09 - AWS Fundamentals

RDS + Aurora + ElastiCache

➔001 Amazon RDS Overview (Relational Database Service)

- عبارة عن service بتقديمها AWS بتسهيلك ال setup, operation, and scaling of relational databases in the cloud

● Key Features

- **Supports multiple database engines**, including Amazon Aurora, PostgreSQL, MySQL, MariaDB, Oracle, and Microsoft SQL Server.
- **Automates tasks** such as hardware provisioning, database setup, patching, and backups.
- Provides **easy scaling** of database instance types and storage, Supports read replicas for read-heavy applications.
- **High Availability and Durability**: Multi-AZ (Availability Zone) deployments for automated failover and redundancy.
- **Automated backups** and **point-in-time recovery** options.

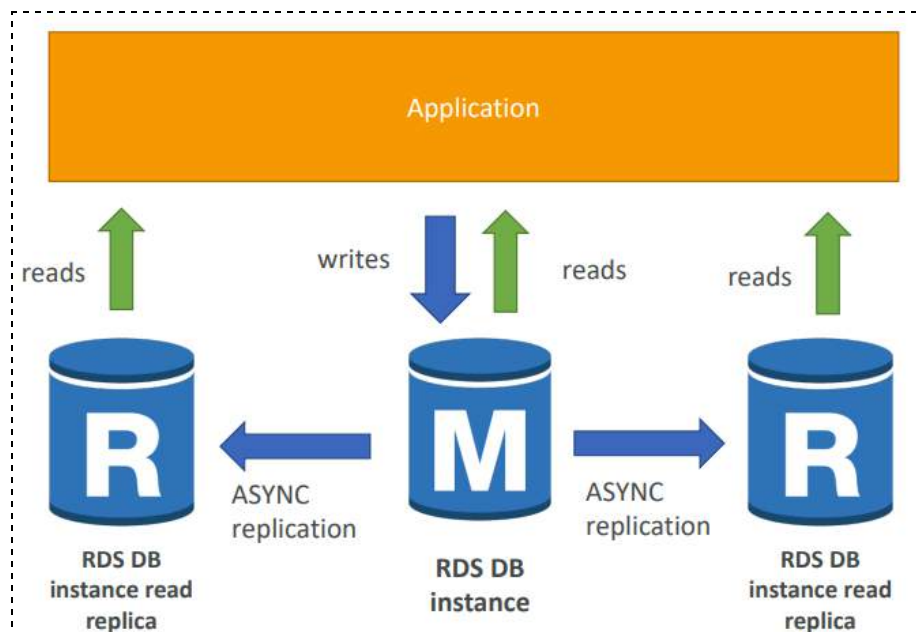


- Amazon **CloudWatch** integration for monitoring database performance and setting alarms.
- **Pay-as-you-go** pricing model.

BUT you can't SSH into your instances 💡

➡ 002 RDS Read Replicas vs Multi AZ

● Overview



- Replicas : نسخ متماثلة
- ASYNC = asynchronous غير متزامن
- SYNC = Synchronization متزامن



- Read Replicas : عبارته عن بديل من ال database الاساسية يحصل عليها read فقط مش read & write زي الاساسية بنستخدمها بحيث نوزع عليها حمل ال read بحيث نخف عن الاساسية.
- لو عندك web app عنده primary database بتهندل ال read , write operations وال app بي grow وبقي بيجيلو heavy traffic وده بيسبب performance issues علشان نحل المشكلة دي بنكريت Read Replicas من ال primary database بحيث انها تاخد ال read traffic وتخفف الضغط عال primary database instance
- ال data عال read replicas بتكون **eventually consistent** "متسقة في النهاية" معناها ان بيكون فيه delay ف ان ال replicas تاخد نفس ال recent data changes عال primary database
- ليك لحد **15 replicas** في ال AZ or Cross AZ or Cross Region

● Connection string

- عندك parameter اسمه **connection string** لازم ت configure it بحيث ان ال read traffic اللي جاي ميروحش عال primary database ولكن يروح عال replicas end point

● Replicas End point

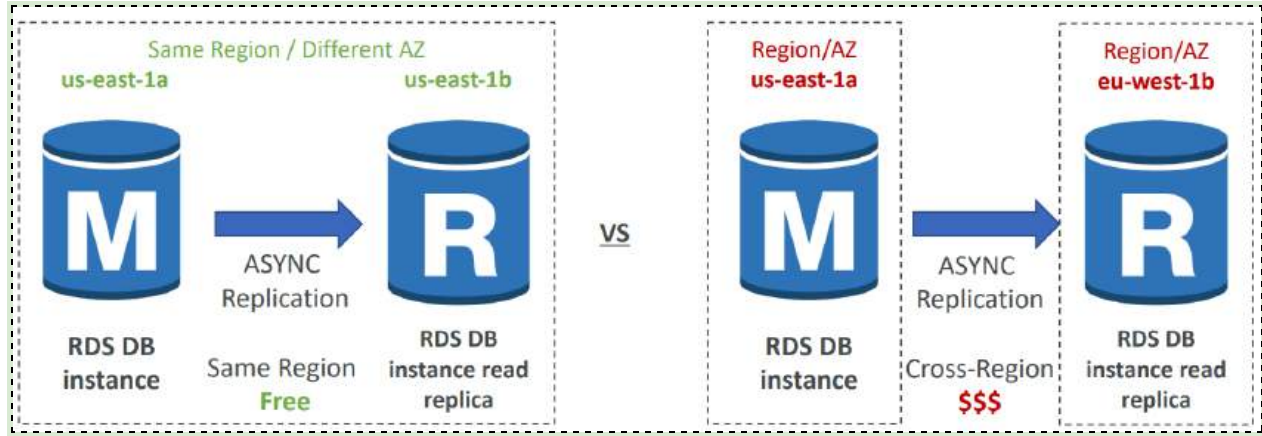
- ال **end point** معناها .. انت لما بتعمل replicas ببيكون ل كل واحدة منها & hostname port دول ال end point
- تقدر تحول ت promote ال replicas دي ل standalone database انها تكون primary ياغني يحصل عليها read & write

● Cost

💡 نقل ال data في الطبيعي من AZ الى AZ تانية بيندفع عليه cost أما في حالة ال

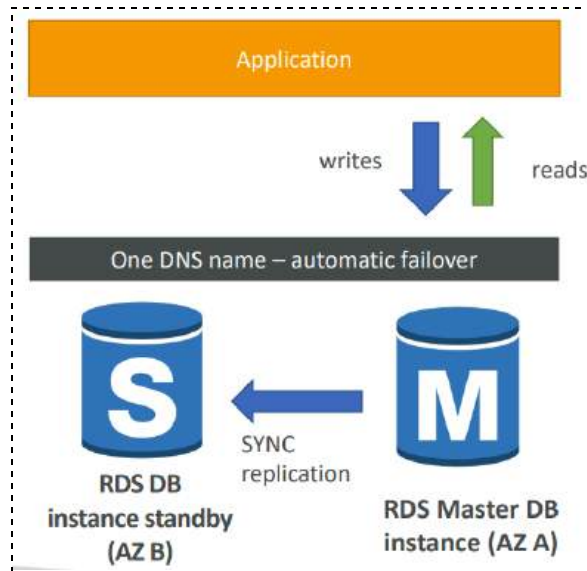
replicas عندك حالتين





- لو هتكرت replicas فى نفس ال Region مش هتدفع
- لو هتكرت replicas فى Regions مختلفة هتدفع

• RDS Multi AZ (Disaster Recovery)



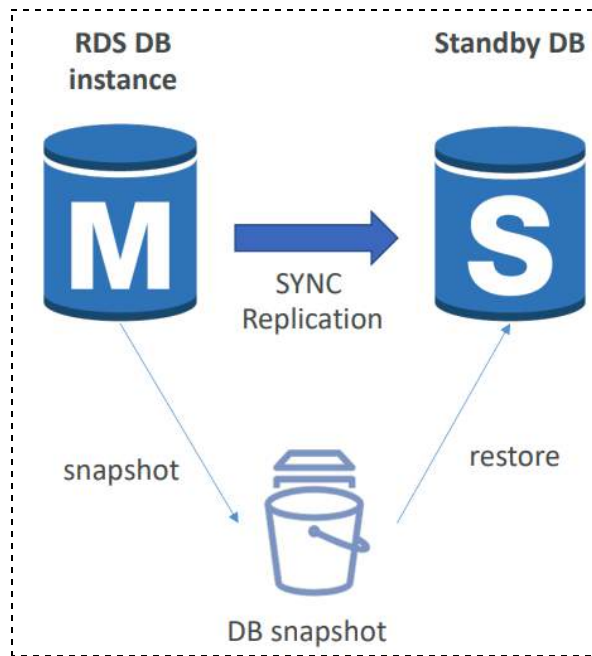
- لو عندك e-commerce application with critical data base علشان تضمن نفسك فى حالة لو ال database دي حصلها مشكلة بت deploy ال data base دي فى multi AZ بحيث لو واحدة وقعت الثانية تستلم مكانها.. بتسمى DB instance Standby
- الاتنين بيكونو مرتبطين بنفس ال DNS Name ف دة شئ كويس وبدون اي تدخل manual منك الموضوع هيتم من غير ماتحس.
- ال read replicas هيا كمان بيتعملها setup as multi az.



I'm Here

- علشان تخلي ال RDS multi AZ :

- فقط بت modify ال database وت enable ال multi AZ ولكن اللي بيحصل فى الخلفيه بيكون بالشكل ده



- بيتاخذ snapshot من ال primary database وبعددين بيحصل restore ليه ف AZ مختلفة وبيحصل بعدها Synchronization بينهم.



003 Amazon RDS Hands On

هل محتاج تحط كل ال configuration ولا محتاج حاجة فى السريع

Choose a database creation method [Info](#)

☒ Standard create

You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ Easy create

Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

ال engines المتاحة

Engine options

Engine type [Info](#)

☐ Aurora (MySQL Compatible) 	☐ Aurora (PostgreSQL Compatible) 
☒ MySQL 	☐ MariaDB 
☐ PostgreSQL 	☐ Oracle 
☐ Microsoft SQL Server 	☐ IBM Db2 

امكانيات ال instance اللى هتقوم عليها ال data base



I'm Here

DB instance size

☒ Production

db.r6g.xlarge
4 vCPUs
32 GiB RAM
500 GiB
1.017 USD/hour

☐ Dev/Test

db.r6g.large
2 vCPUs
16 GiB RAM
100 GiB
0.231 USD/hour

☐ Free tier

db.t3.micro
2 vCPUs
1 GiB RAM
20 GiB
0.020 USD/hour

1 primary & 2 standby **or** 1 primary & 1 standby **or** 1 primary only

Availability and durability

Deployment options [Info](#)

The deployment options below are limited to those supported by the engine you selected above.

☒ Multi-AZ DB Cluster

Creates a DB cluster with a primary DB instance and two readable standby DB instances, with each DB instance in a different Availability Zone (AZ). Provides high availability, data redundancy and increases capacity to serve read workloads.

☐ Multi-AZ DB Instance

Creates a primary DB instance and a standby DB instance in a different AZ. Provides high availability and data redundancy, but the standby DB instance doesn't support connections for read workloads.

☐ Single DB Instance

Creates a single DB instance with no standby DB instances.

ال storage اللى هتحتاجها نوعها اية وحجمها .. gp3 suitable for free tier

Storage

Storage type [Info](#)

Provisioned IOPS SSD (io2) storage volumes are now available.

General Purpose SSD (gp3)

Performance scales independently from storage

Allocated storage [Info](#)

20

GiB

Minimum: 20 GiB. Maximum: 65,536 GiB

i After you modify the storage for a DB instance, the status of the DB instance will be in storage-optimization. Your instance will remain available as the storage-optimization operation completes. [Learn more](#)

علشان ميكونش فيه تكلفه اختار النوع دة لازم db.t2.micro حتي لو ظهرلك تكلفه تحت ف

عادي لازم بس تختار النوع دة



I'm Here

Instance configuration

The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class [Info](#)

▼ Hide filters

☐ Show instance classes that support Amazon RDS Optimized Writes [Info](#)
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

☐ Include previous generation classes

☐ Standard classes (includes m classes)

☐ Memory optimized classes (includes r and x classes)

☒ Burstable classes (includes t classes)

db.t3.micro

2 vCPUs 1 GiB RAM Network: 2,085 Mbps

Estimated Monthly costs

DB instance	12.41 USD
Storage	2.30 USD
Total	14.71 USD

This billing estimate is based on on-demand usage as described in [Amazon RDS Pricing](#). Estimate does not include costs for backup storage, I/Os (if applicable), or data transfer.

Estimate your monthly costs for the DB instance using the [AWS Simple Monthly Calculator](#).

ان ربطتها ب instance موجودة مش محتاج تعمل network configuration هتلاقي ال security groups linked والامور تمام



I'm Here

Connectivity [Info](#)



Compute resource

Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☐ **Don't connect to an EC2 compute resource**
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☒ **Connect to an EC2 compute resource**
Set up a connection to an EC2 compute resource for this database.

EC2 Instance [Info](#)

Choose the EC2 instance to add as the compute resource for this database. A VPC security group is added to this EC2 instance. A VPC security group is also added to the database with an inbound rule that allows the EC2 instance to access the database.

Choose an EC2 instance



لو دخلت عال security جروب دة هتلاقى فى ال inbound rules بتاعته ال income مش
من anywhere ف هنعدل دة



I'm Here

Connectivity & security

Endpoint & port

Endpoint: database-1.cjuoe2eg2lwn.us-east-1.rds.amazonaws.com

Port: 3306

Networking

Availability Zone: us-east-1f

VPC: vpc-0799daa55c9fb6493

Subnet group: default-vpc-0799daa55c9fb6493

Subnets: subnet-0b033c07c2fc2d1b, subnet-0a3d455d7730dc9f2, subnet-017c65ab00faaf2af, subnet-0d0cb74f903576b2b, subnet-00f797a262b242263, subnet-09a2ba7c68db5b25e

Security

VPC security groups: demo-database-sg (sg-0804f2ab32b4f4091)

Active

Publicly accessible: Yes

Certificate authority: rds-ca-rsa2048-g1

Certificate authority date: May 26, 2061, 02:34 (UTC+03:00)

DB instance certificate expiration date: August 02, 2025, 03:40 (UTC+03:00)

Inbound rules (1)

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sg-0adfb00ce6778c56b	IPv4	MySQL/Aurora	TCP	3306	156.211.40.141/32	-

Inbound rules

Security group rule ID: sg-0adfb00ce6778c56b

Type: MySQL/Aurora

Protocol: TCP

Port range: 3306

Source: Anywhere-IPv4

Description - optional:

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Cancel Preview changes Save rules

بعد ما بتكرت ال database بيكون ليها ال endpoint اللي اتكلنا عنها من خلالها هنتصل عليها وده هيكون باستخدام sqlectron عبارته عن GUI سهله ف جوجل هتلاقه

sqlectron-1.38.0.tar.gz	99.4 MB
sqlectron-1.38.0.tar.xz	67.2 MB
sqlectron-1.38.0.x86_64.rpm	67.7 MB
sqlectron-Setup-1.38.0.exe	146 MB
sqlectron-Setup-1.38.0.exe.blockmap	157 KB
sqlectron_1.38.0_amd64.deb	67.6 MB



```
Sudo -i dpkg sqlectron_1.38.0_amd64.deb  
sudo apt remove sqlectron
```

بعد ما تفتحه هت test وبس

بالنسبة لل initial database/keyspace هتلاقية في ال configuration وانت بتعمل
ال database او تقدر بعد ما عملتها ترجع تشوفه من ال configuration tab

Server Information

Connection Test
Successfully connected

Name: RDS-Demo Database Type: MySQL SSL: ☐

Server Address: database-1.cjuoe2og2iwn.us-east-1:3306 Domain: Unix socket path:

User: admin Password: Initial Database/Keyspace: mydb Initial Schema: Schema

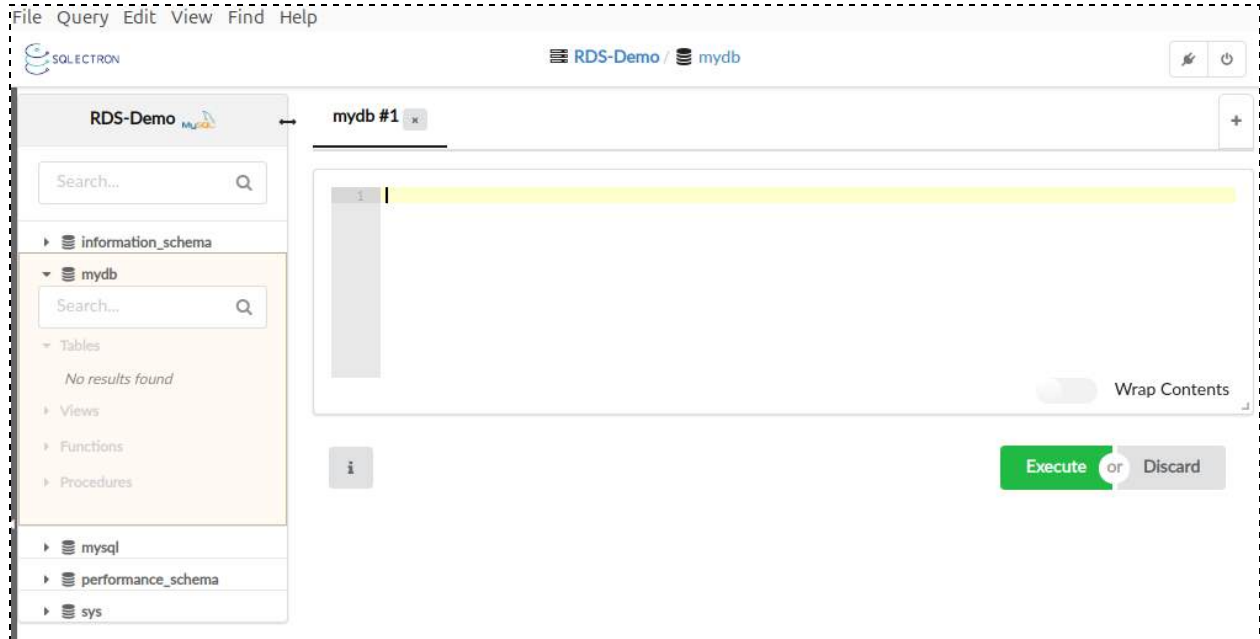
URI: mysql://admin:*****@database-1.cjuoe2og2iwn.us-east-1.rds.amazonaws.com/mydb
Make the password visible in order to change the database credentials through the URI format.

SSH Tunnel: ☐ Filter: ☐

Test Cancel Save



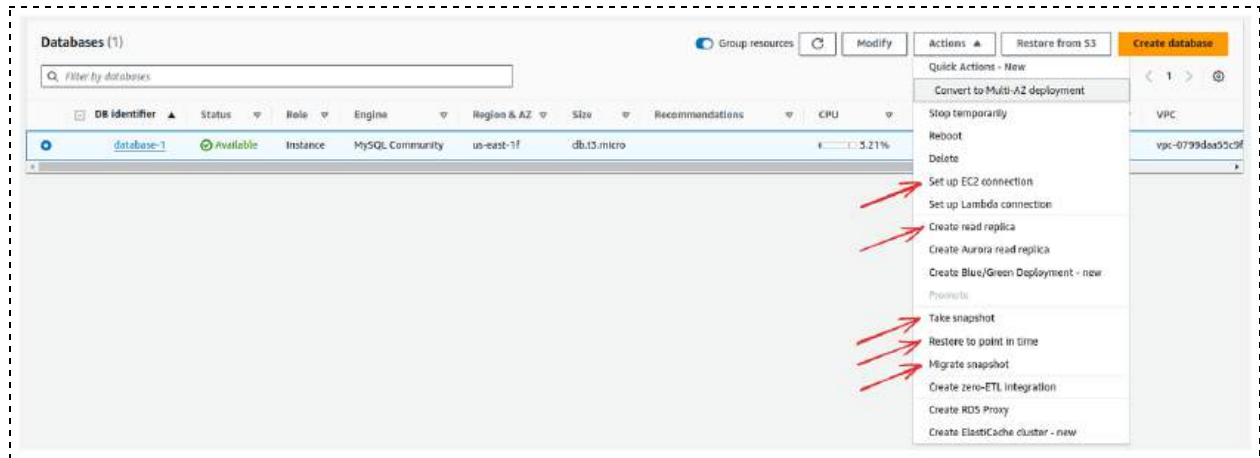
بعد ما ت save خش ب connect وتقدر بقا تعدل عليها ب SQL عادي.



من action تقدر ت create replica وال setting بسيطة.



من action هتلاقي كل ال options اللي ممكن تحتاجها



بعد ما تخلص امسحها ولو كنت مفعّل ال Deletion protection ارجع الغيها.



I'm Here

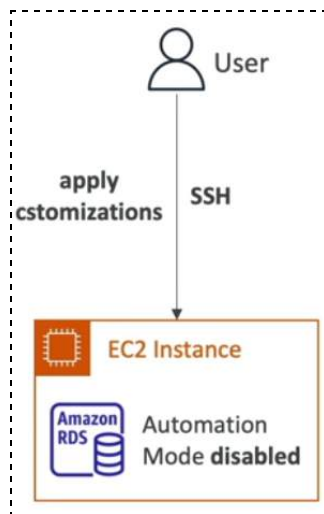
004 RDS Custom for Oracle and Microsoft SQL Server

- **RDS vs. RDS Custom**

- **RDS:** entire database and the OS to be managed by AWS

ال OS و database كل حاجة By AWS managed

- **RDS Custom:** full admin access to the underlying OS and the database



- من خلال RDS Custom تقدر يكون ليك full access عال OS و database وال instance ومنتساش تقفل ال Automation mode



005 Amazon Aurora

Relational				Non - Relational	
ID	Product Name	Size	Price	Key	Value
1	T-Shirt	S	\$20	1	Name: Bob Email: bob@example.com Birthday: Feb 18, 1983
2	Shorts	M	\$22	2	Name: Grace Address: 23 Street 2 Favorite Movie: Movie-x
3	Shirt	L	\$25		

● Relational Databases:

- من اسمها فيه علاقه بين ال tables جوا نفس ال database وال schema fixed
- اللغة اللي ب query بسال بيها عن ال data بتاعتي اسمها SQL Structured Query Language
- عكسها ال non-relational database

● Overview:

- عبارته عن relational database service و managed by AWS ومصممه علشان توفرلك high performance, high availability, and scalability و compatible مع MySQL & PostgreSQL

● Key Features

1. High Performance:

- **Optimized Storage:** Aurora uses a distributed, fault-tolerant, self-healing storage system that automatically grows in increments of 10 GB, up to 128 TB per database instance.



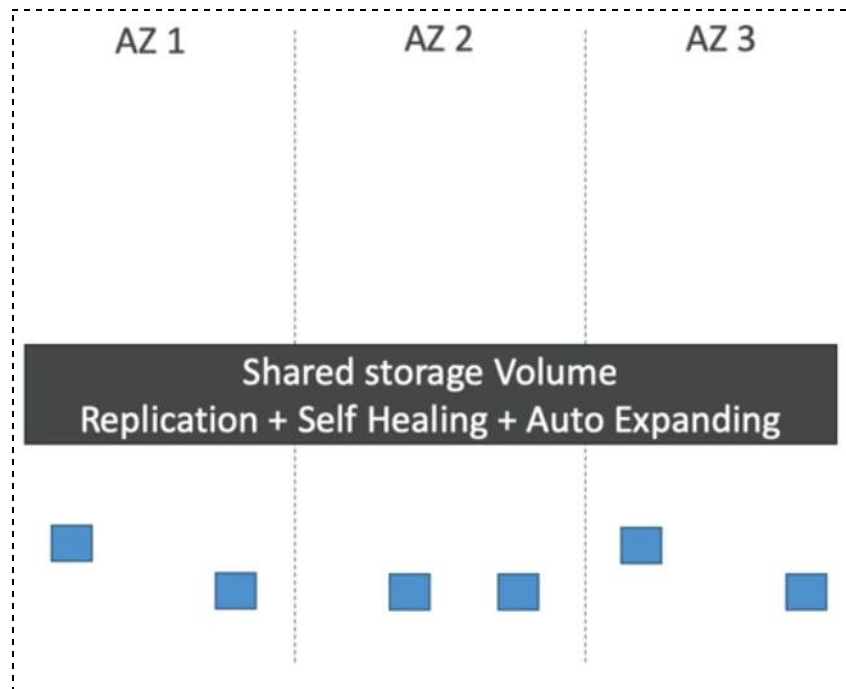


- ال Self healing يعني لو فيه مشكلة حصلت فى الداتا corrupted تلفت هيا هتصلحها من نفسها .. عندك نسخ منها فى اماكن تانيه ف بتاخذ منهم تحط مكان اللى تلف مش بتخترع جديد.

- **Low Latency:** It provides low-latency reads and writes, with the ability to deliver up to five times the throughput of standard MySQL and three times the throughput of standard PostgreSQL databases.

2. High Availability and Durability:

- **Fault-Tolerant Storage:** Data is replicated across six copies across three Availability Zones (AZs), ensuring data durability and high availability.



- لو عملت write ل some data هتلاقى عندك 6 نسخ من ال data على AZ 3..يبقى ال

aurora شبه ال RDS multi AZ

- **Automatic Failover:** Aurora automatically detects and handles instance failures, with failover typically completing within 30 seconds.
- **Multi-AZ Deployment:** Supports Multi-AZ deployments for automatic failover and increased availability.



I'm Here

3. Scalability:

- **Aurora Replicas:** Supports up to 15 low-latency read replicas, allowing you to scale read operations.

- متاح ليك 15 replicas خلال ال AZs بجانب ال Main database

- **Aurora Serverless:** Automatically scales the database capacity up or down based on application needs, allowing you to pay only for the resources you use.

4. Scalability:

- **Encryption:** Data is encrypted at rest using AWS Key Management Service (KMS) and in transit using SSL.
- **Network Isolation:** Integration with Amazon VPC allows you to isolate your database within a virtual network.
- **IAM Integration:** Allows fine-grained access control and authentication using AWS Identity and Access Management (IAM).

5. Backup and Recovery:

- **Automated Backups:** Continuous backups to Amazon S3 with point-in-time recovery.
- **Snapshot Backups:** Manual snapshots can be taken and restored as needed.

6. Compatibility:

- **MySQL-Compatible:** Aurora MySQL is compatible with MySQL 5.6, 5.7, and 8.0, allowing you to use existing MySQL tools and applications.
- **PostgreSQL-Compatible:** Aurora PostgreSQL is compatible with PostgreSQL 9.6, 10.x, 11.x, and 12.x, providing similar compatibility benefits.

- أغلي بحوالي 20% من ال RDS الباقيين.. ولكن هي more efficient



• What if the master database failed for RDS and Aurora:

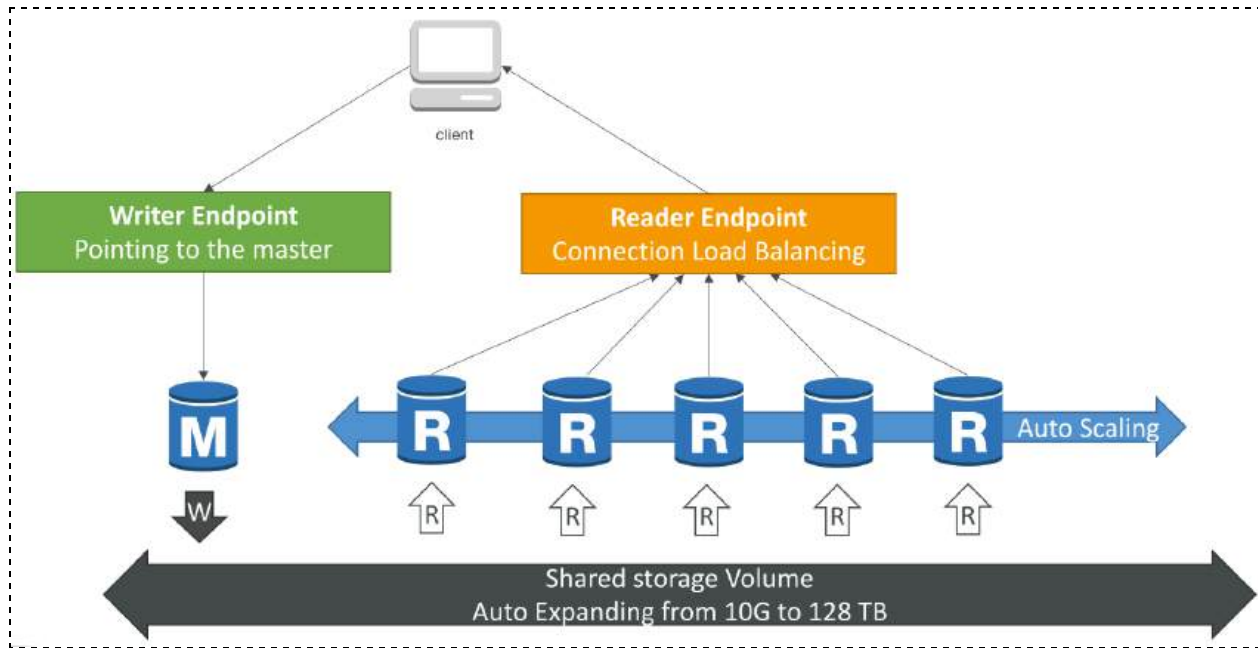
• Aurora

- كذا كذا عندك 15 replicas متاحين اي واحدة منهم تقدر ت promote to be the master ودة بيحصل automatic فى خلال 30ث

• Amazon RDS

- Have RDS Multi AZ
 - لو عندك زي ما شرحنا هيكون عندك بجانب ال master هتلاقى standby instance هتاخذ الدور لو ال master failed
- Don't Have RDS Multi AZ
 - لو معندكش بقا ال Down time هيطول على ما ت restore بقا من خلال backup ولا تشوف المشكلة فين وتصلحها

• Aurora DB Cluster



- ال master DB بتبقى مرتبطة ب writer endpoint عباره عن DNS name وحتى لو ال master حصلها مشكله ال client بيتوجه عال writer endpoint دي وبتوجهه عال instance اللى حصلها promote انها تبقى ال master



I'm Here

- قولنا ان بيكون عندنا 15 replicas ف صعب عليك ت track ال 15 replicas دول خصوصها انهم بيعملو auto scaling

● The diff between Aurora Server less and normal one

● Server less

- مناسبة اكثر لل unpredictable workloads ونظام الدفع pay as you go

➡ 006 Amazon Aurora - Hands On

هيكون مكلف ال hands on دة.

يعتبر كل ال configuration اتكلمنا فيها.

بعد ما كريتناها .. هتلاقها بالشكل دة

- ال Regional Cluster اللي بيوضحك انت ف انهي region

- Writer instance ودة ال main data base اللي بيحصل عليه read & write

عادي وبيكون عدده 1 فقط

- Reader Instance ودة بيحصل عليه read فقط وبيبقى فيه لحد 15

Databases Group resources Modify Actions Restore from S3 Create database					
Filter by databases					
< 1 > ⚙					
DB identifier	Role	Engine	Region & AZ	Size	
database-2	Regional cluster	Aurora MySQL	eu-central-1	2 inst	
database-2-instance-1	Writer instance	Aurora MySQL	eu-central-1b	db.t3	
database-2-instance-1-eu-central-1a	Reader instance	Aurora MySQL	eu-central-1a	db.t3	



I'm Here

بعد ما ت select ال Regional Cluster هتلاقي ال endpoints الخاصة بال reader & writer وال ports وكدا. وممكن تحدد اي instance لوحدها وهيظهرلك برودو ال endpoint

Connectivity & security	Monitoring	Logs & events	Configuration	Maintenance & backups	Tags
Endpoints (2)					
Filter by endpoint Actions Create custom endpoint					
Endpoint name	Status	Type	Port		
database-2.cluster-ro-c2wzuohjkj1v.eu-central-1.rds.amazonaws.com	Available	Reader instance	3306		
database-2.cluster-c2wzuohjkj1v.eu-central-1.rds.amazonaws.com	Available	Writer instance	3306		

عندك Actions كتير تقدر تعملها

Reader instance = replica

- تزود reader instance فى AZ مختلفه فى نفس ال Region

- تزود replica فى Region مختلفه

- حصل مشكلة فى ال main مثلا ف ت restore to point in time

- تضيف Replica ل Auto scaling

database-2

Modify

Actions

Related

Filter by databases

DB identifier	Role	Engine
database-2	Regional cluster	Aurora MySQL
database-2-instance-1	Writer instance	Aurora MySQL
database-2-instance-1-eu-central-1a	Reader instance	Aurora MySQL

Stop

Delete

Add AWS Region

Add reader

Create cross-Region read replica

Create clone

Promote

Restore to point in time

Backtrack

Add replica auto scaling



I'm Here

Replica to Auto Scaling Group دلوقتي هنضيف

بنكریت policy وظيفتها ان معناها لو ال CPU Utilization زاد عن حد معين وليكن 50% او

عدد ال connection زاد عن رقم كذا عال Replicas اللى already موجودة .. كريتللي

replica اكثر

Policy details

Policy name
A name for the policy used to identify it in the console, CLI, API, notifications, and events.

Policy name must be 1 to 256 characters.

IAM role
The following service-linked role is used by Aurora Auto Scaling.

AWSServiceRoleForApplicationAutoScaling_RDSCluster

Target metric
Only one Aurora Auto Scaling policy is allowed for one metric.

☒ Average CPU utilization of Aurora Replicas [View metric](#)

☐ Average connections of Aurora Replicas [View metric](#)

Target value
Specify the desired value for the selected metric. Aurora Replicas will be added or removed to keep the metric close to the specified value.

50 %

► Additional configuration

Maximum Replicas Number وبتحدد عدد ال

Cluster capacity details

Configure the minimum and maximum number of Aurora Replicas you want Aurora Auto Scaling to maintain.

Minimum capacity
Specify the minimum number of Aurora Replicas to maintain.

1 Aurora Replicas

Maximum capacity
Specify the maximum number of Aurora Replicas to maintain. Up to 15 Aurora Replicas are supported.

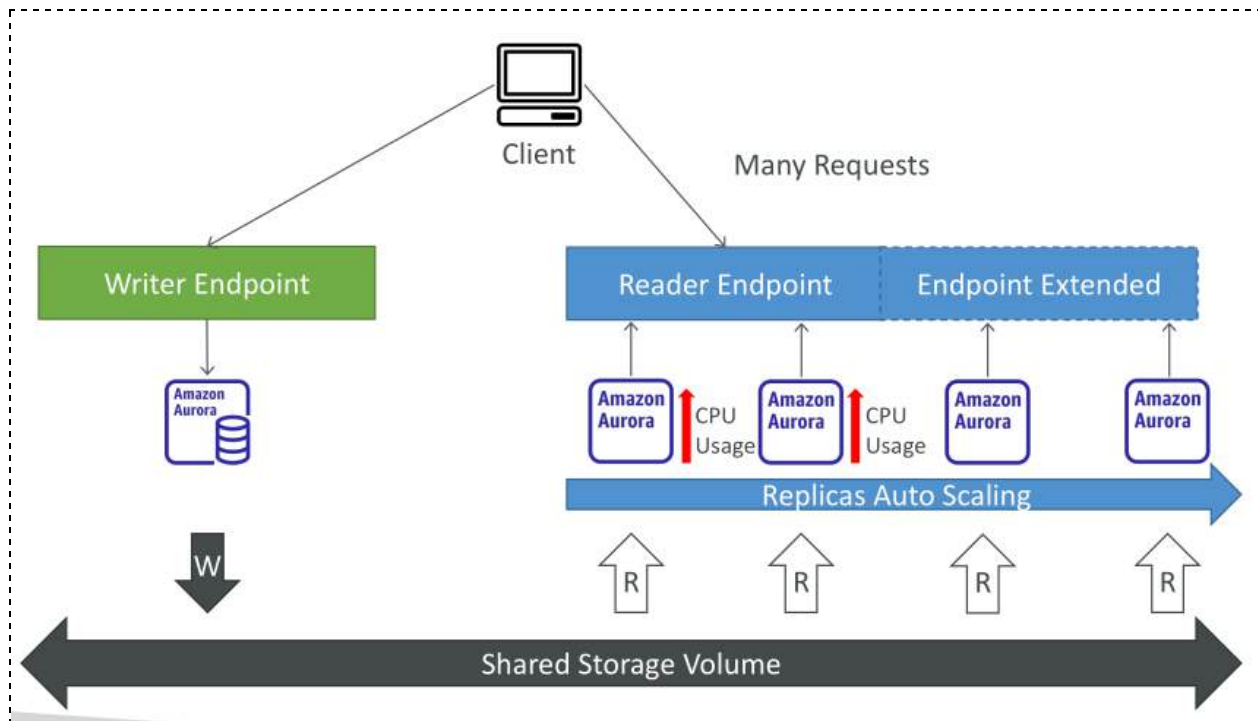
15 Aurora Replicas



I'm Here

007 Amazon Aurora - Advanced Concepts

- Aurora Replicas - Auto Scaling

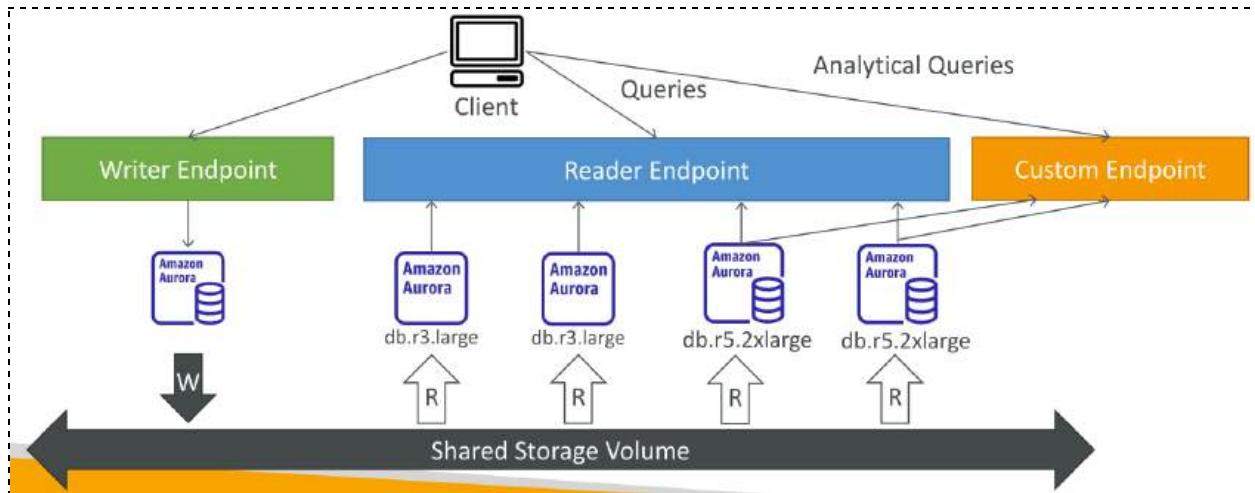


- بعد ما ال CPU utilization يزيد ال replicas بتزيد ب auto scaling وال reader endpoint هيا كمان بت extend معاهم.



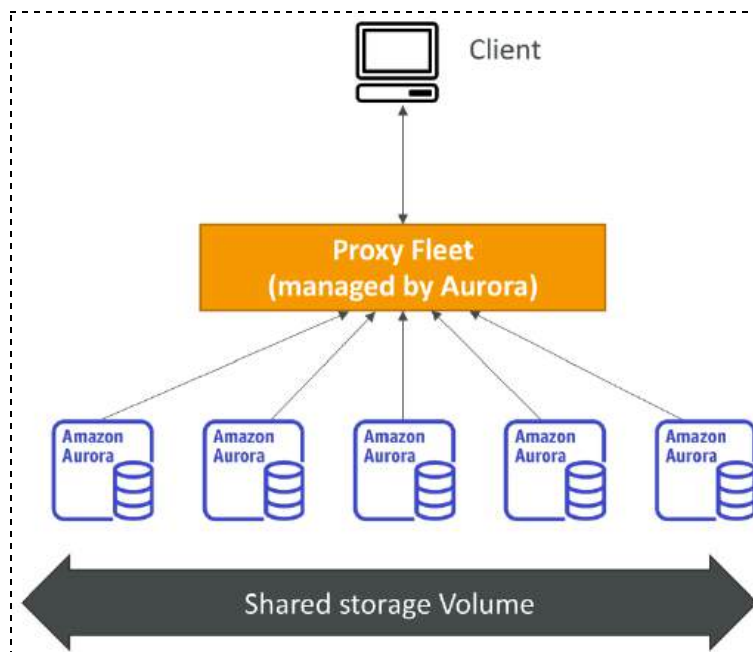
I'm Here

• Aurora – Custom Endpoints



- بنكرت ال custom endpoint علشان نوزع traffic معين على instance معينة .. بمعنى انك مثلا علوز ال Analytical queries traffic يروح على instance تستحمل الشغل ده تكون اقوي من ال replicas العادية ف بنعمل isolation بين ال reader & custom endpoint

• The diff between Aurora Server less and normal one



- مناسبة اكثر لل unpredictable workloads او ال infrequent access ونظام الدفع pay as you go ومش محتاج تعمل capacity planning

I'm Here



- ال data bases جاهزة بال auto scaling بناءا عال usage بتاعك.
- يبقى ال client بي connect ل proxy fleet ودة managed by Aurora وبيتوزعه عال instances اللى بيكونو محطوطيين فى auto scaling وعلى حسب بقا ال usage ال instances بتزيد او بتقل.

● Global Aurora

The screenshot shows the AWS Aurora console interface. At the top, there's a 'database-2' header with 'Modify' and 'Actions' buttons. Below this is a 'Related' section with a search bar 'Filter by databases'. A table lists the database components:

DB identifier	Role	Engine
database-2	Regional cluster	Aurora MySQL
database-2-instance-1	Writer instance	Aurora MySQL
database-2-instance-1-eu-central-1a	Reader instance	Aurora MySQL

The 'Actions' menu is open, showing the following options: Stop, Delete, Add AWS Region, Add reader, Create cross-Region read replica, Create clone, Promote, Restore to point in time, Backtrack, and Add replica auto scaling.

- من **Action** زي مشوفنا تقدر كمان تضيف **AWS Region** بس لازم تكون مختار Aurora Version بيدعم ال Global database feature ونوع instance يكون compatible مع ال Global database feature
- إستخدامها : بيكون عندك ال main + reader replicas فى ال primary region ولكن انت مثلا محتاج ت serve عملاء اوربيين مثلا ف عاوز يبقى عندك replicas ف region تانيه بحيث يكون الموضوع اسرع ف بالتالي بت add region .. يبقى انت كنت ف وليكن us-east-1 (N. Virginia) region

eu-west-1 (Ireland) to serve European users with low-latency read access and to have a disaster recovery solution.



- **Aurora Global Database (recommended):**

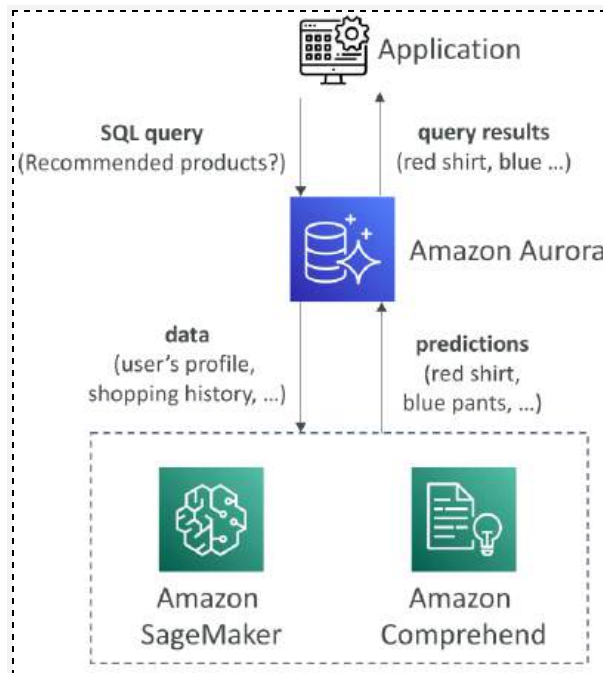
- دة اللى تقدر يكون عندك:

1. **Primary region** >> read & write
 2. **Up to 5 secondary (read-only) regions**, replication lag is less than 1 second
 3. Up to 16 Read Replicas per secondary region
- ال 15 read replicas primary ياغني + 1 main ياغني توتال 16 .. ف عندك برودو فى ال 5 regions الاضافيه اللى تقدر تضيفهم كل واحدة يكون فيها 16 replicas
- Promoting another region (for disaster recovery) has an RTO of < 1 minute
 - Typical cross-region replication takes less than 1 second



I'm Here

● Aurora Machine Learning



- عبارة عن integration بين Aurora و AWS Machine Learning Services
- عندك مثلاً Aurora متصلة ب ML Services معينة ف تقدر انت من خلال SQL simple Query ي run it ال application وليكن مثلاً اية ال recommended products ف Aurora بما انها database تبعت شوية داتا لل ML services زي ال shopping profile , shopping history وهكذا وتشتغل ال ML services دي عالداتا دي وتعرف اية ال recommended products .. وتبعت ال query results دي لل Aurora اللي هتبعها لل APP.



➡ 008 RDS & Aurora - Backup and Monitoring

● RDS Backup:

● Automated Backup

- يحصل backup كل 5 دقائق بيتجدد يعني لو عملت restore ل any point in time هت restore اخر نسخة اللي معموله من حوالي 5 دقائق .. وال backups دي بتفضل من 0 : 35 يوم

● Manual DB Snapshots

- دي بتعملها manual والفرق بينها وبين ال automated ان دي بتفضل موجودة لحد ما انت تمسحها الثانية اخرها 35 يوم

💡 لو مش عاوز costs عالية .. لو بتشتغل مثلا على database ساعتين فقط في الشهر ..

بدل ما تشغل ال Automated Backup ويعد عليك كل ساعة ال data بتاعة ال backup

دي موجودة فيها .. خد افضل snapshot بعد ما تخلص ساعتين الشغل دول وابقى اعمالها

restore دة ارخص بكثير.. لانك حتي لو عامل stop لل data base بعد ما تخلص الساعتين

شغل ف انت هتدفع عليها حتي لو واقفة.

● Aurora Backup:

● Automated Backup

- يحصل من 1 : 35 يوم ومتقدرش توقف ال backup دي اقل حاجة هتفضل ال data يوم.. وليك point in time recovery في اي وقت خلال المدة دي.

● Manual DB Snapshots

- زيها زي اللي فوق .. بتعملها manual وبتخليها موجودة زي مانت عاوز وتمسحها وقت ما تعوز.



● RDS & Aurora Restore options

- لما بت restore ال backup سواء من snapshots , Automated , فبتعمل new database بيها.
- تقدر ت restore database من S3 ودي عبارة عن service بنخزن بيها ال objects على AWS Cloud هنتكلم عنها بالتفصيل قدام .

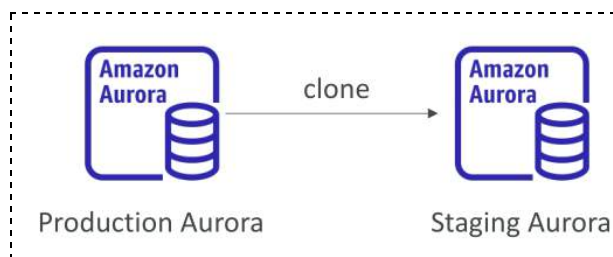
• Restoring MySQL RDS database from S3

- Create a backup of your on-premises database
- Store it on Amazon S3 (object storage)
- Restore the backup file onto a new RDS instance running MySQL

• Restoring MySQL Aurora cluster from S3

- Create a backup of your on-premises database using Percona XtraBackup
- Store the backup file on Amazon S3
- Restore the backup file onto a new Aurora cluster running MySQL

● Aurora Database Cloning



- بت clone بتنسخ ال main database اللي عال production لو محتاج تعمل test عليها ف حاجة معينة علشان متضرهاش هيا .. ف بتعمل ال test بتاعك عال cloned database ودة اسرع من انك تاخذ snapshot وت restore it بتستخدم protocol اسمه copy-on-write



➡ 009 RDS & Aurora Security

● Encryption at Rest

- لما بت enable ال Automatic Encryption لل RDS or Aurora ف كل ال data المتخزنة بتكون encrypted دة بيشمل ال database instance, automated backups, read replicas, and snapshots.
- مينفعش ت enable ال encryption على database مش معمولها encryption الحل انك بتاخذ snapshot وتعملها restore as new encrypted instance

● Encryption in Transit

- RDS & Aurora بي support encryption in transit using SSL/TLS.
- بت configure ال database connection to use SSL/TLS بحيث تضمن ان ال data اللي بتنتقل بينك وبين ال app encrypted

● IAM Database Authentication

- ممكن ت create IAM Policies تسمحلك بانك ت access ال RDS & Aurora
- ممكن تستخدم IAM Tokens بدل من ال data base passwords
- IAM authentication relies on temporary authentication tokens generated by AWS. These tokens are valid for 15 minutes and can be used in place of a password when connecting to the database.

● Security Groups

- تقدر ت control the network access to your database من خلال انك تقدر تتحكم ف مين ي access ال database ممكن ip معين او ports معينه ..

● SSH Access

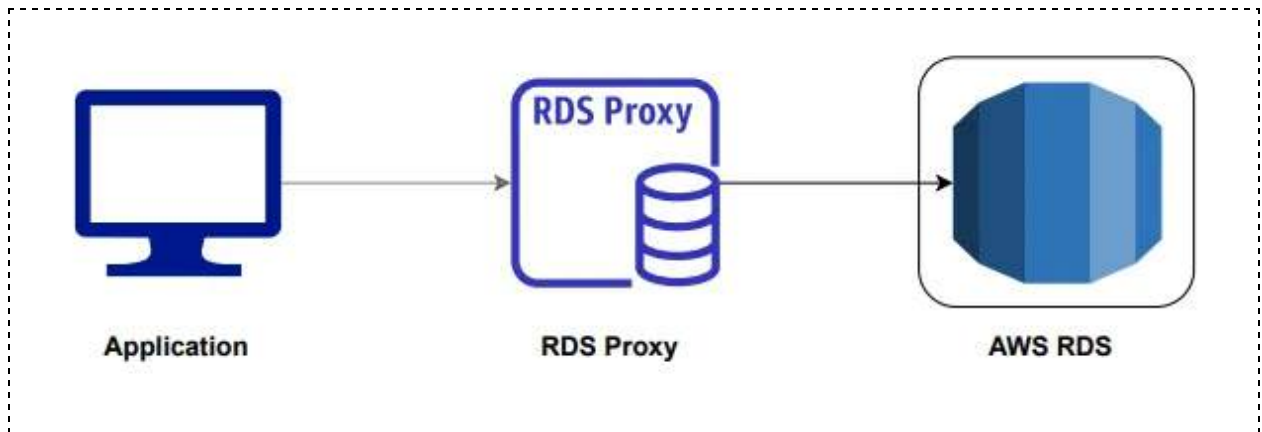
- متقدرش ت ssh عال databases دي لانها managed serves by AWS ولكن ممكن من خلال RDS Custom اللي اتكلمنا عليها قبل كدا.



● Audit logs

- فيها كل حاجة حصلت على ال data bases وكل ال queries ولكن مش بتفضل موجودة لوقت طويل .. ف علشان كذا تقدر تبعت ال audit logs دي لل cloud watch علشان تحفظها هناك.

➡ 010 RDS Proxy



- **Proxy** باختصار عبارته عن وسيط بين ال client & Server في حالتنا ال application , database server

● Use Case

- طب هو وسيط بين ال database , client في الحالة دي ليه محتاجينه طالما احنا نقدر ن connect directly to the database ؟ في عندنا e-commerce app مثلا واثنا ال sale days فيه ضغط كبير و connections كتيرة رايحة عال database ف دة طبعا ممكن يوقعها ف علشان كذا بنستخدم ال RDS Proxy بحيث يخف الضغط عنها وهو بيكون عبارته عن serverless و multi AZs وبي scale up and down بناء على حجم ال connection وبي direct the connections بسرعه سواء ل standby database او replica



I'm Here

- **Supports** RDS (MySQL, PostgreSQL, MariaDB, MS SQL Server) and Aurora (MySQL, PostgreSQL)

- **RDS Proxy Enhanced Security**

- لو عندك web app وعاوز ي connect to a RDS :

- **Without RDS Proxy**

- فى الغالب بت save the user name & password فى ال app code ودة خطر.

- **With RDS Proxy**

- تقدر تستخدم IAM roles فى انك تتحكم ف who can connect to the database

- او بي retrieve ال database credentials من ال AWS Secrets Manager وبي

Update it automatically ف ب كدا مش محتاج تحط ال password فى ال application

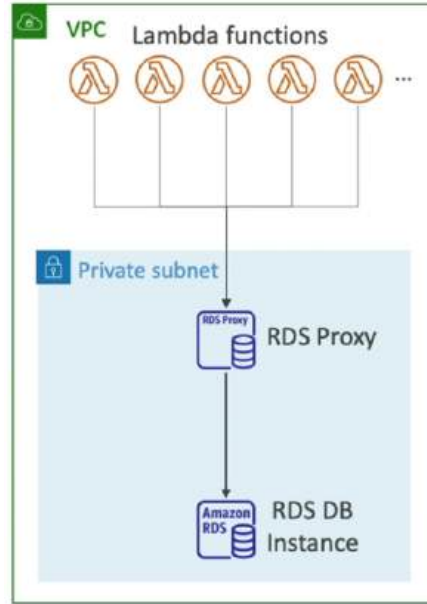
- **AWS Secrets Manager**: عبارته عن service بيتخزن فيها ال sensitive database passwords زي ال informations

💡 RDS Proxy لازم يحصله access فقط من خلال ال VPC مش من ال public ..

يعني ال resources فقط اللي جوا ال VPC تقدر تأكسسه غير كدا لا

- من اكثر ال services اللي بتستخدم RDS Proxy هي Lambda function وهنتكلم عنها بالتفصيل بعدين ..





➡011 ElastiCache Overview

● Diff Between Cookie & Cache

● Cookie

- غرضها انها تخزن ال data اللى ليها علاقة بال user login session وال preferences بتاعته شوية داتا بسيطة ياعني .. وبتتخذ عند ال user's browser وزى ما شرحنا قبل كذا بتتبع فى ال requests اللى بين ال server وال user .. وال cookie دي ال expiration date بتاعها بنعملو set من خلال ال server ولو انتهى او عملت مسح لل cookies اللى فى ال browser مثلا تلاقي ان ال login sessions عملت log out وكذا زى ما بيحصل ..

- Cookies are sent with every HTTP request to the server.

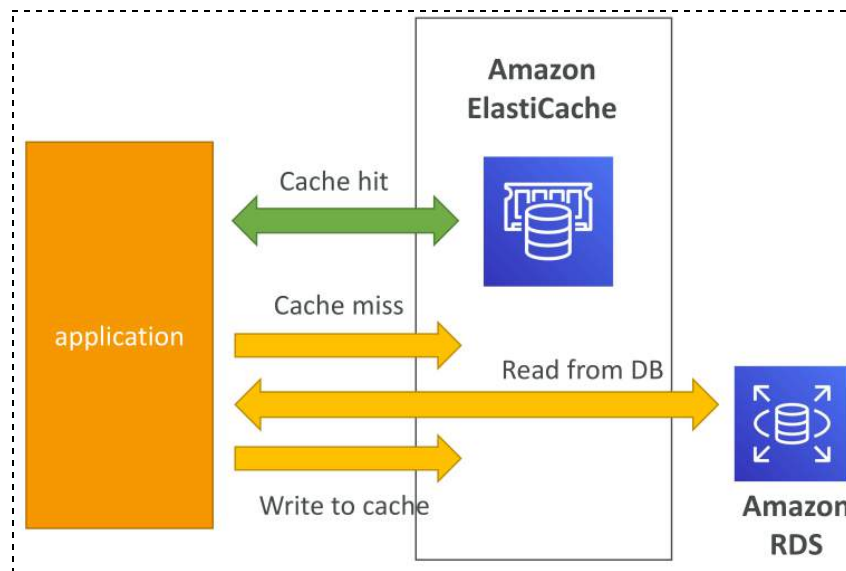


- **Cookie**

- أكبر شوية من اللي فانتت عبارته عن web resources غرضها تقليل ال load times عال server او ال database وتقليل ال bandwidth .. عبارته عن files زي HTML pages, images, CSS stylesheets, JavaScript files, and other multimedia content بتتخزن locally بردو عال user's computer by the browser ويحصلها access لو جيت تزور نفس ال website او ال page .. ليها بردو expiration date محطوط في ال cache headers الكلام ده اتعمله set at the server
- Cached resources are accessed locally by the browser without additional HTTP requests to the server.

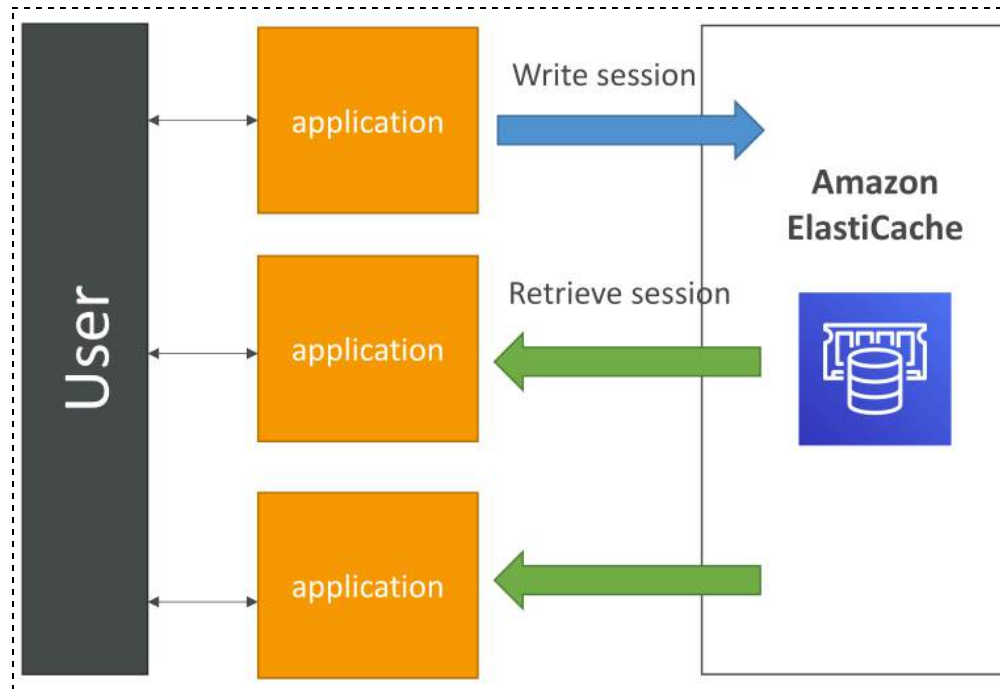
- **Amazon ElastiCache**

- عبارته عن Managed Service By AWS



- لو ال application عمل ElasticCache query لو ال queries دي already موجودة ومتخزنة في ال AWS ElasticCache ف ده اسمه **Cache Hit**
- لو ال caches دي مش موجودة في ال AWS ElasticCache ف هيروح ال App يجيبها من ال DataBase





- مثال كمان : ال user هي login ف اي واحد من ال Apps بتوعك فال App هي Write ال Session data دي في ال ElasticCache .. ف لو ال user دة حصلو redirecting لاي instance تانية شايله ال App بتاعك ف ال instance دي بت retrieve علطول ال session data دي من ال ElasticCache ف بيحصل ال user logged in ومش بيحتاج ي log in again دة كذا اسمه ان ال Application بتاعك **Stateless**

- **Amazon ElastiCache Engines**

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